



Department of Information Science & Engineering

Term: 03rd Oct 2024 – 25th Jan 2025

Semester: 5

Subject Code: IS53

Credits: 4:0:0

Subject Name: Computer Networks

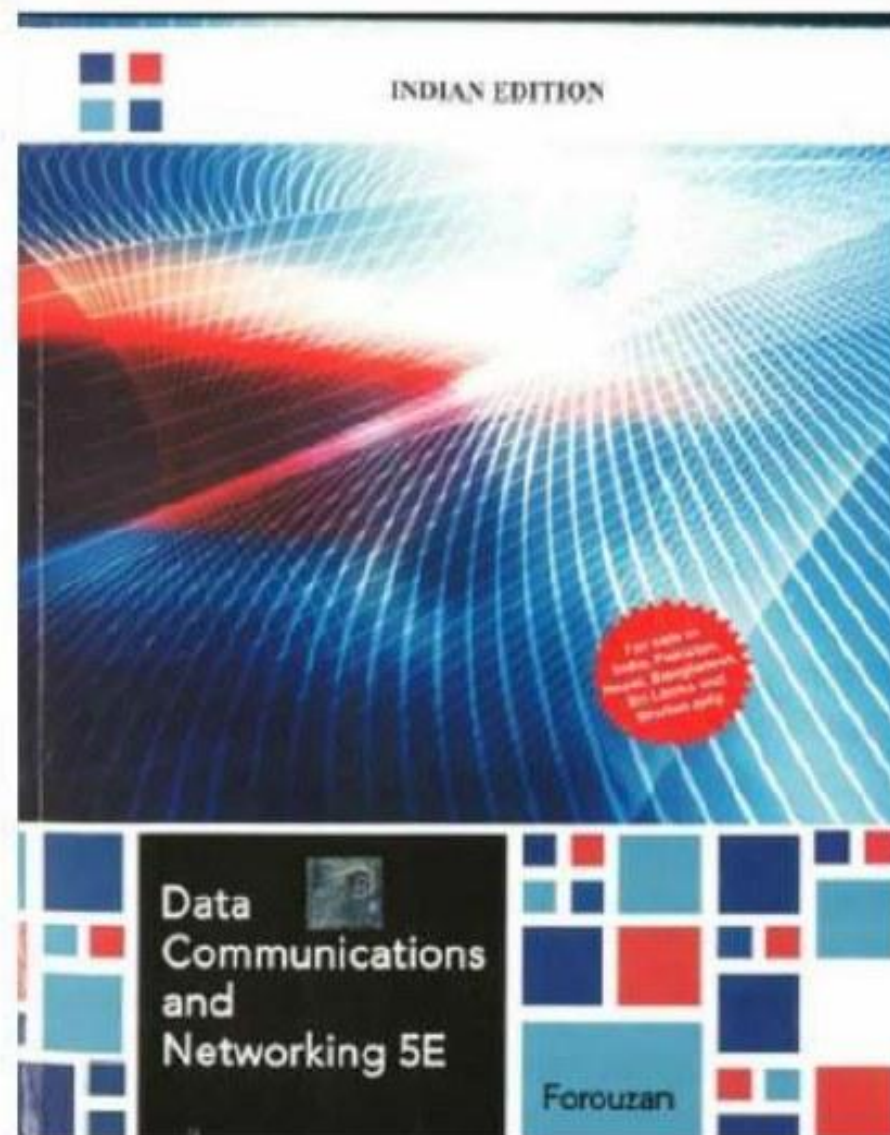
Computer Networks

Text Book:

Behrouz A. Forouzan,
Data Communications
and Networking,
Fifth Edition,
Tata McGraw-Hill, 2013

Faculty handling this subject

Prof. Suresh Kumar K R



DISCLAIMER

- Majority of the contents used in this presentation is from the Text Book “Data Communications and Networking, 5th Edition, Tata McGraw-Hill, 2013” by Behrouz A. Forouzan.
- Few images and contents used in this presentation is taken from various websites, textbooks and from the presentations of different authors.
- This presentation is purely used for learning and teaching purpose in an educational institution.
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UNIT-I

- **Data communication Fundamentals** - Introduction, Networks, Network types (LAN, WAN, MAN, Internet), Protocol Layering, TCP/IP protocol suite, OSI Model.
- **Physical layer** - Data and Signals, Basics of Periodic analog signals and digital signals, Transmission impairments, Data rate limits, Line coding schemes (unipolar, polar, bipolar) in digital-to-digital conversion

UNIT-II

- **Data Link layer** - Introduction, Link layer addressing – Three types, ARP, Error detection and correction: Types of errors, redundancy, error detection vs error correction, block coding, cyclic codes – CRC and polynomials, checksum, Data link control: Framing, Media Access Control: Random Access Protocols, Controlled Access Protocols, Channelization.

UNIT-III

- **Network layer** - Network layer services, Packet switching, Network-Layer Performance, IPV4 Addressing, Internet protocol (IP) datagram format, fragmentation, IP options, Routing algorithms: Distance Vector Routing, Link State Routing, Path Vector Routing, Multicast Distance Vector (DVMRP).

UNIT-IV

- **Network Layer** - IPV6 addressing, The IPV6 Packet format, Extension header, Transition from IPV4 to IPV6
- **Transport Layer** - Introduction, Transport layer Protocols, Simple Protocol, Stop-and-Wait Protocol, Go-Back-N Protocol (GBN), Selective-Repeat Protocol, Bidirectional Protocols: Piggybacking.

UNIT-V

- **Transport Layer** - Services, Port Numbers, User Datagram Protocol (UDP) – user datagram, services and applications, Transmission control Protocol: TCP Services and features, Segment, TCP Connection, Flow control, Error control, and TCP Congestion control; Domain Name System (DNS) – Purpose and Resolution.

Computer Networks Assessment

30 Marks

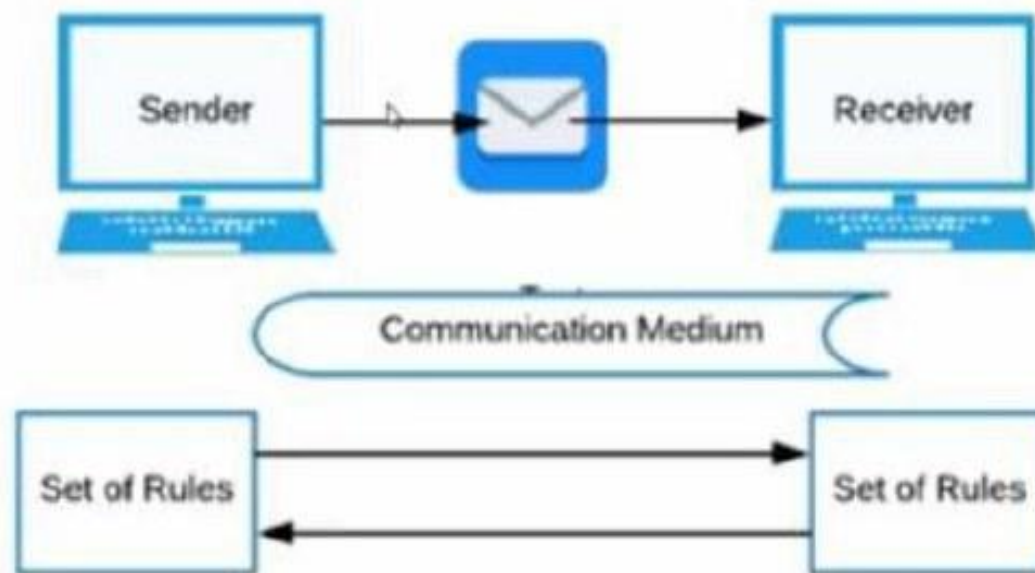
2 tests each for 30 marks & Average of those 2 taken

20 Marks

NS2 project

DATA COMMUNICATIONS

*The term **telecommunication** means communication at a distance. The word **data** refers to information presented in whatever form is agreed upon by the parties creating and using the data. **Data communications** are the exchange of data between two devices via some form of transmission medium.*



Simple Data Communication Model

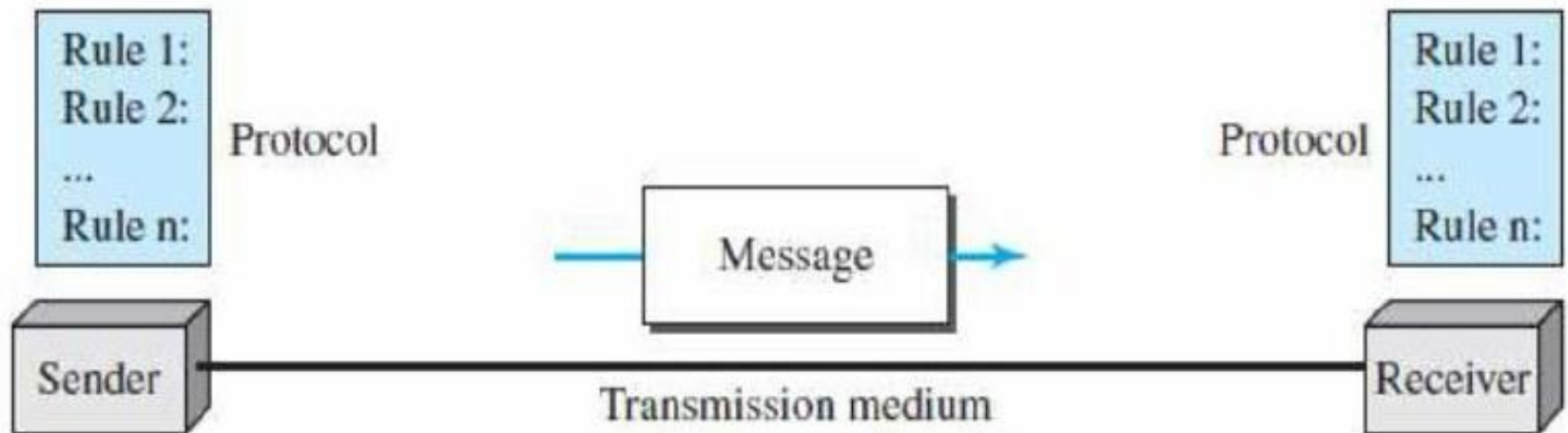
Fundamental Characteristics

- ✓ **Delivery**
- ✓ **Accuracy**
- ✓ **Timeliness**
- ✓ **Jitter**

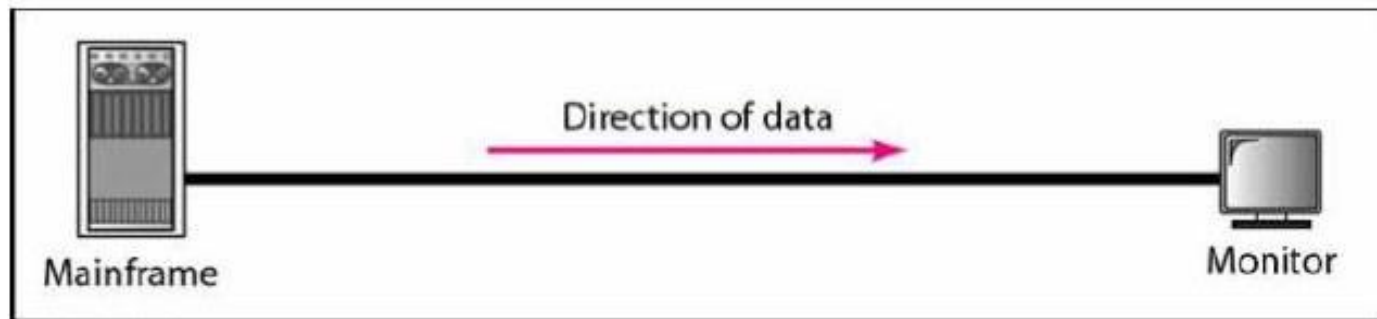
Data Representation

- ✓ **Text**
- ✓ **Numbers**
- ✓ **Images**
- ✓ **Audio**
- ✓ **Video**

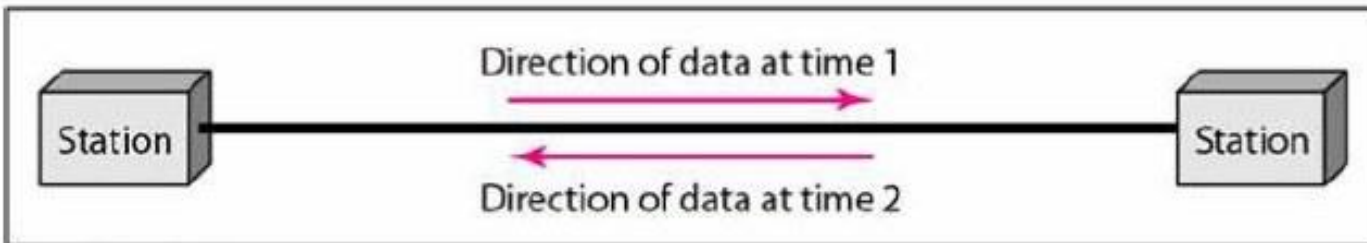
Five components of data communication



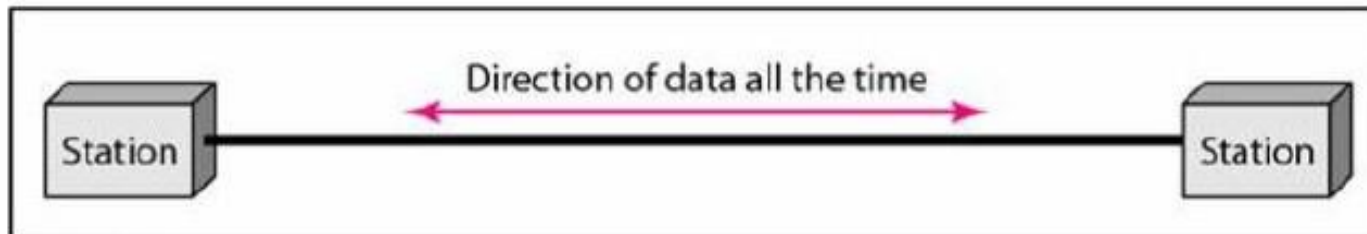
Data flow (simplex, half-duplex, and full-duplex)



a. Simplex



b. Half-duplex

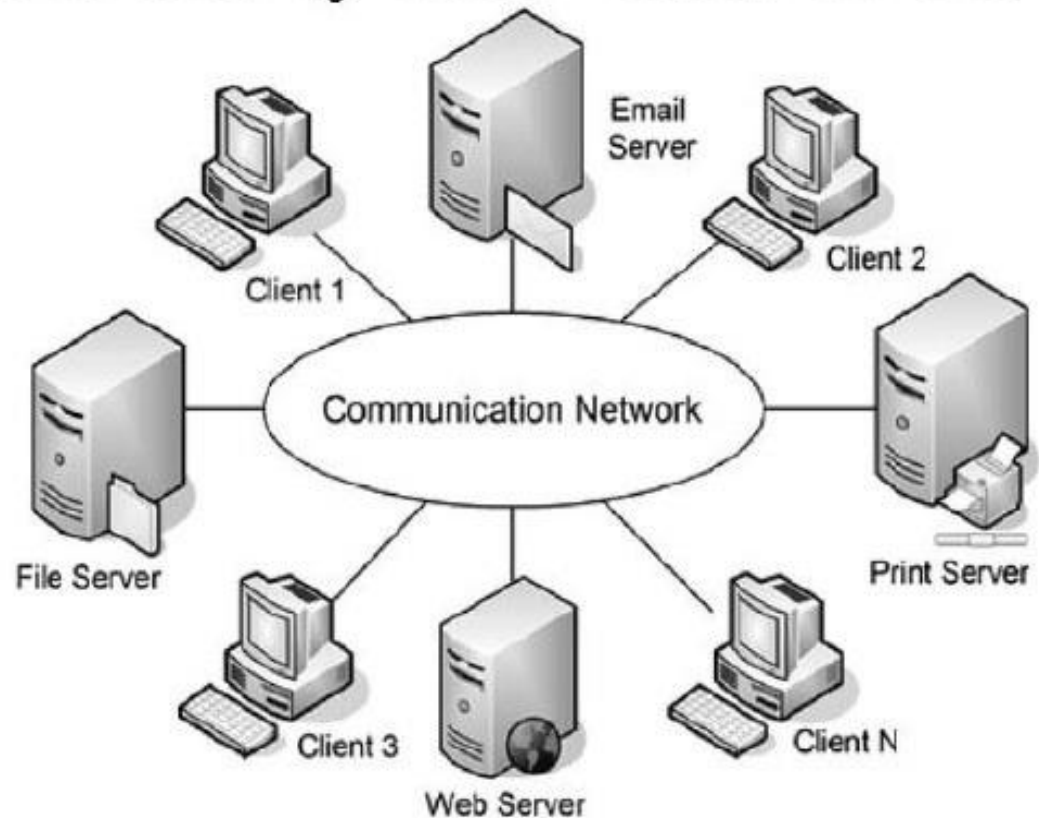


c. Full-duplex

NETWORKS

*A **network** is a set of devices (often referred to as **nodes**) connected by communication **links**. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.*

Network is a group of devices that can communicate with each other over links. Each device is called a host or node. Each host has a unique address.



NETWORK CRITERIA

✓ Performance

- ✓ Transit time

- ✓ Response time

- ✓ Throughput

- ✓ Delay

- ✓ Other factors – number of users, type of transmission medium, capability of the connected hardware and the efficiency of the software

✓ Reliability

✓ Security

PHYSICAL STRUCTURES

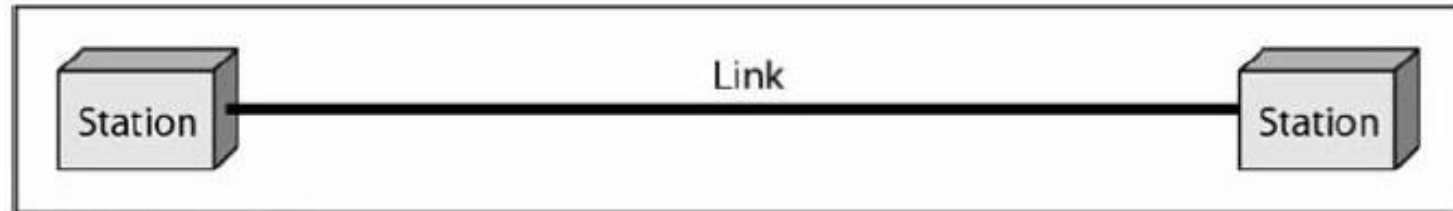
✓ **Type of Connection**

- ✓ A network is two or more devices connected through links
- ✓ A link is a communications pathway that transfers data from one device to another

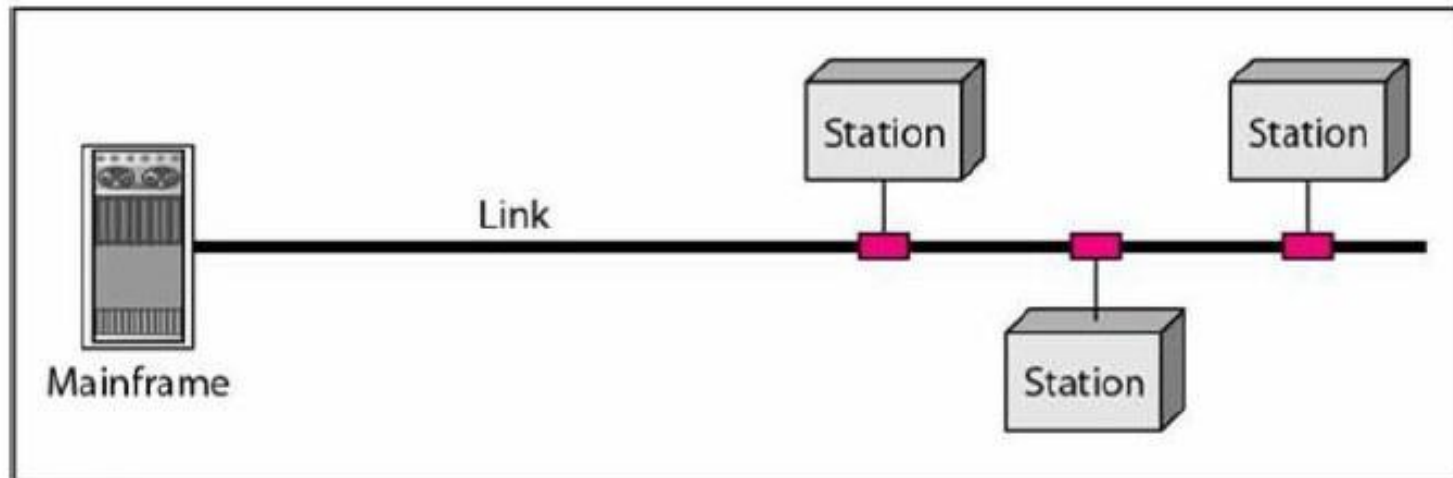
✓ **Physical Topology**

- ✓ The term physical topology refers to the way in which a network is laid out physically.
- ✓ Two or more devices connect to a link; two or more links form a topology.
- ✓ The topology of a network is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another

Types of connections: point-to-point and multipoint

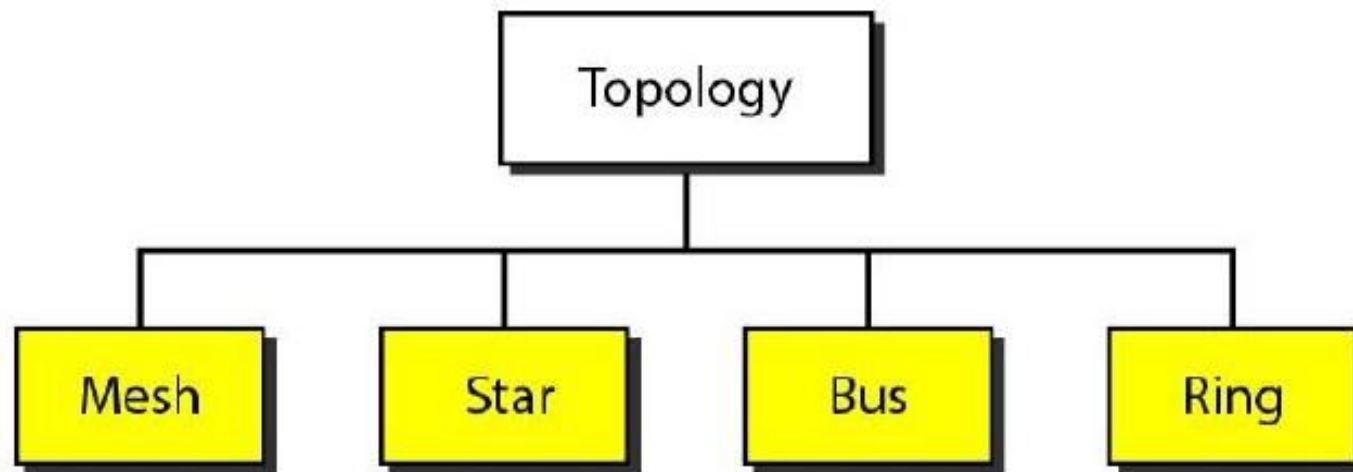


a. Point-to-point

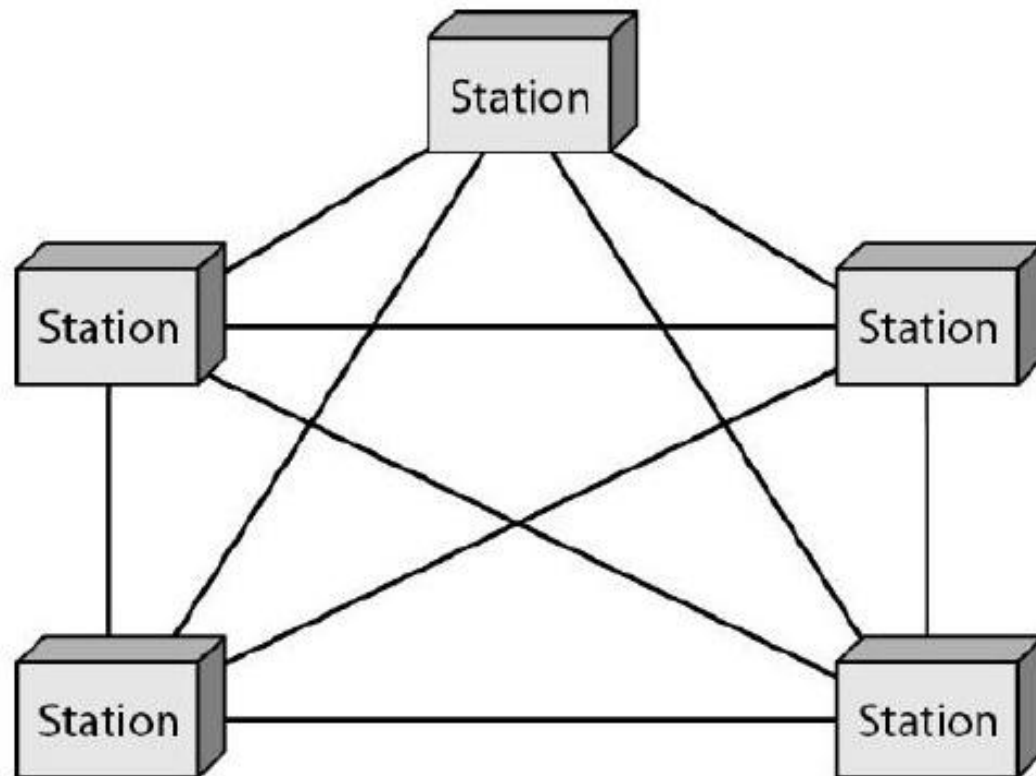


b. Multipoint

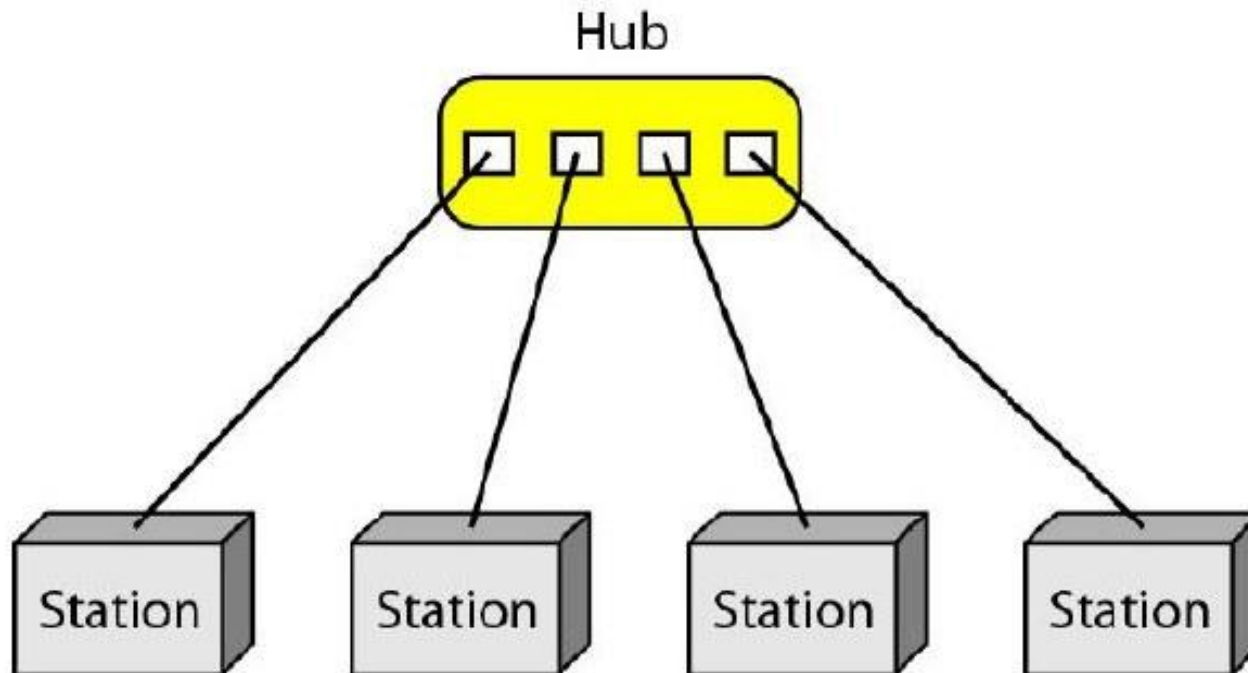
Categories of topology



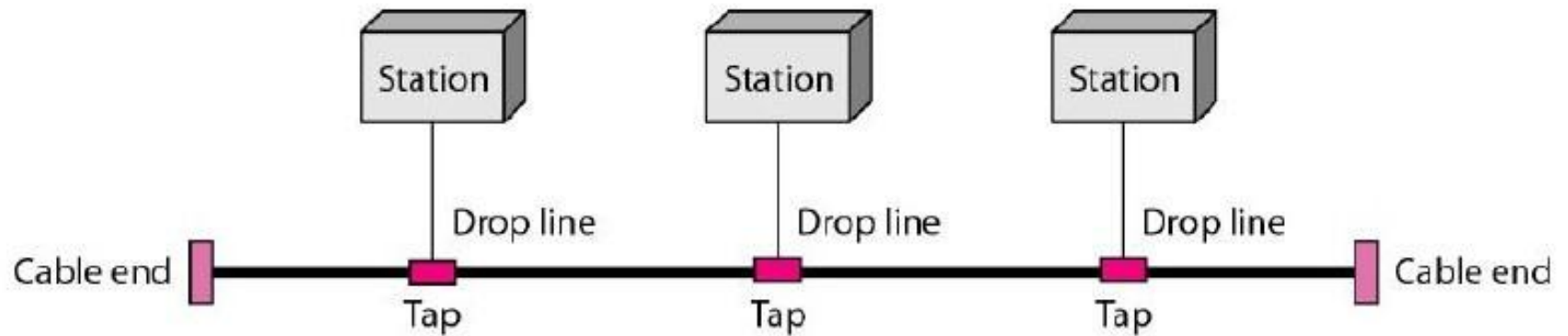
A fully connected mesh topology (five devices)



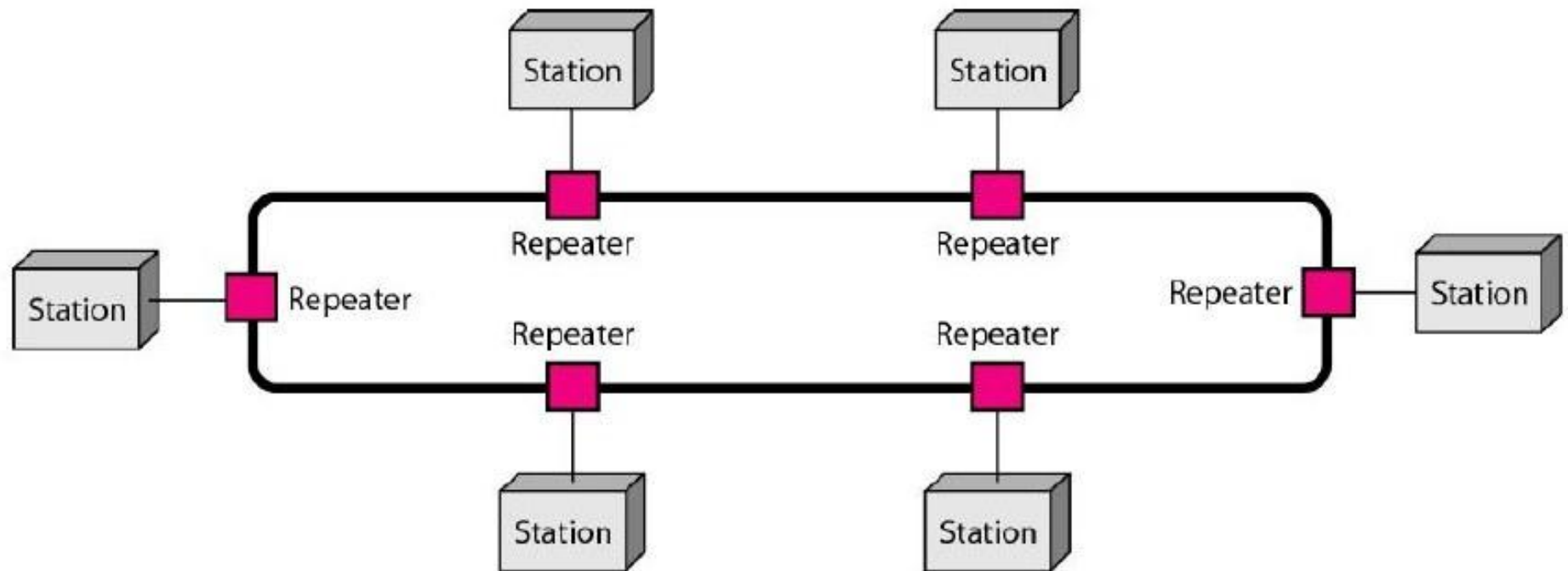
A star topology connecting four stations



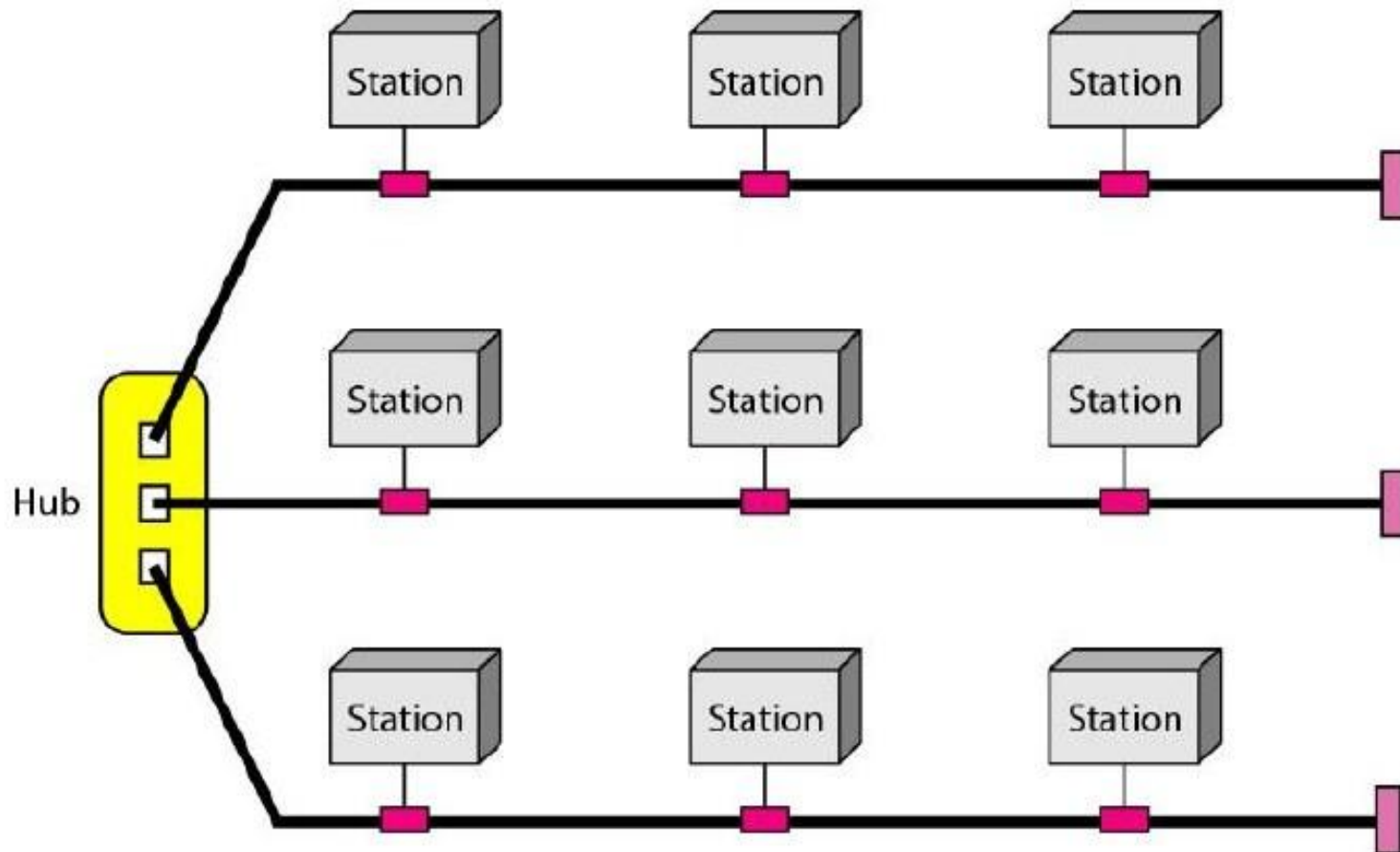
A bus topology connecting three stations



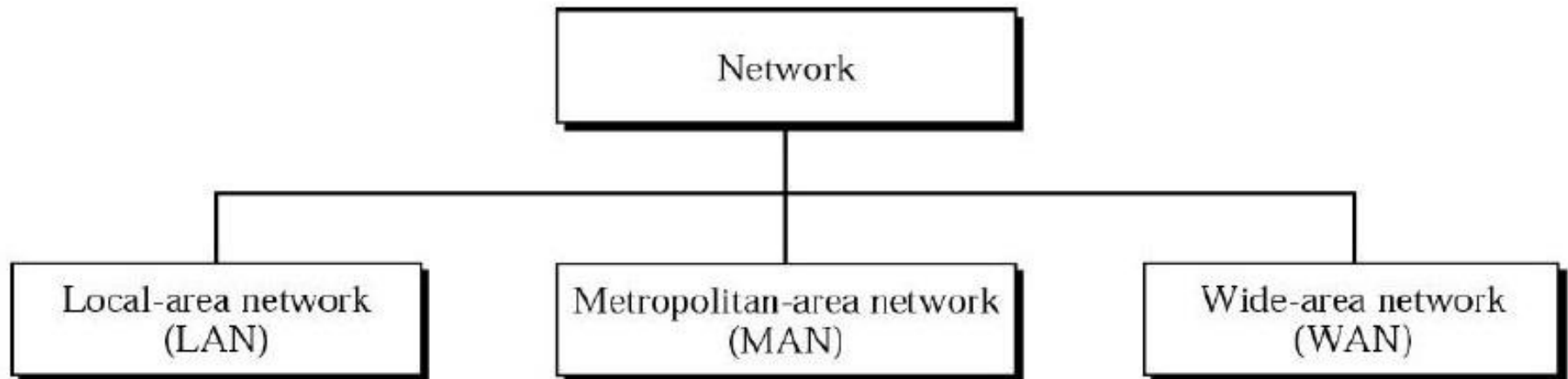
A ring topology connecting six stations



A hybrid topology: a star backbone with three bus networks



Categories of networks



PAN PERSONAL AREA NETWORK

Network that is used on a personal level.



PANs use wireless technologies such as:

Bluetooth Infrared NFC



Can also connect by using a USB cable.

Image Source: PowerCert Animated Videos

LAN

LOCAL AREA NETWORK

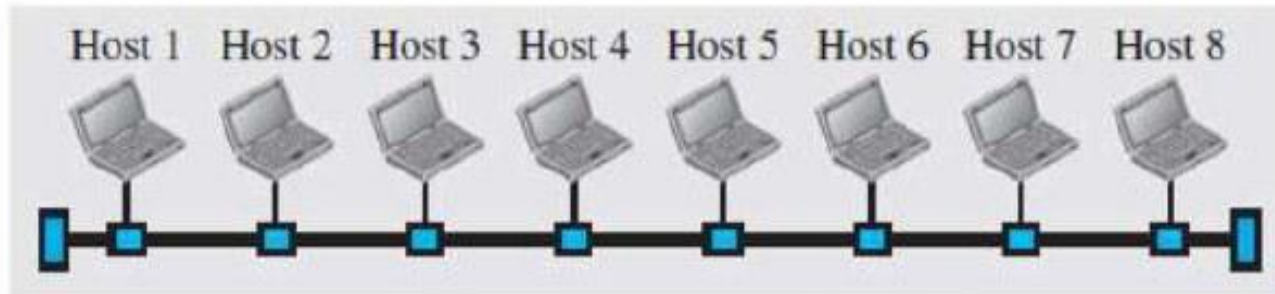
A group of devices (computers, servers, switches, & printers) that are located in the same building.

In close proximity to each other.

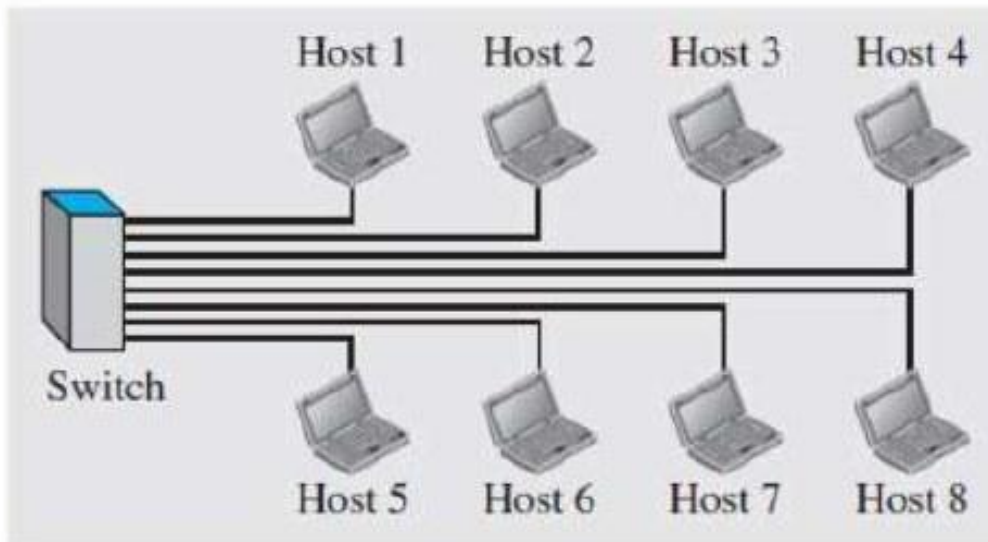
The most common type of LAN is an Ethernet LAN.



LAN

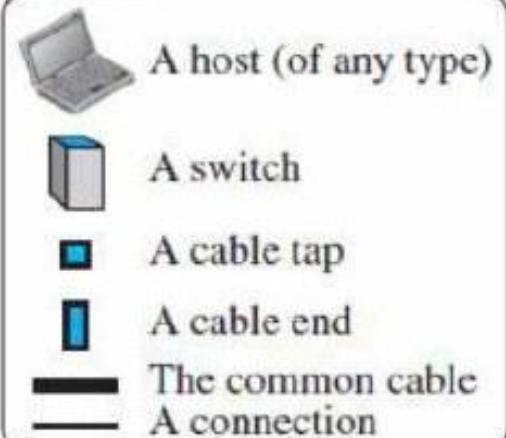


a. LAN with a common cable (past)



b. LAN with a switch (today)

Legend



WLAN

WIRELESS LOCAL AREA NETWORK



Uses wireless communication instead of wired communication.

A WLAN is a LAN but without using cables.

CAN CAMPUS AREA NETWORK

A network that joins 2 or more LANs together within a limited area.

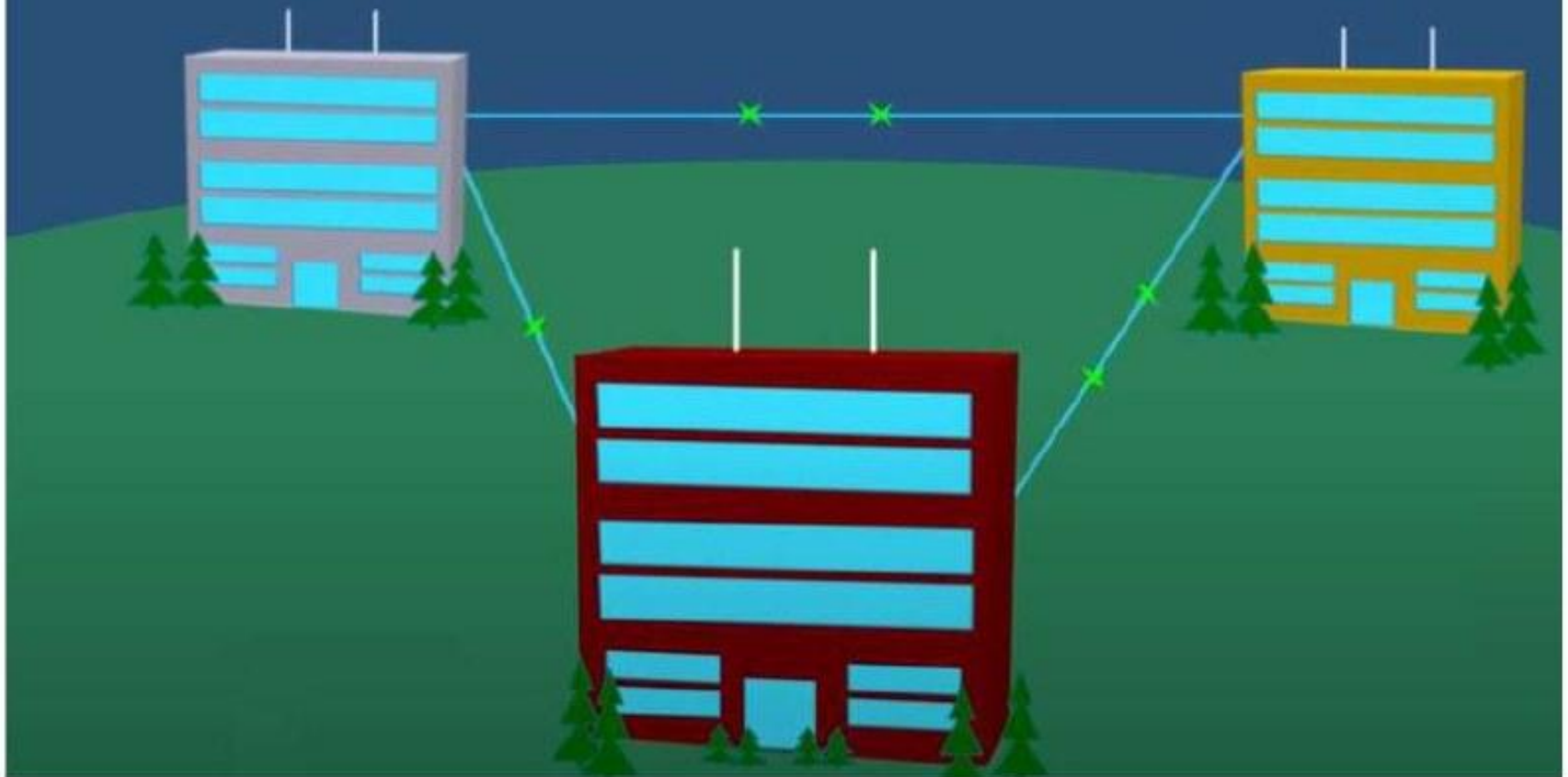
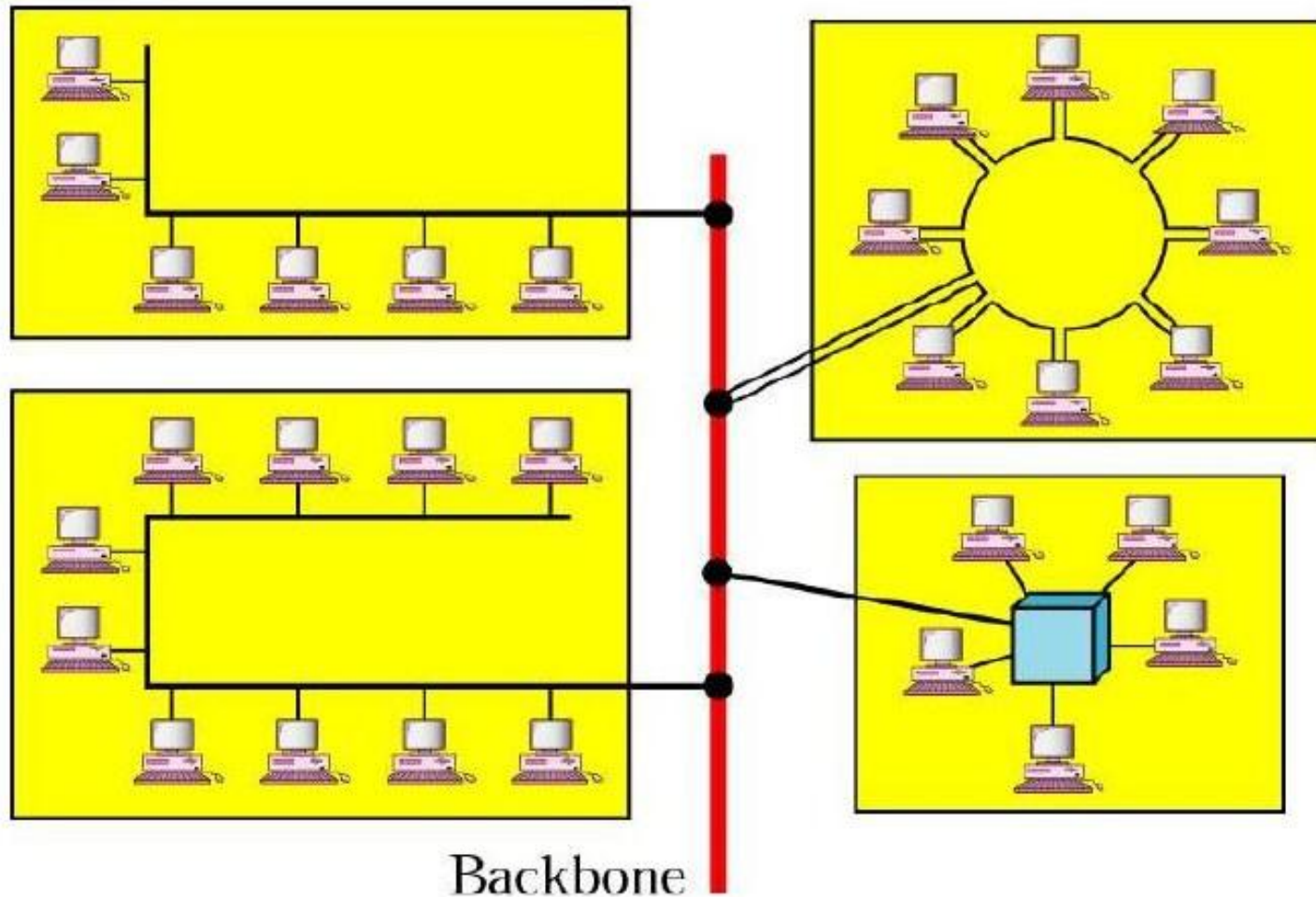


Image Source: PowerCert Animated Videos

LAN (Continued)



b. Multiple-building LAN

MAN

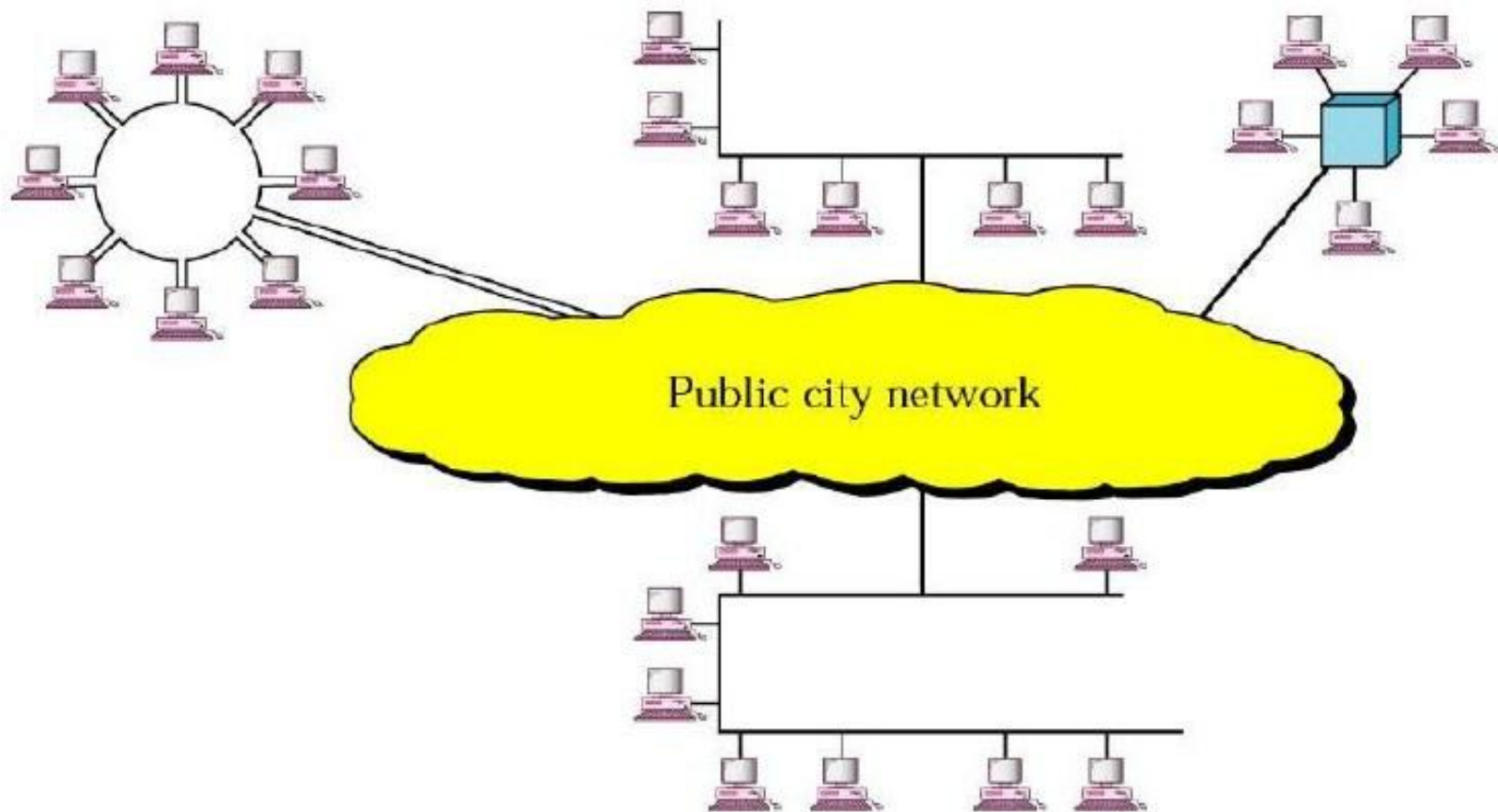
METROPOLITAN AREA NETWORK

Larger than a CAN.

Spans over several buildings in a city or town.



MAN



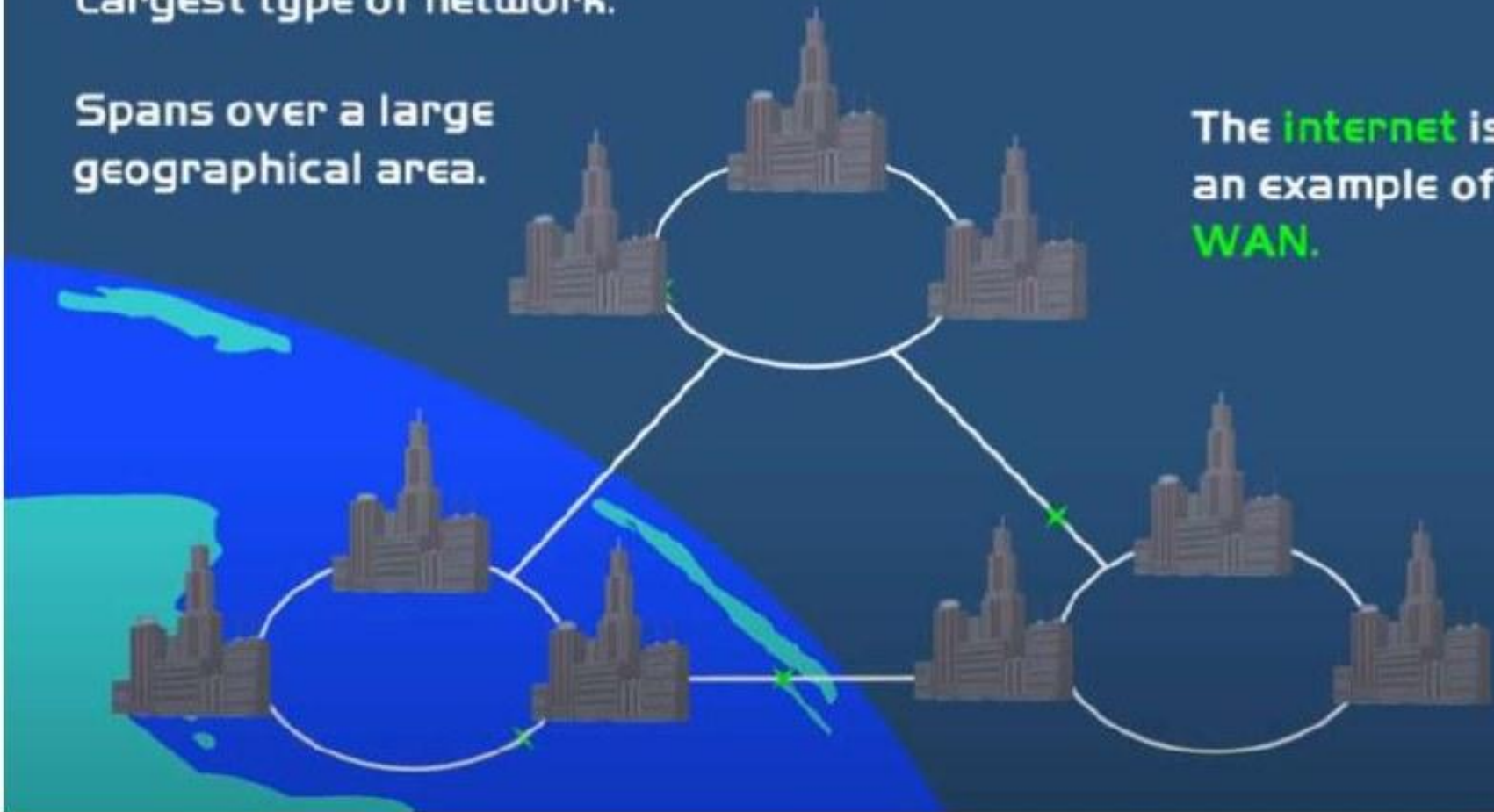
WAN

WIDE AREA NETWORK

Largest type of network.

Spans over a large geographical area.

The **internet** is
an example of a
WAN.

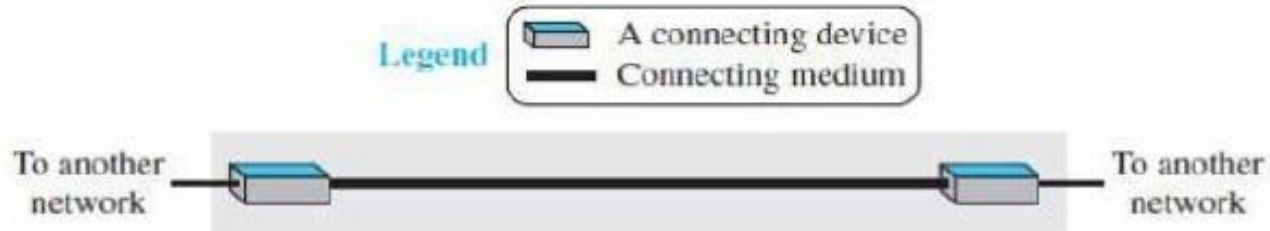


WAN

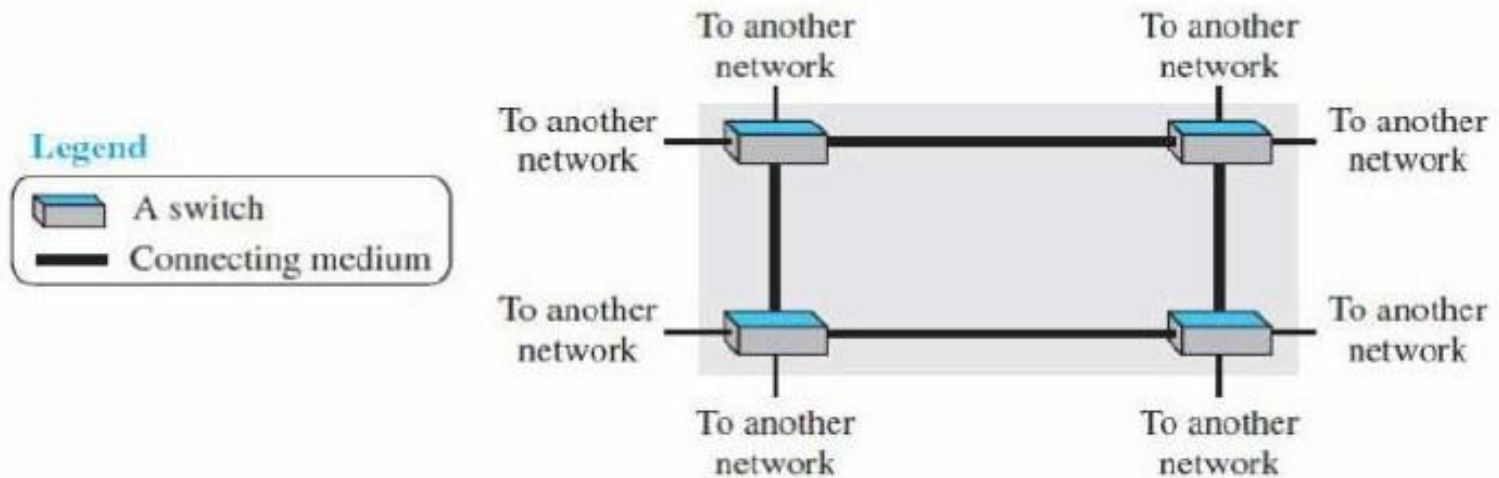


WAN

A point-to-point WAN

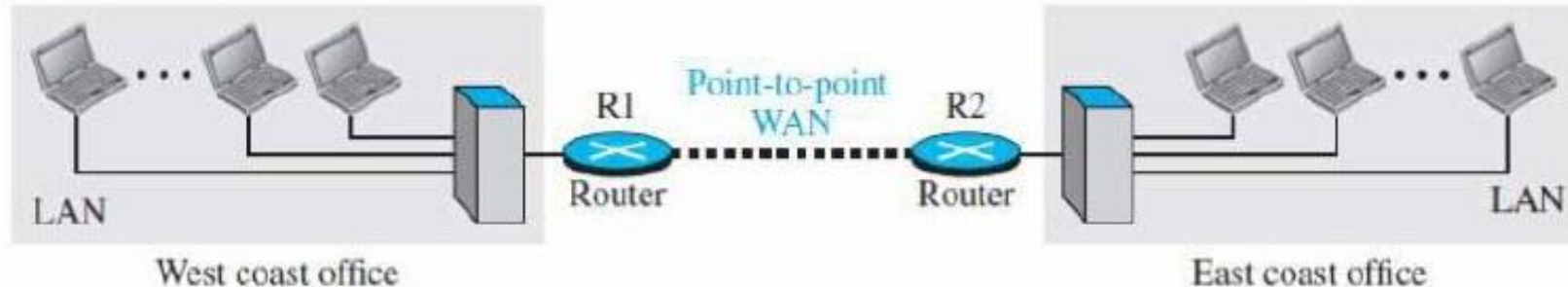


A switched WAN



Internetwork

An internetwork made of two LANs and one point-to-point WAN



Today, it is very rare to see a LAN or a WAN in isolation; they are connected to one another. When two or more networks are connected, they make an **internetwork**, or **internet**.

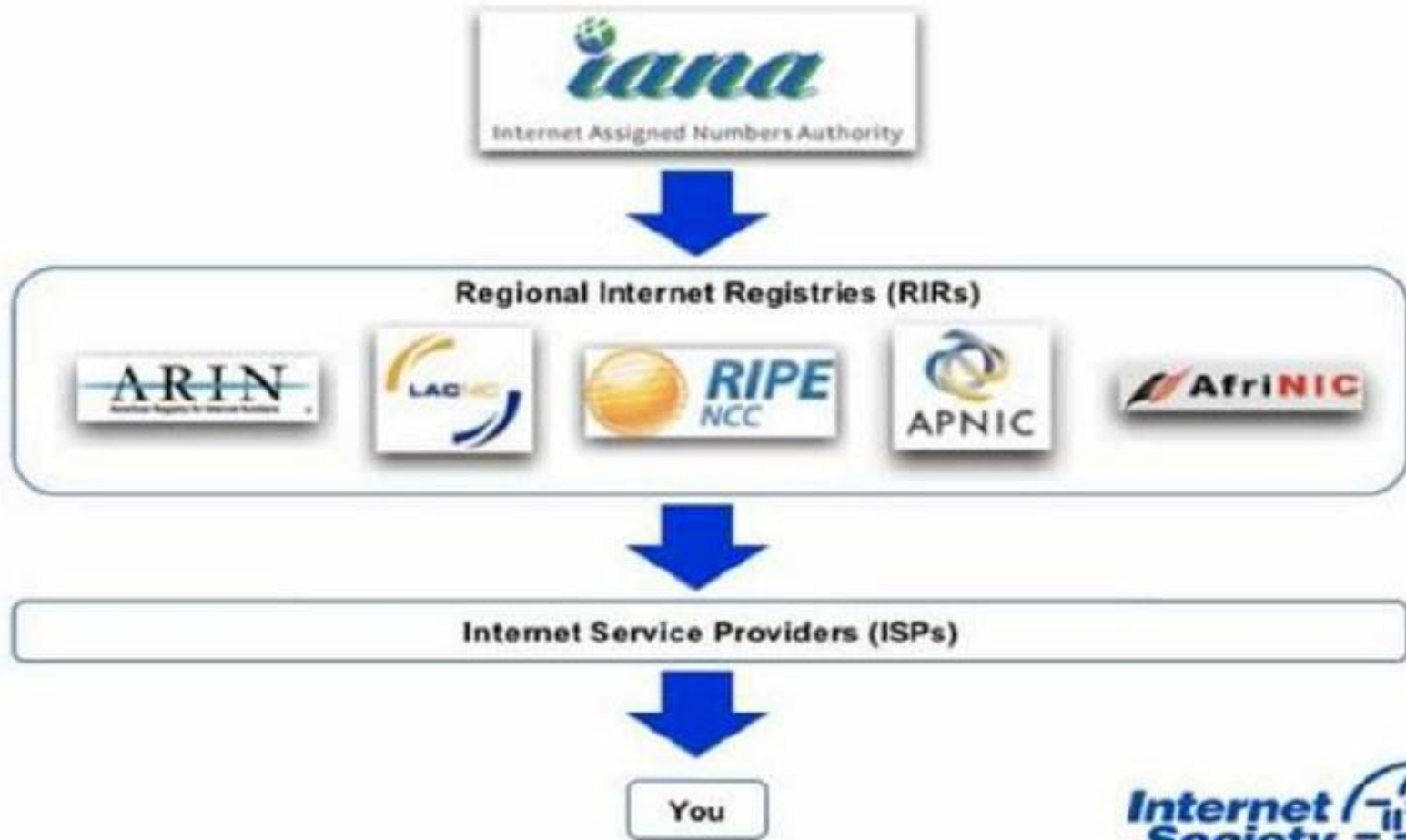
THE INTERNET

An internet (note the lowercase i) is two or more networks that can communicate with each other.

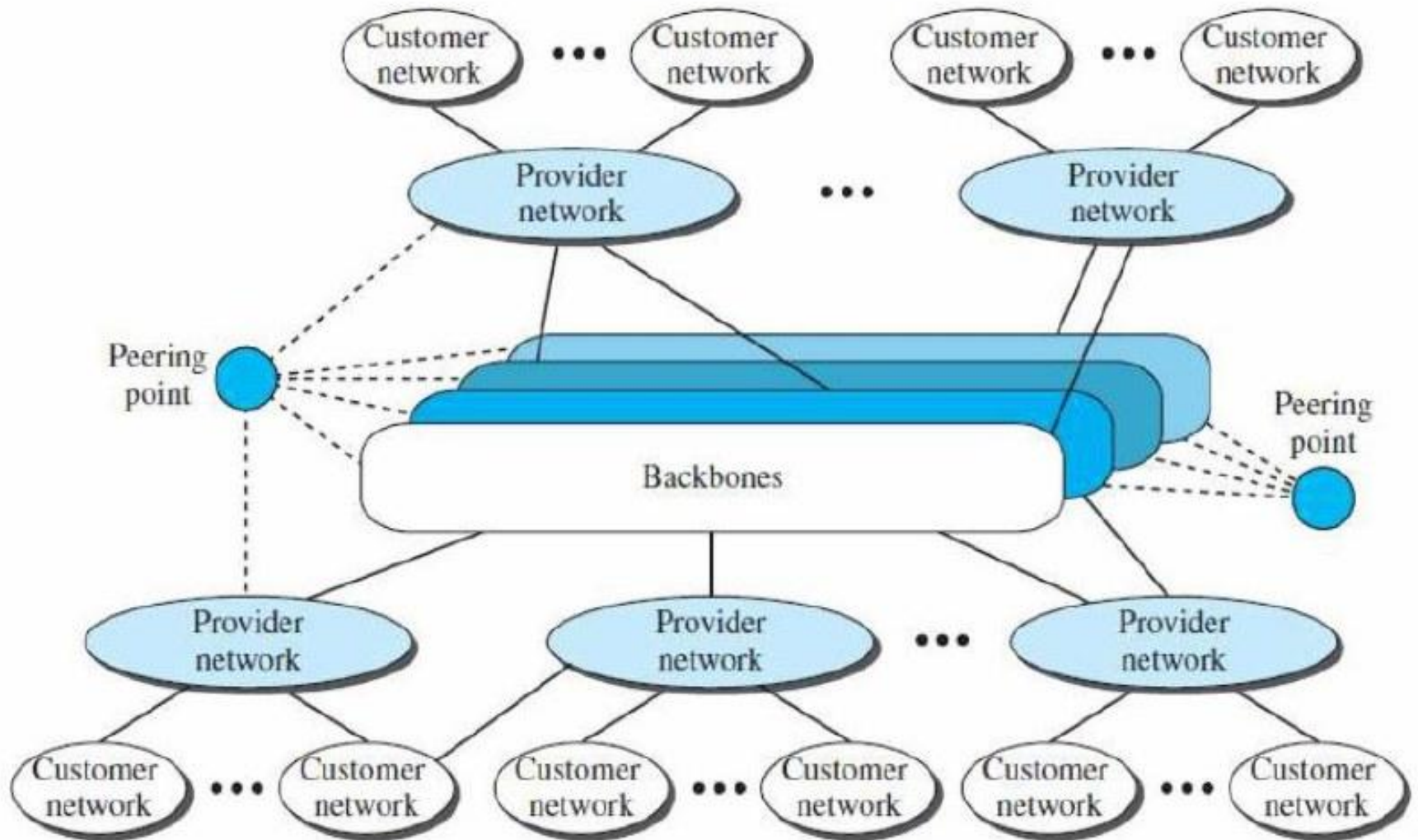
The most notable internet is called the Internet (uppercase I), and is composed of thousands of interconnected networks.

Requires that each host to have a unique IP address and are provided by ISPs

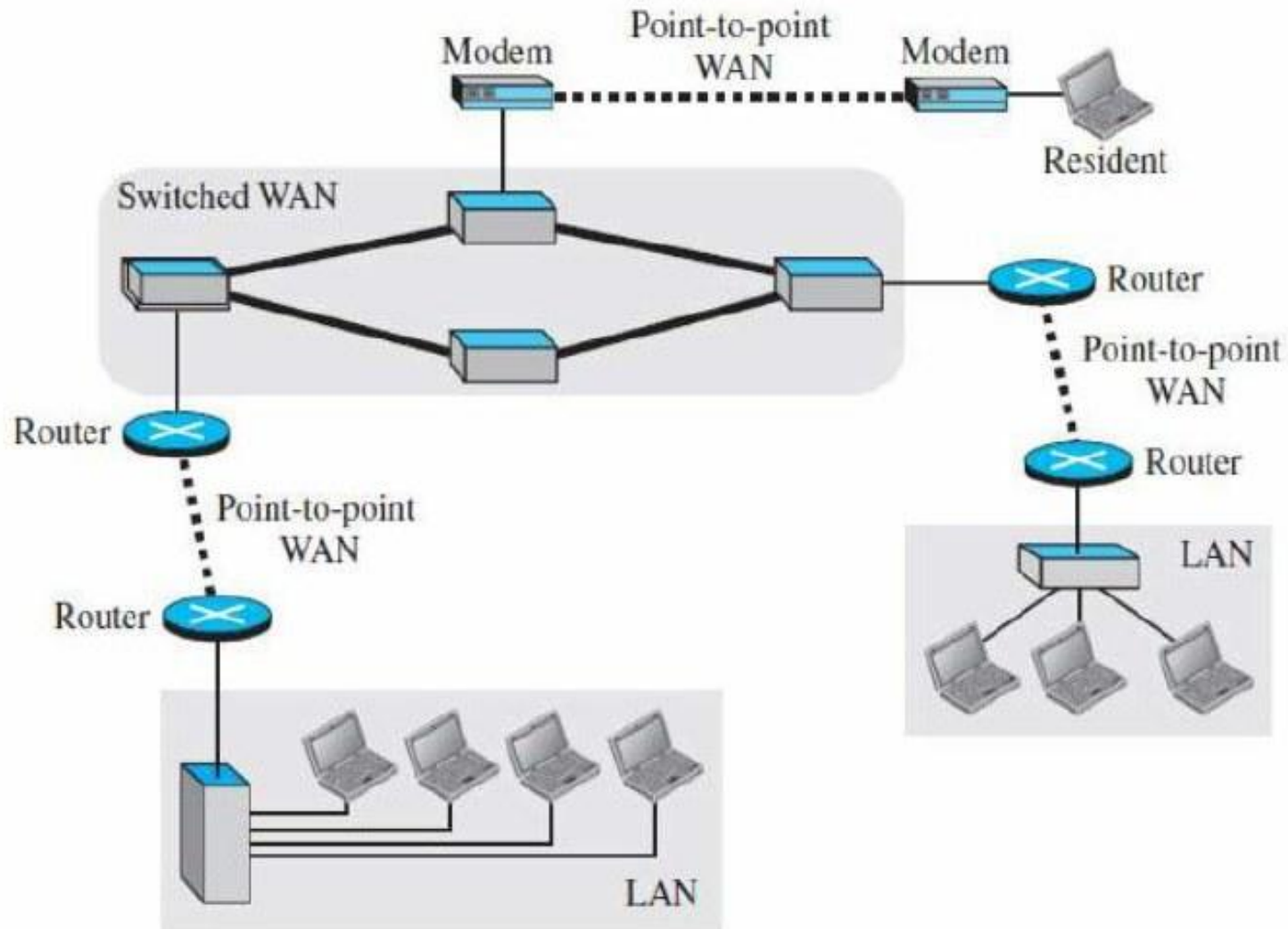
How IP Address Allocation Works



The Internet today



A heterogeneous network made of four WANs and three LANs



Switching

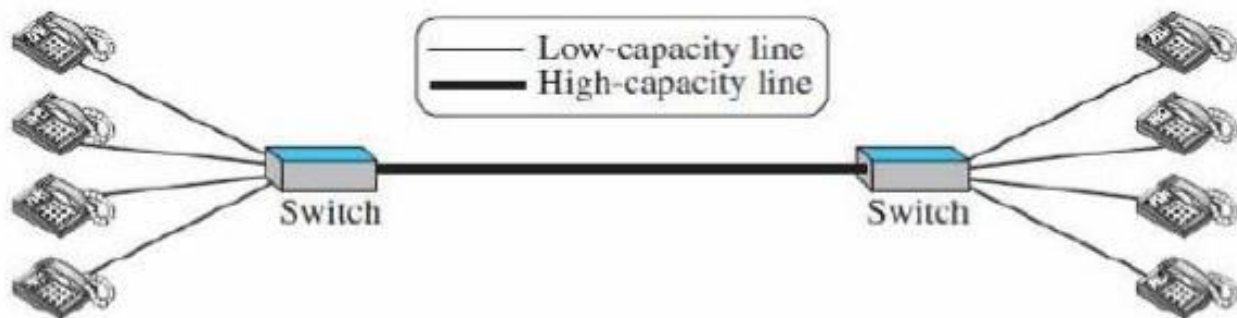
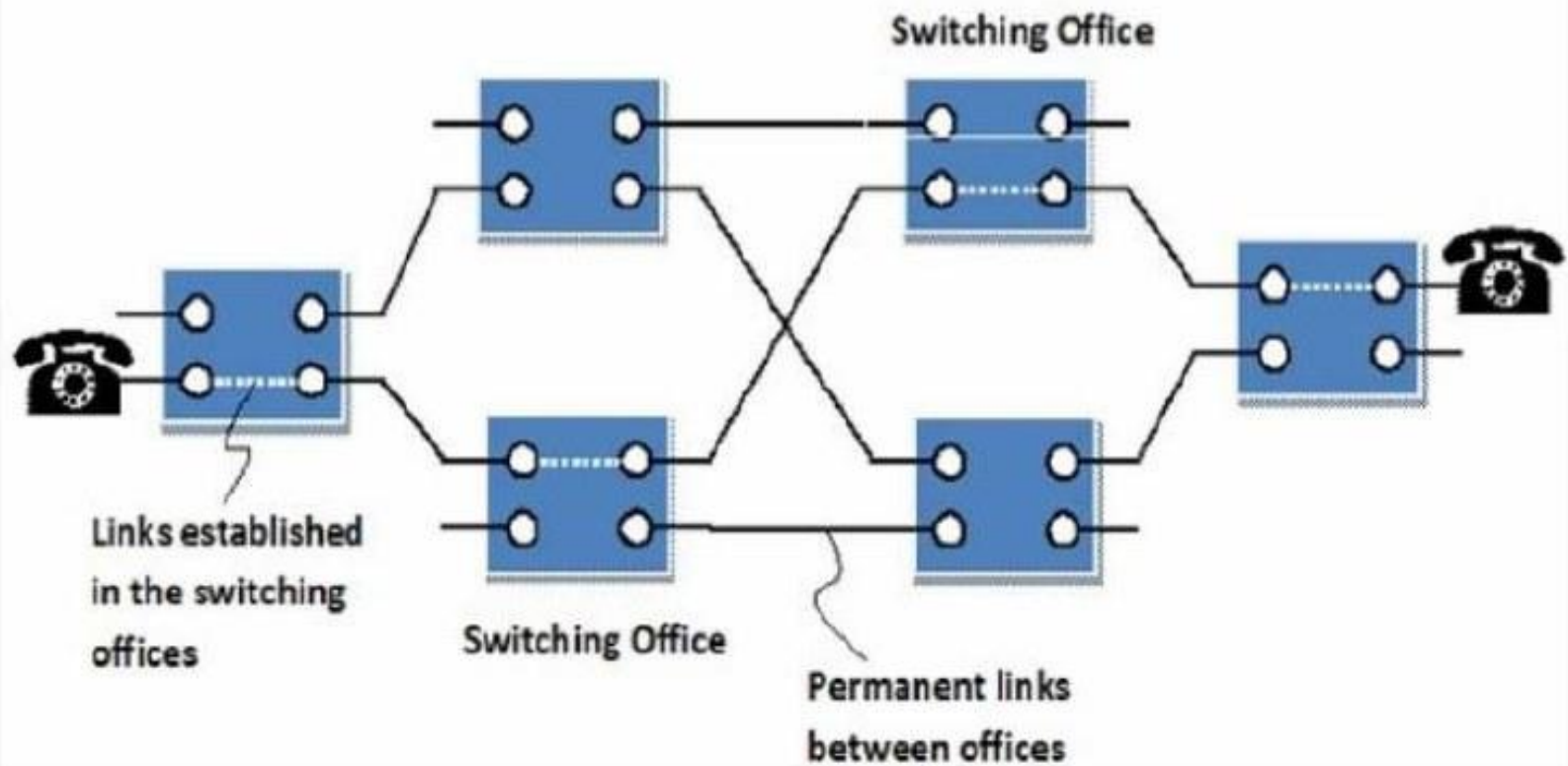
An internet is a **switched network** in which a switch connects at least two links together.

A switch needs to forward data from a network to another network when required.

The two most common types of switched networks

- ✓ Circuit-switched networks
 - ✓ Packet-switched networks
-

Circuit Switching

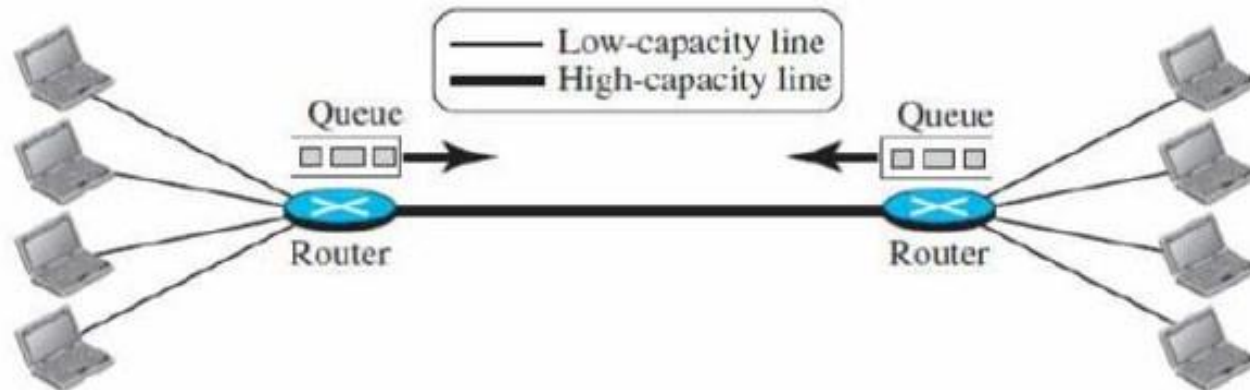


Sender

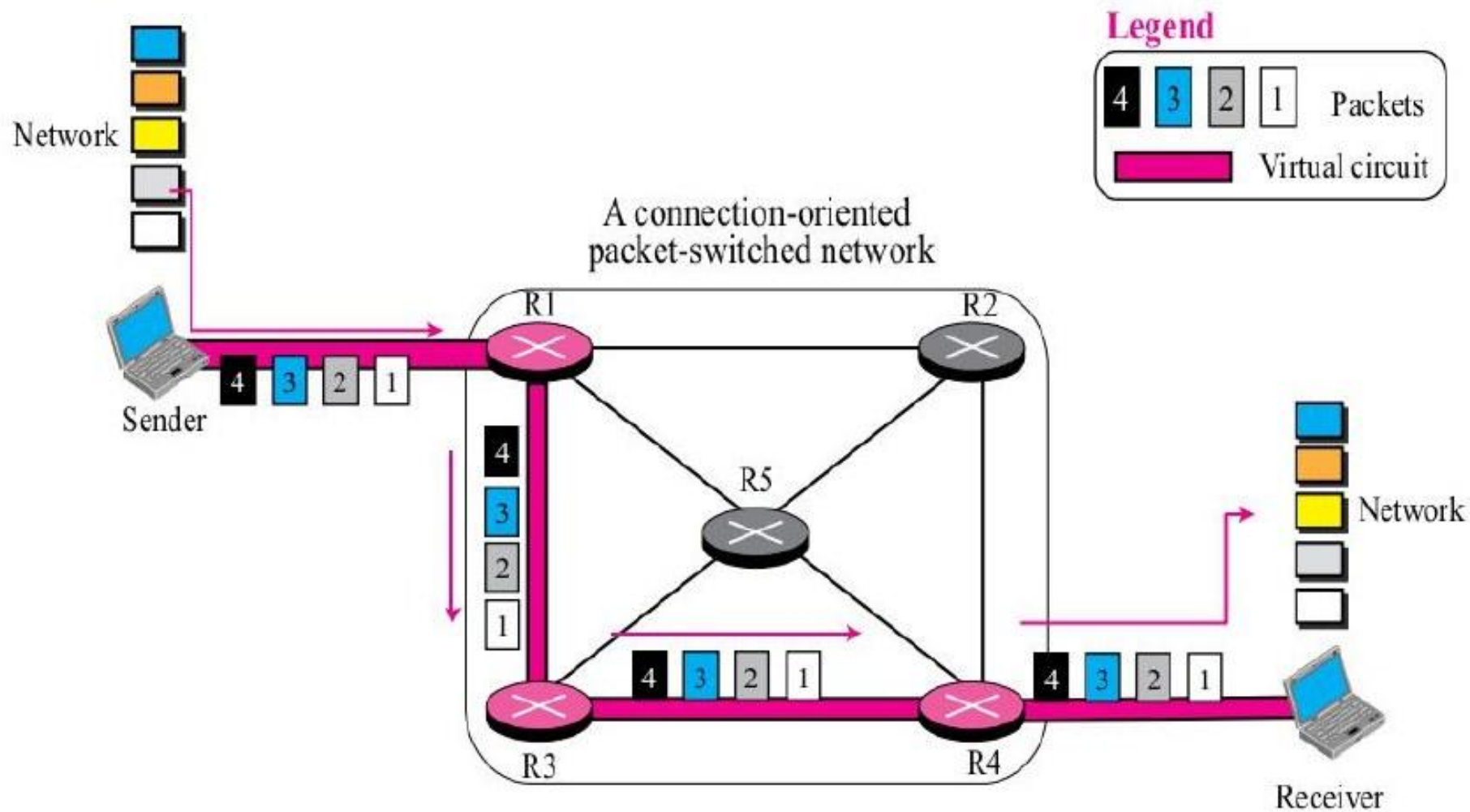
Packet Switching

Intermediate node

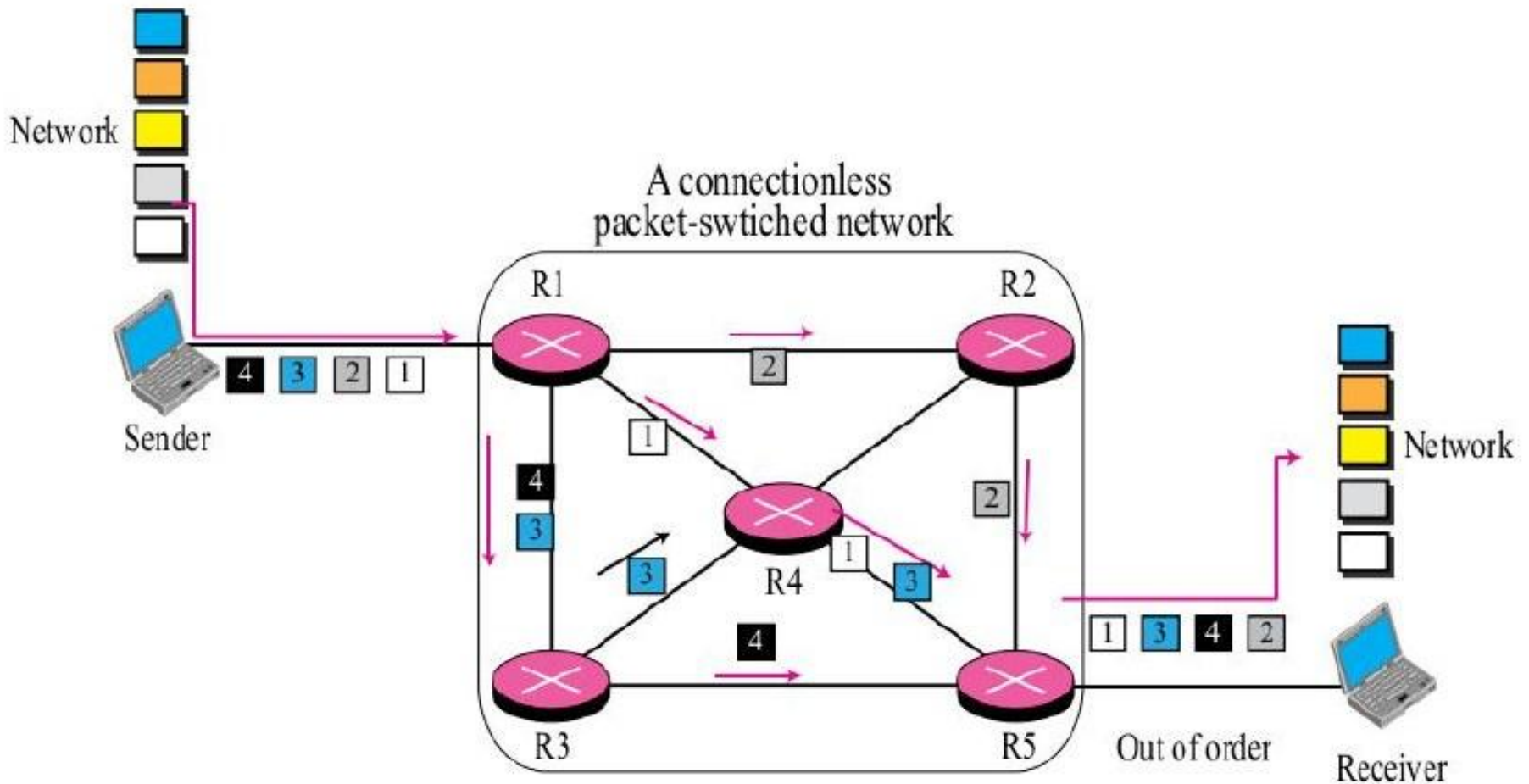
Receiver



A connection-oriented network

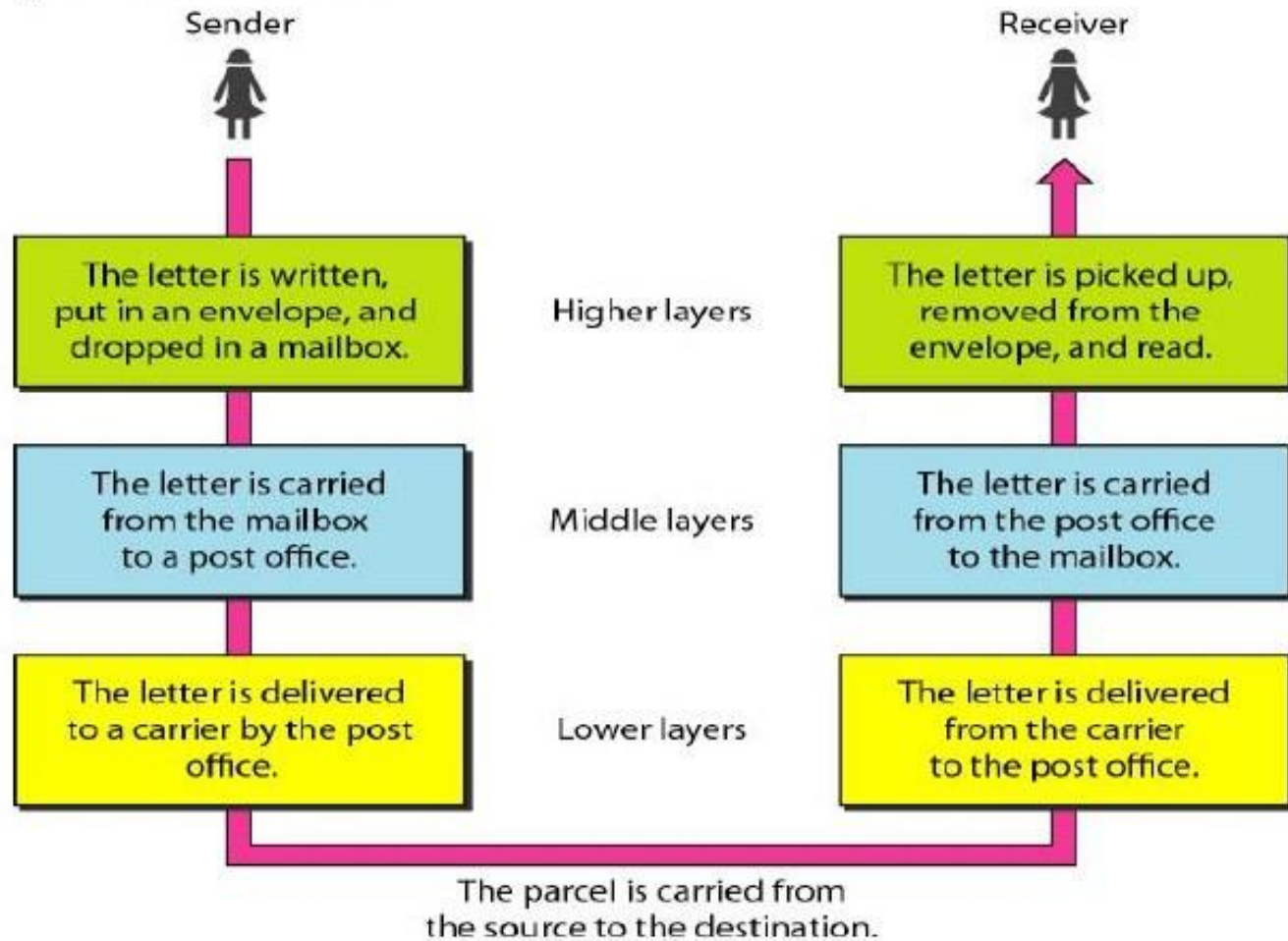


A connectionless network



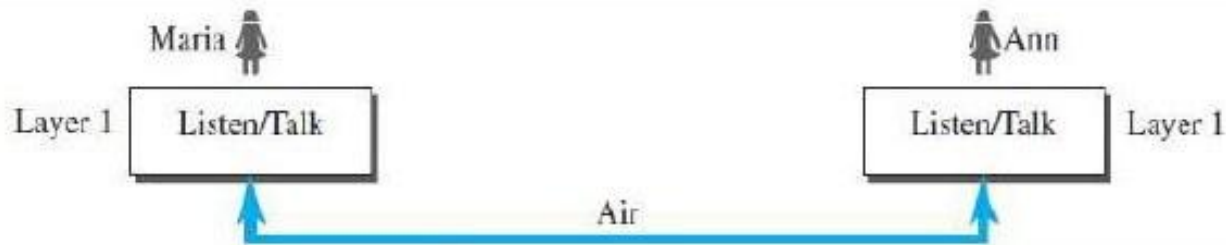
LAYERED TASKS

*We use the concept of **layers** in our daily life. As an example, let us consider two friends who communicate through postal mail.*

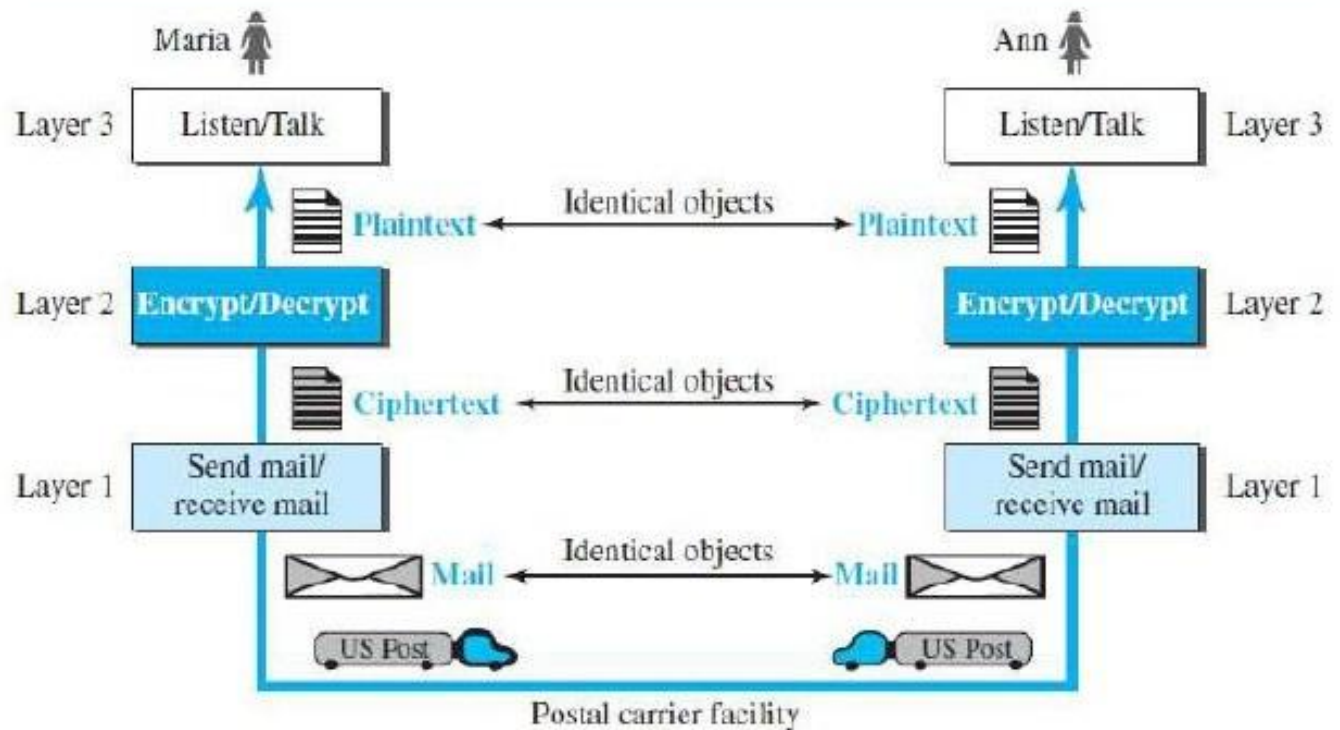


PROTOCOL LAYERING

A single-layer protocol



A three-layer protocol



PRINCIPLES OF PROTOCOL LAYERING

First Principle

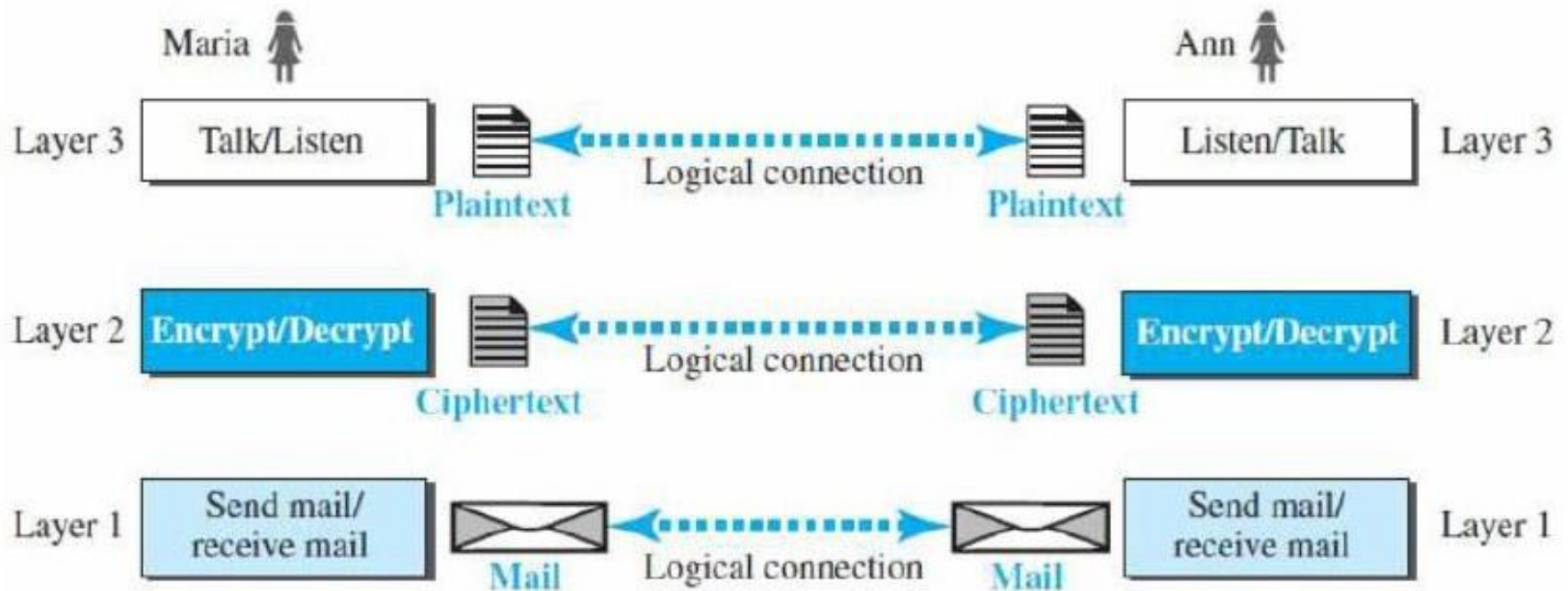
- ✓ The first principle dictates that if we want bidirectional communication, we need to make each layer so that it is able to perform two opposite tasks, one in each direction.

Second Principle

- ✓ The second principle that we need to follow in protocol layering is that the two objects under each layer at both sites should be identical.

LOGICAL CONNECTIONS

Logical connection between peer layers



THE OSI MODEL

- ✓ Established in 1947, the International Standards Organization (**ISO**) is a multinational body dedicated to worldwide agreement on international standards.
- ✓ An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (**OSI**) model and was first introduced in the late 1970s.

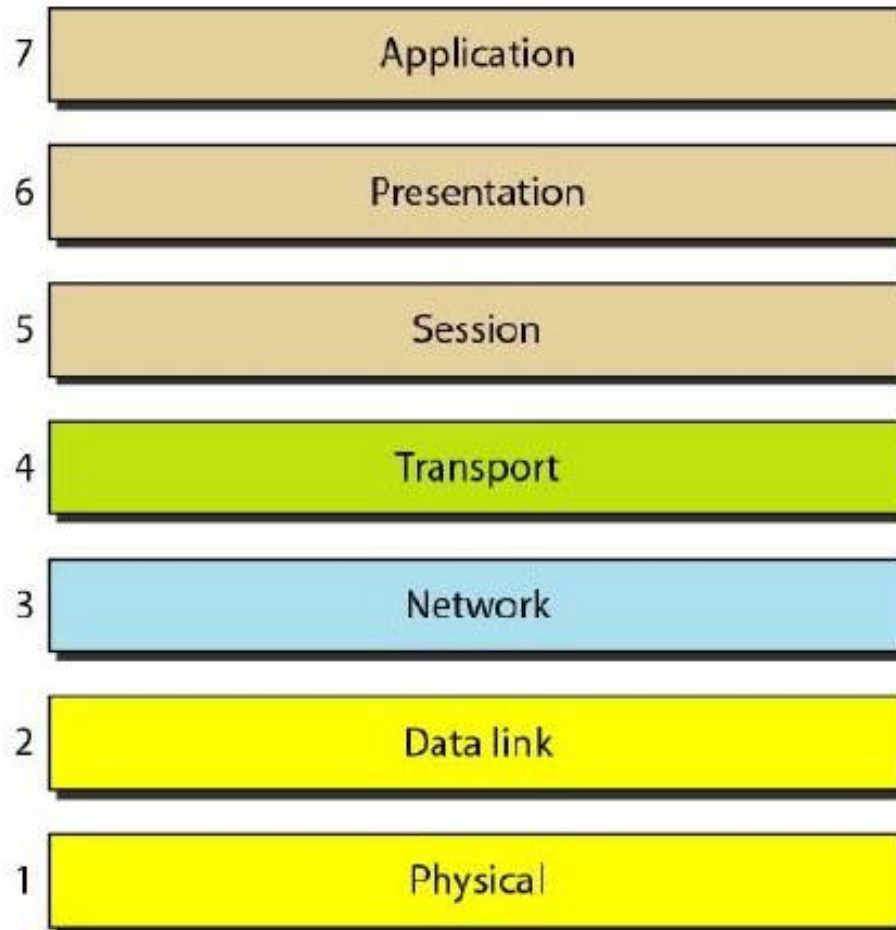
Note

**ISO is the organization.
OSI is the model.**

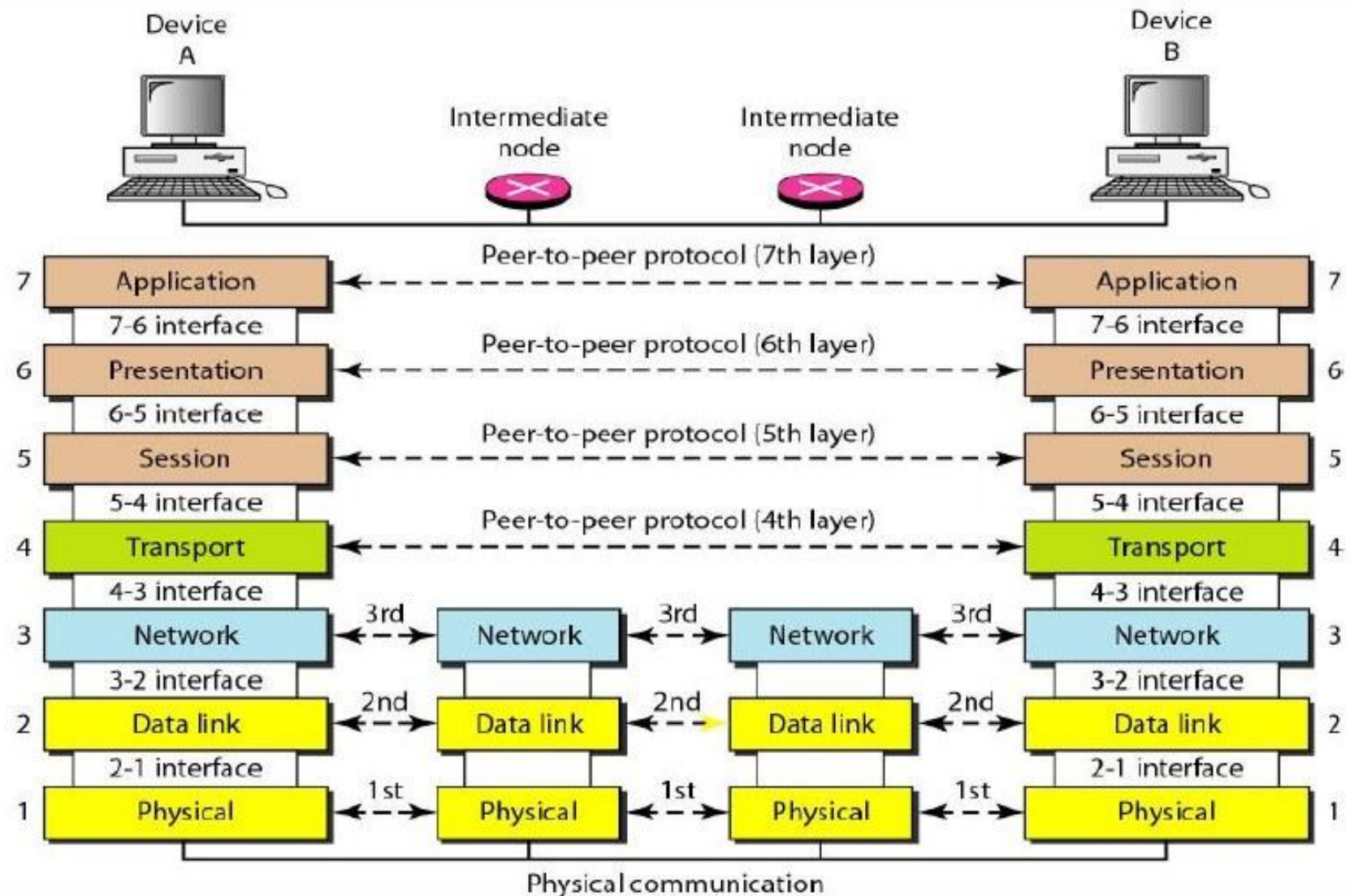
THE OSI MODEL

- ✓ The OSI model is not a protocol; it is a model for understanding and designing a network architecture.
- ✓ OSI model is a layered framework for the design of network systems that allows communication between all types of computer systems.
- ✓ OSI model consists of seven separated layers, each of which defines a part of the process of moving information across a network.

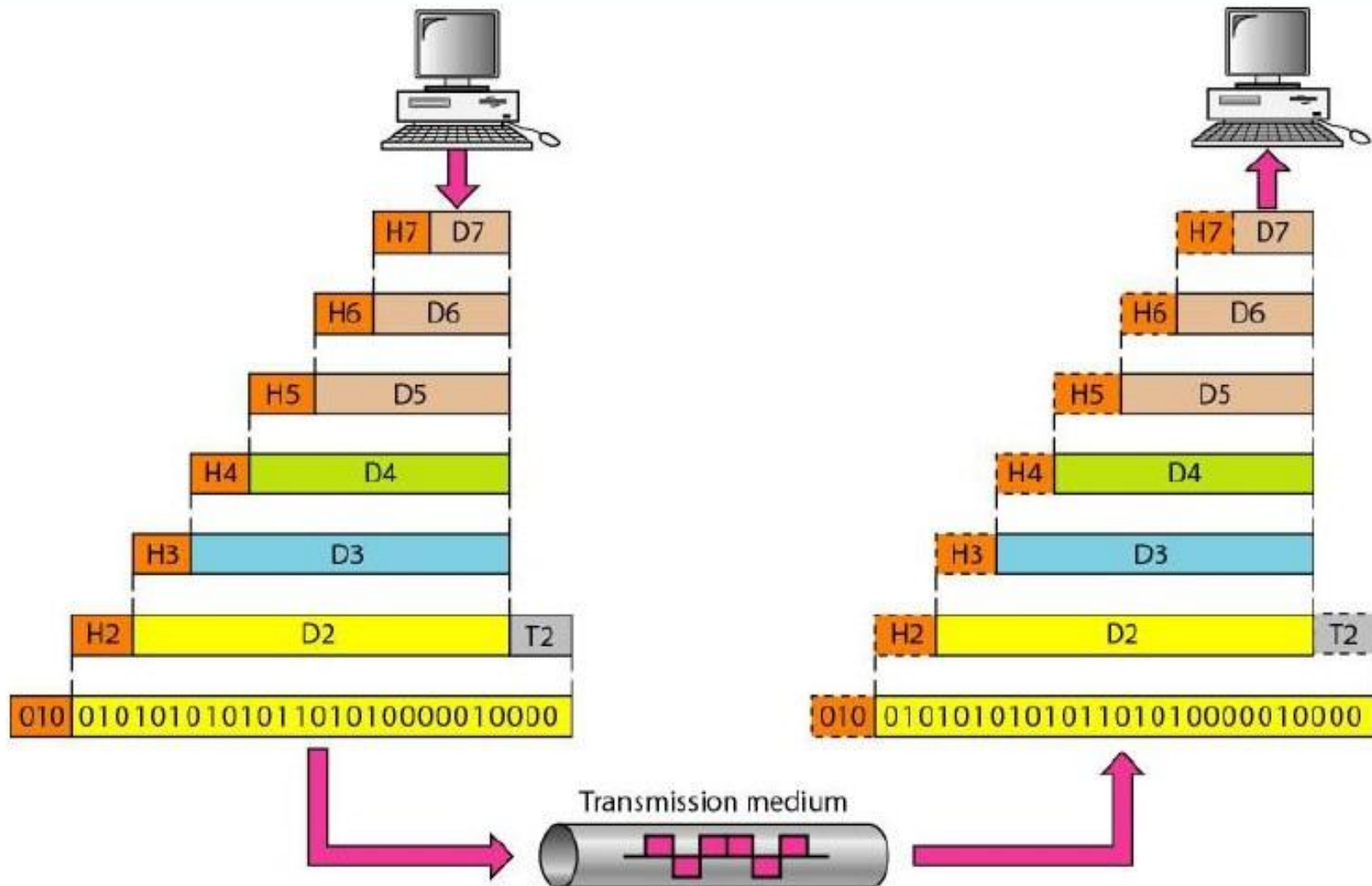
Seven layers of the OSI model



The interaction between layers in the OSI model



Data exchange using the OSI model



LAYERS IN THE OSI MODEL

In this section we briefly describe the functions of each layer in the OSI model.

Topics discussed in this section:

Physical Layer

Data Link Layer

Network Layer

Transport Layer

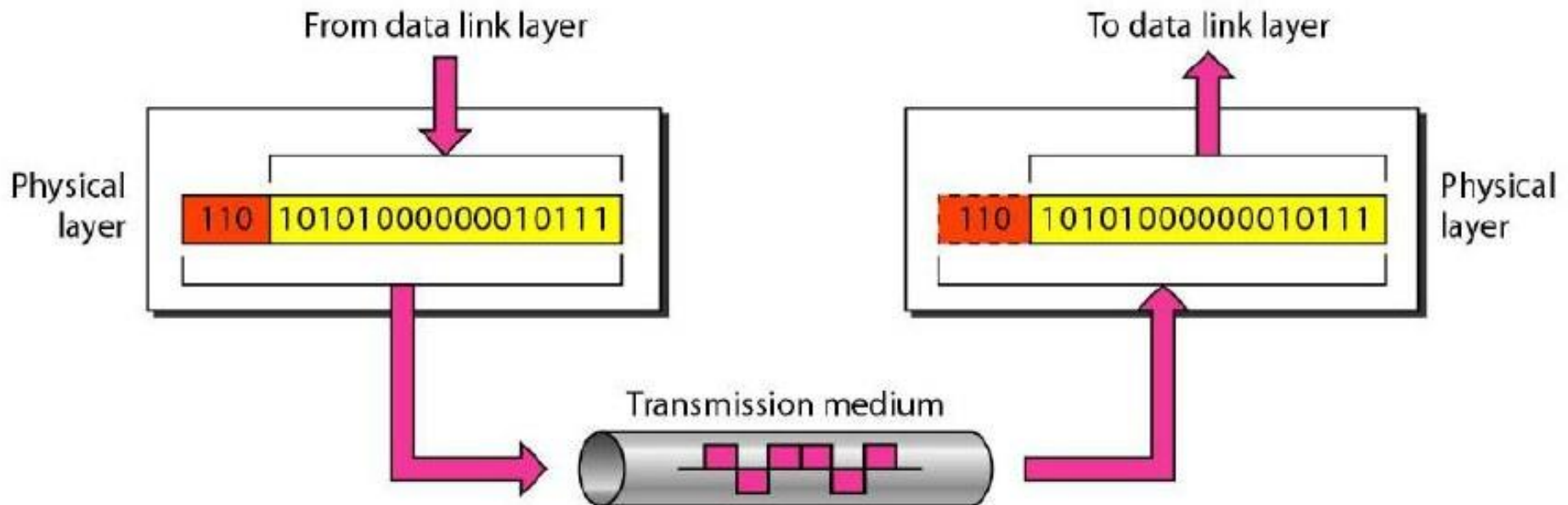
Session Layer

Presentation Layer

Application Layer

PHYSICAL LAYER

The physical layer is responsible for movements of individual bits from one hop (node) to the next.



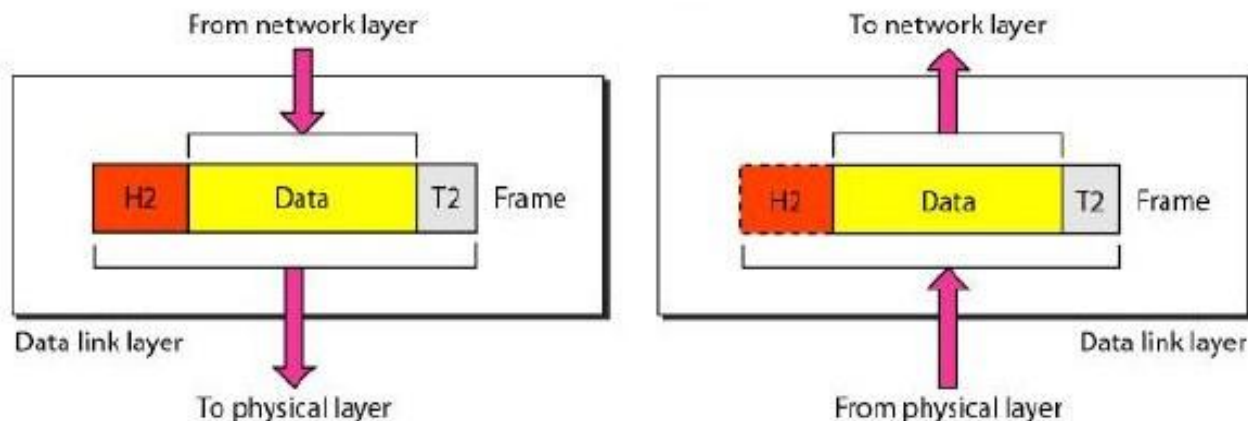
Responsibilities / Functions / Services of Physical Layer

- Physical characteristics of interfaces and medium
 - Representation of bits
 - Data rate
 - Synchronization of bits
 - Line configuration
 - Physical topology
 - Transmission mode
-

DATALINK LAYER

- Links network layer and physical layer
- Makes the physical layer to appear error free to the network layer
- Responsible for hop to hop (node to node) delivery

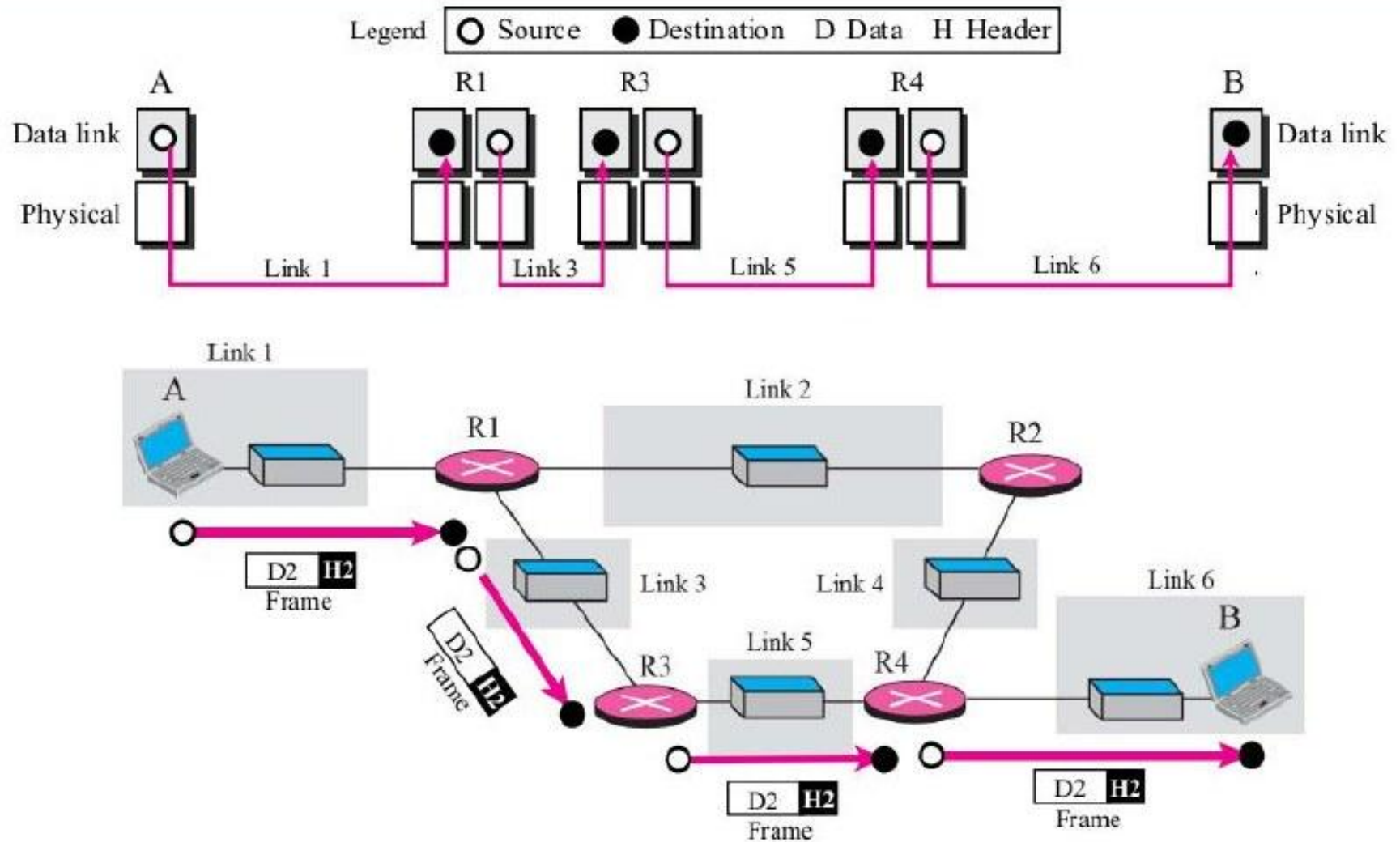
The data link layer is responsible for moving frames from one hop (node) to the next.



Responsibilities / Functions / Services of Data Link Layer

- Hop to hop delivery
 - Framing
 - Physical addressing
 - Flow control
 - Error control
 - Access control
-

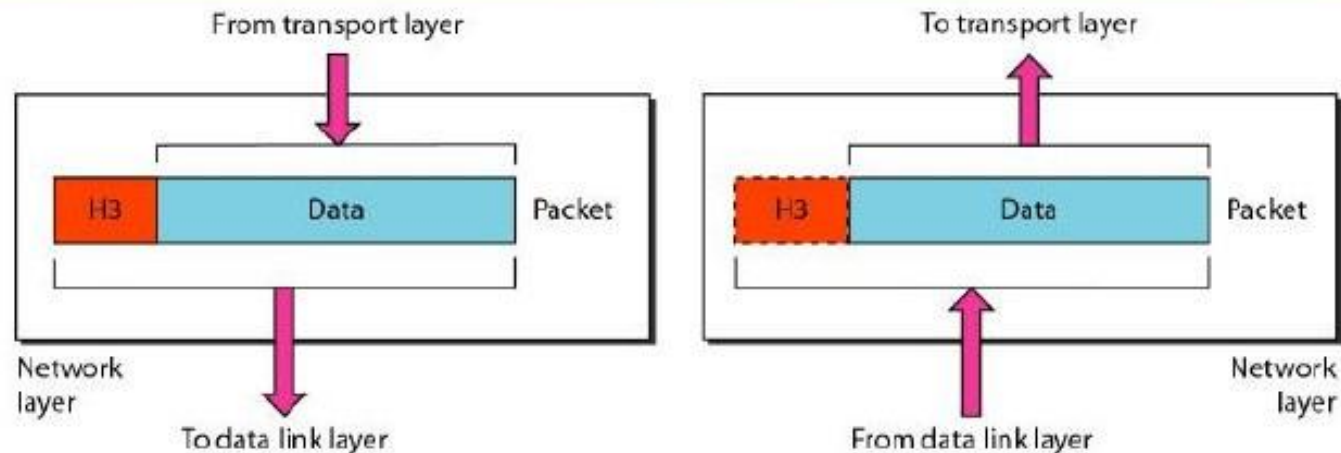
Hop-to-hop delivery



NETWORK LAYER

- Data link layer operates on the same network(link)
- Network layer operates on different networks(links)
- Responsible for source to destination delivery

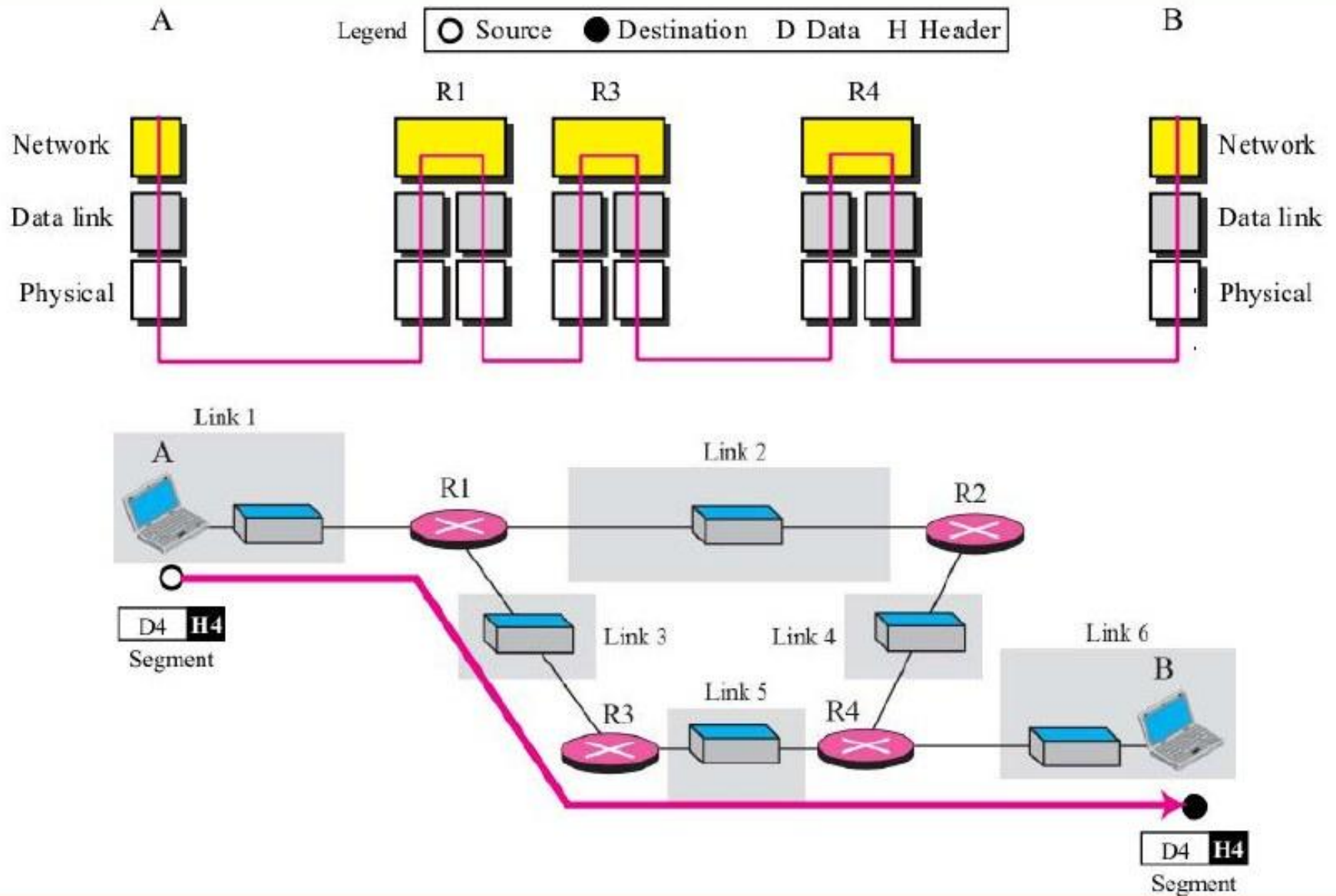
The network layer is responsible for the delivery of individual packets from the source host to the destination host.



Responsibilities / Functions / Services of Network Layer

- Source to destination delivery
 - Logical addressing
 - Routing
-

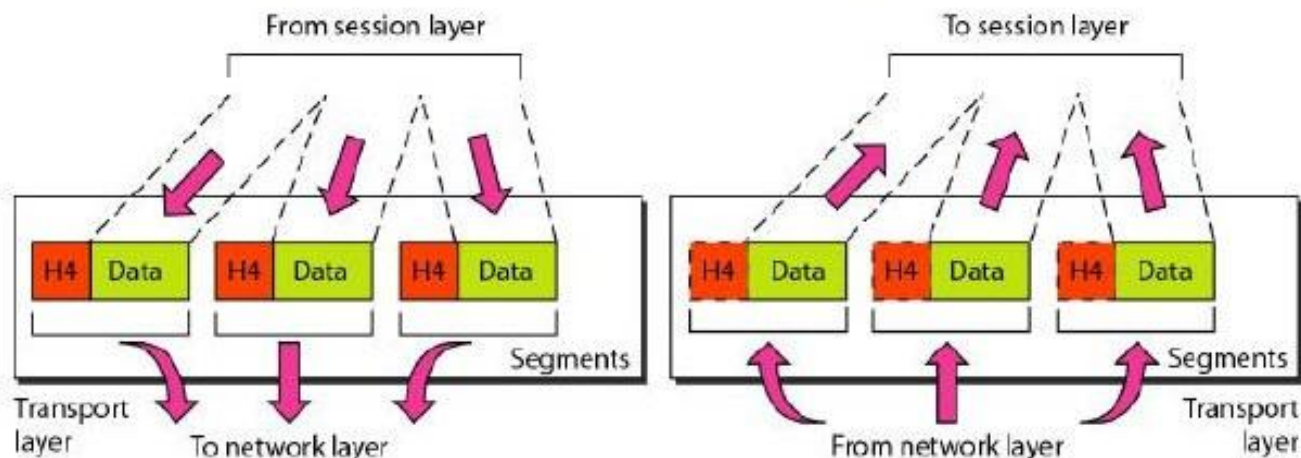
Source-to-destination delivery



TRANSPORT LAYER

- Network layer concentrates on individual packets
- Transport layer concentrates on the entire message
- Responsible for process to process delivery
- Process is an application program running on a host

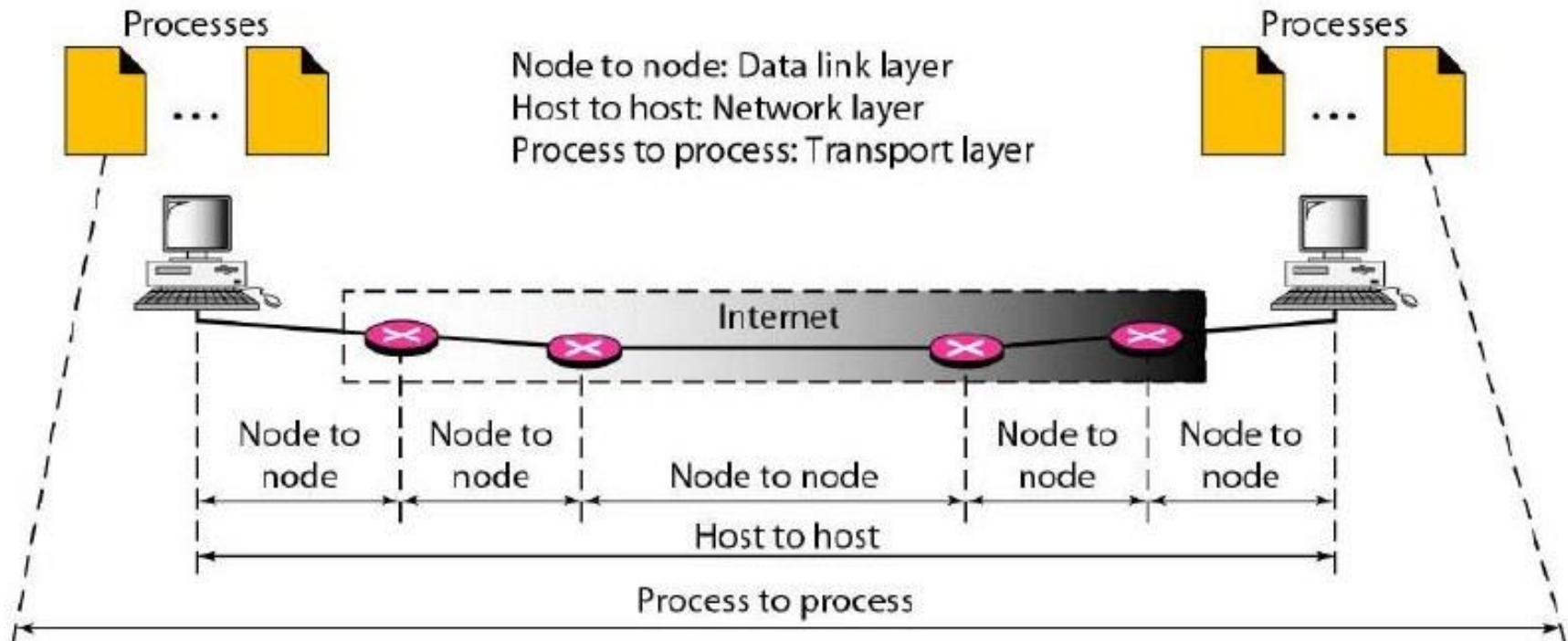
The transport layer is responsible for the delivery of a message (segments) from one process to another.



Responsibilities / Functions / Services of Transport Layer

- Process to process delivery
 - Service point addressing (Port addressing)
 - Segmentation and reassembly
 - Connection control
 - Flow control
 - Error control
-

Types of data deliveries

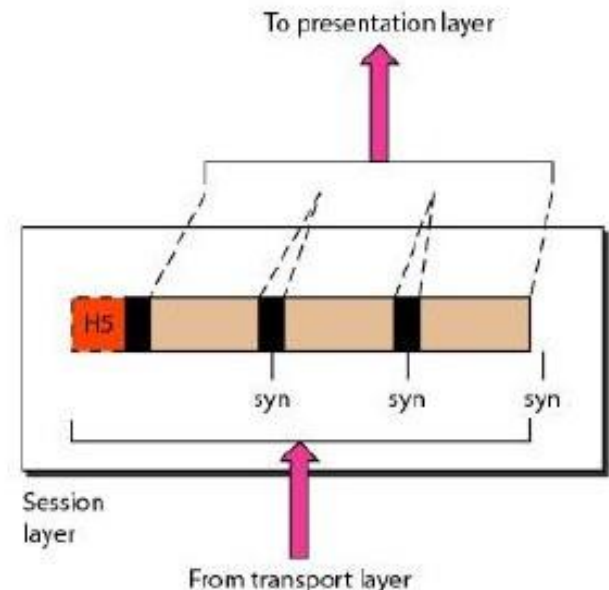
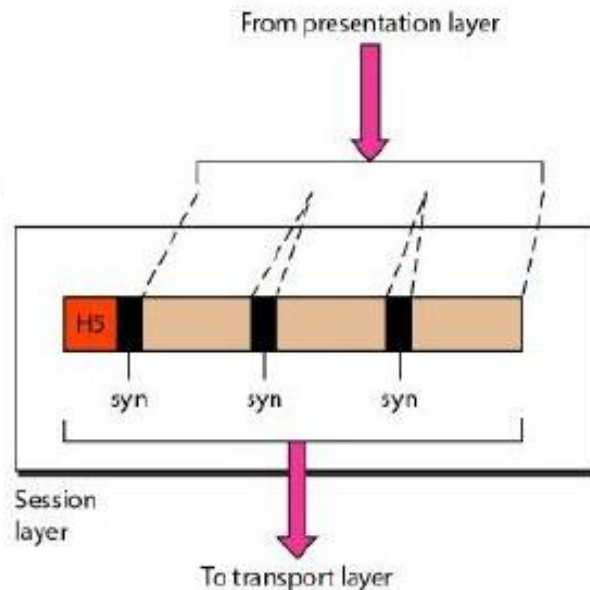


SESSION LAYER

- It is the network dialog controller
- Establishes, maintains and synchronizes the interaction among communicating systems

Responsibilities / Functions / Services of Session Layer

- Dialog control
- Synchronization



PRESENTATION LAYER

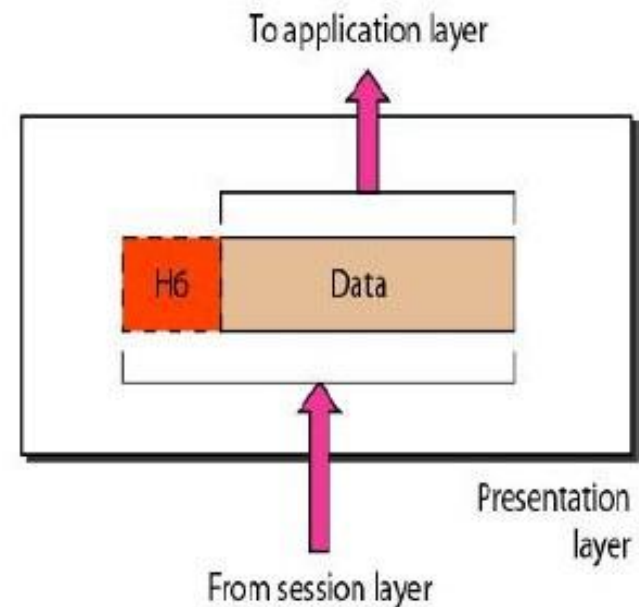
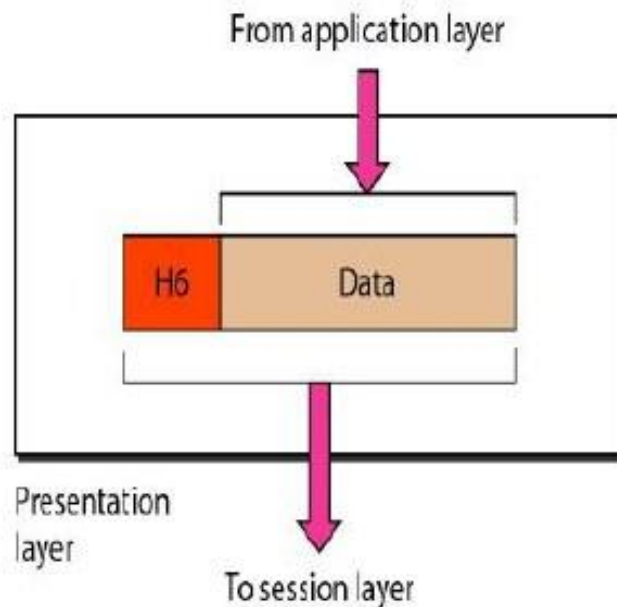
➤ Concerned with syntax and semantics of the information exchanged between two systems

Responsibilities / Functions / Services of Presentation Layer

➤ Translation

➤ Encryption

➤ Compression

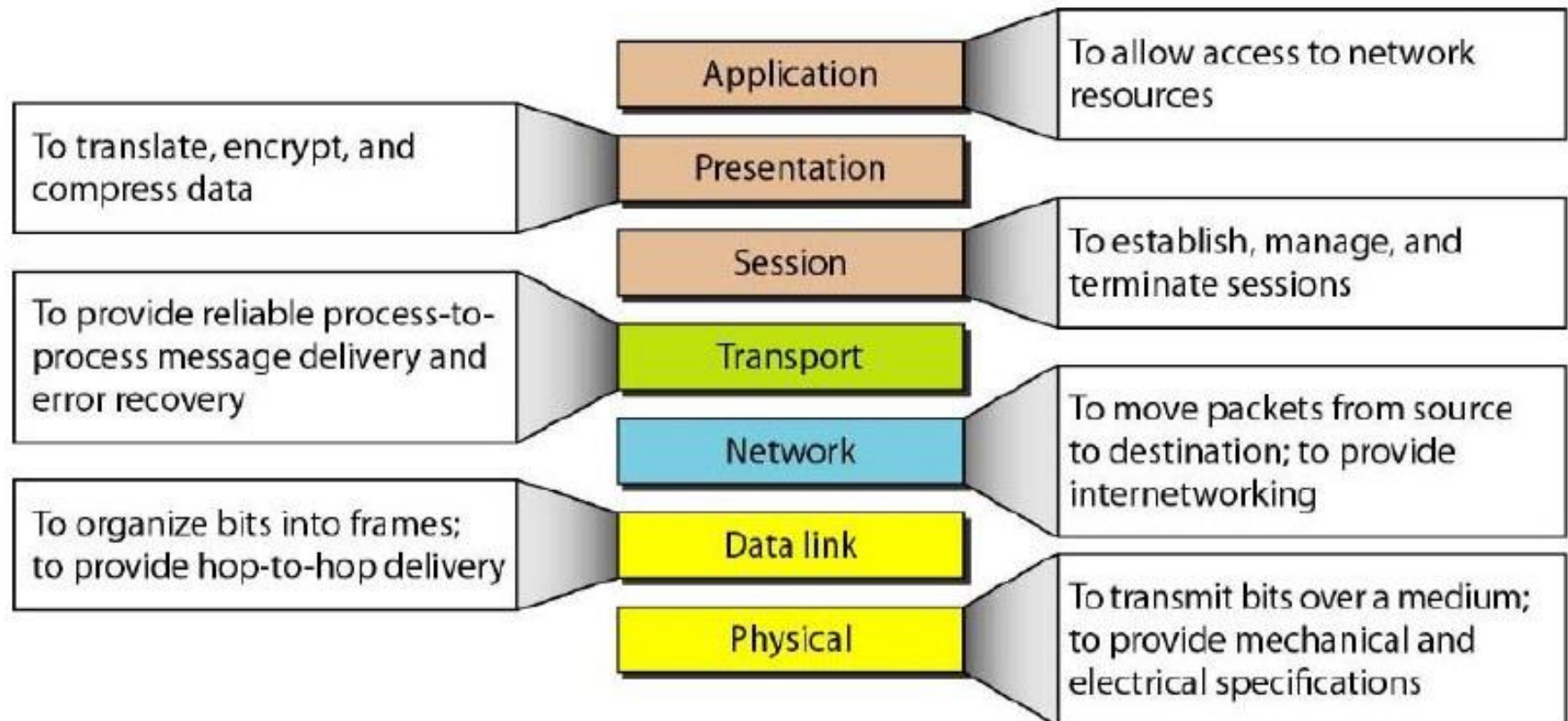


APPLICATION LAYER

- Enables the user (human or software) to access the network
- Provides user interfaces and support for services such as
 - E-mail
 - Remote file access and transfer
 - Shared database management
 - Other types of distributed information services

The application layer is responsible for providing services to the user.

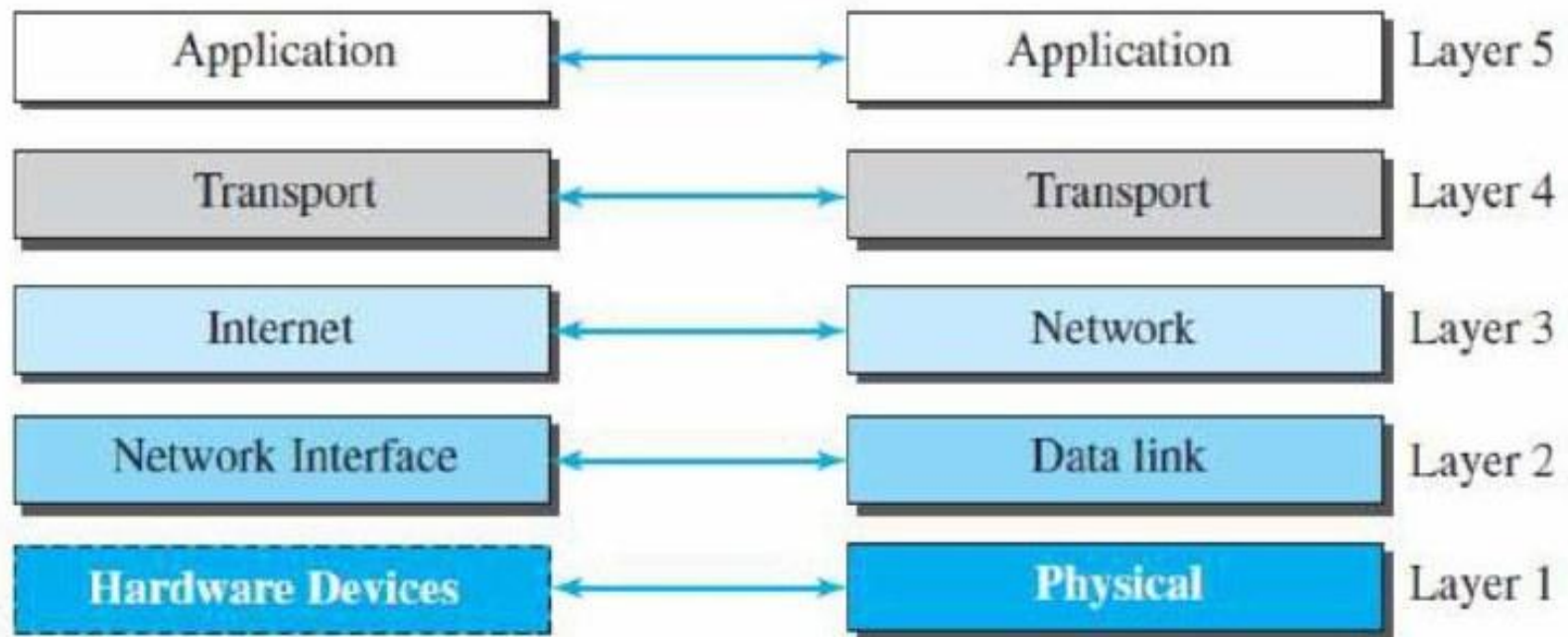
Summary of layers



TCP/IP PROTOCOL SUITE

*The layers in the **TCP/IP protocol suite** do not exactly match those in the OSI model..*

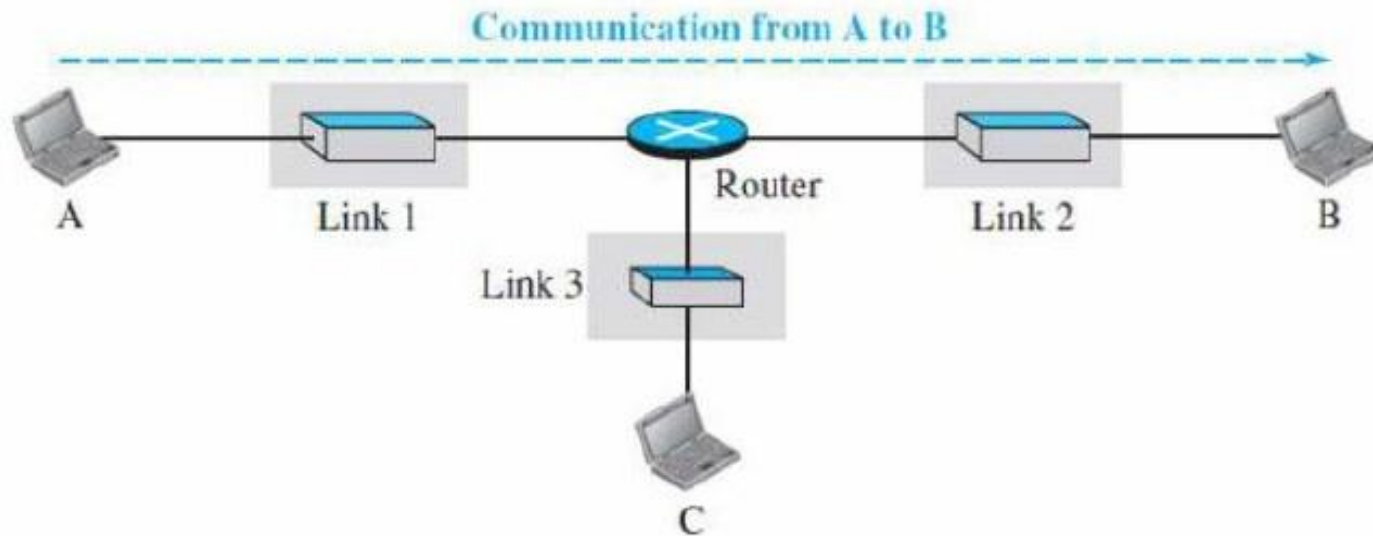
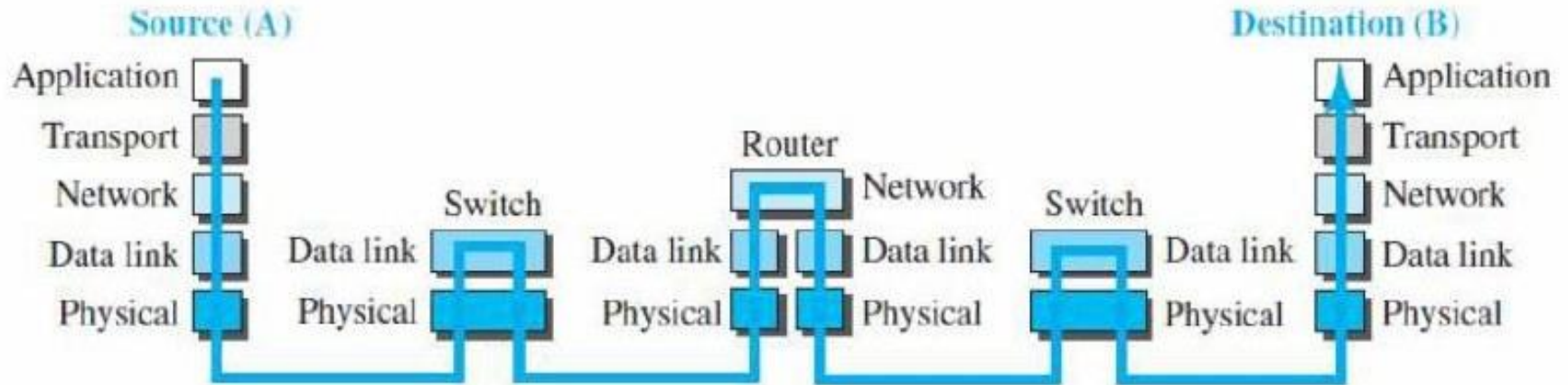
Layers in the TCP/IP protocol suite



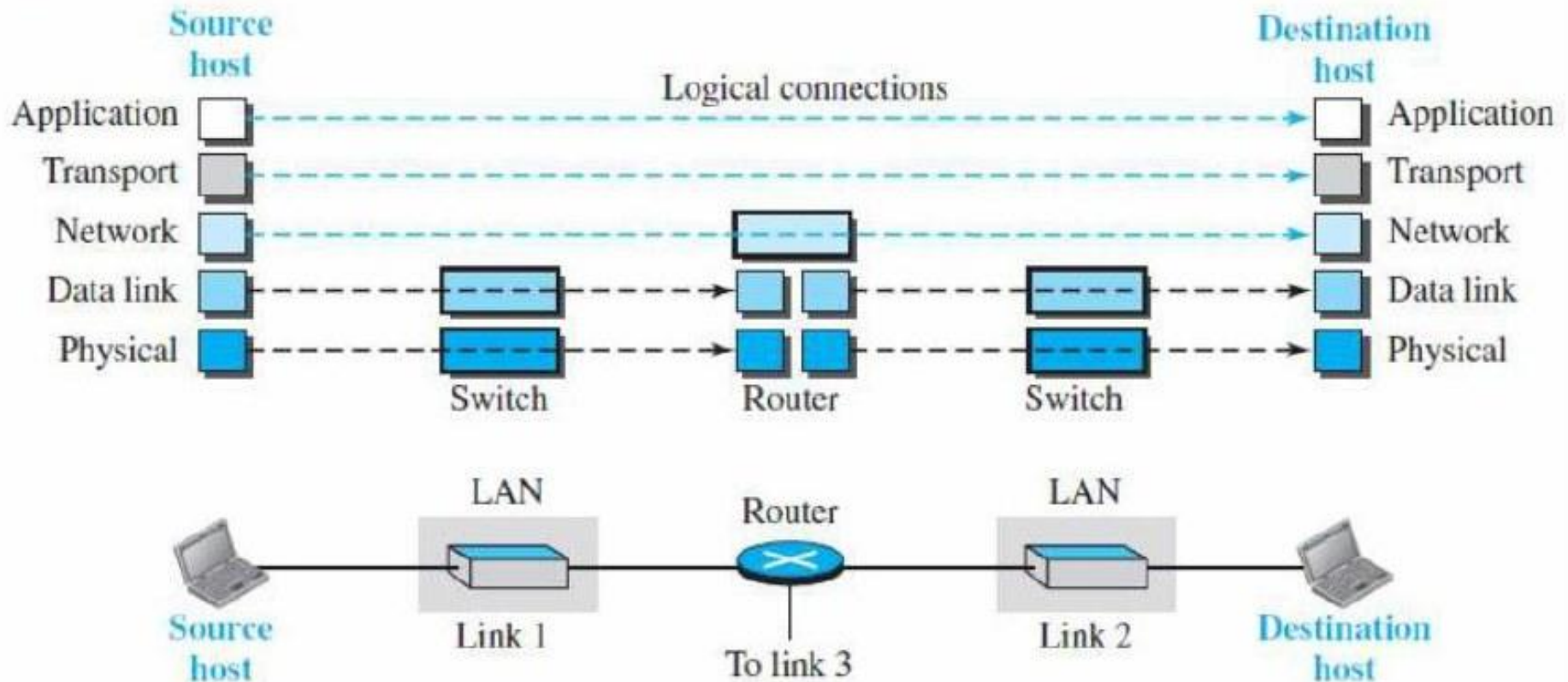
a. Original layers

b. Layers used in this book

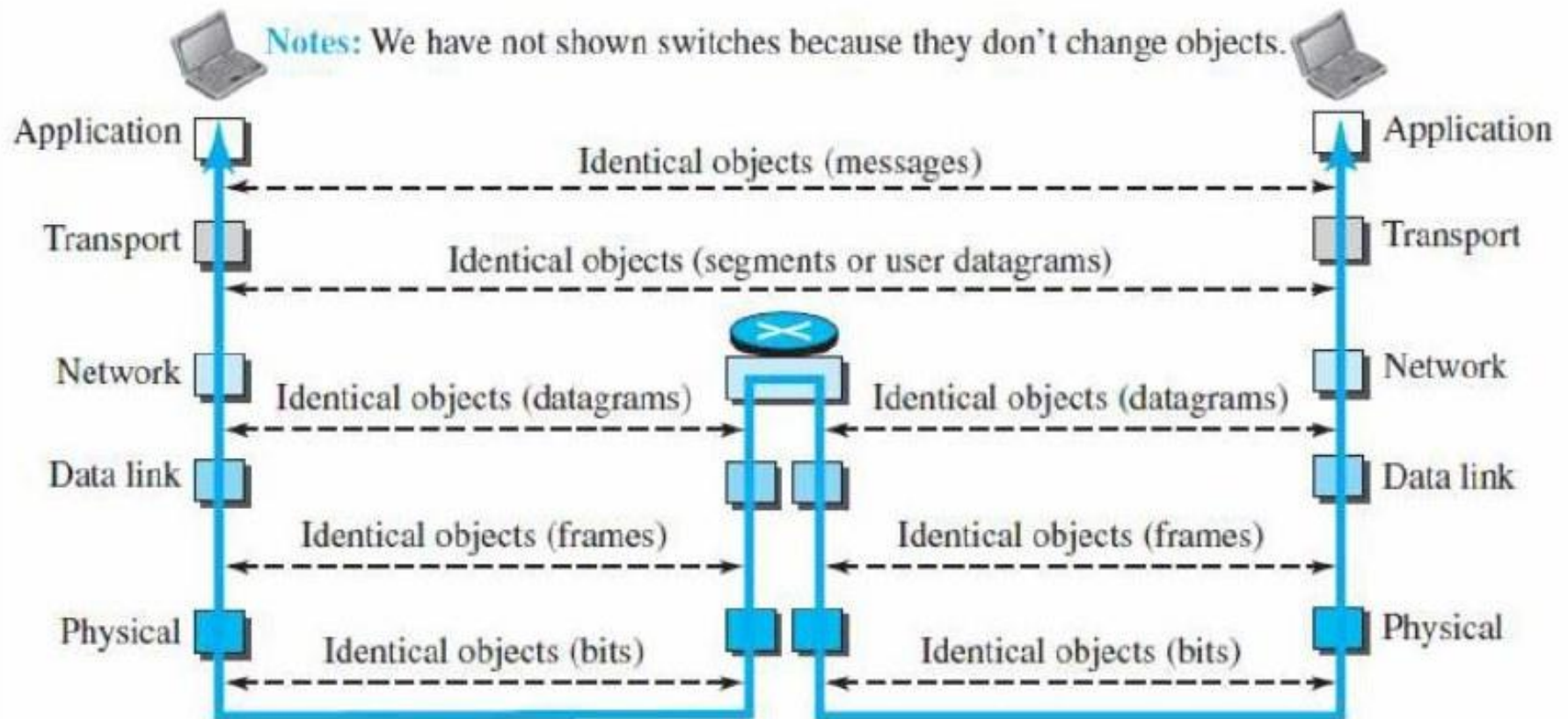
Communication through an internet



Logical connections between layers of the TCP/IP protocol suite



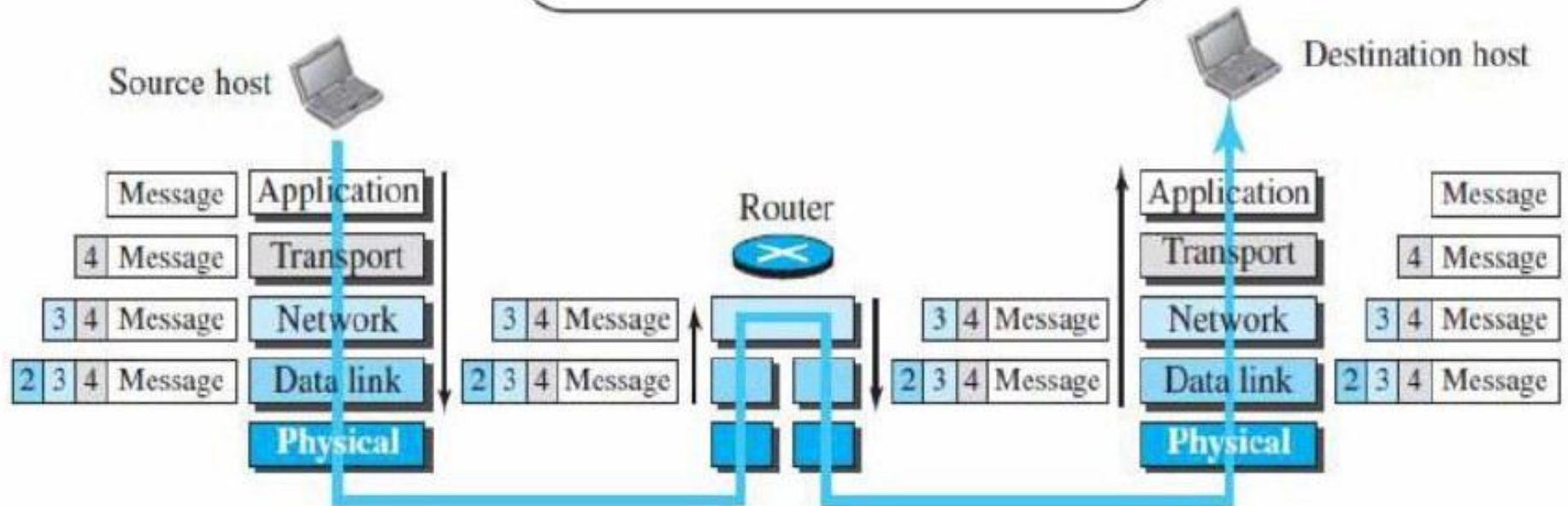
Identical objects in the TCP/IP protocol suite



Encapsulation/Decapsulation

Legend

- 4 Header at transport layer ↓ Encapsulate
- 3 Header at network layer ↑ Decapsulate
- 2 Header at data-link layer ↑ Decapsulate



Relationship of layers and addresses in TCP/IP

Packet names

Layers

Addresses

Message

Application layer

Names

Segment / User datagram

Transport layer

Port numbers

Datagram

Network layer

Logical addresses

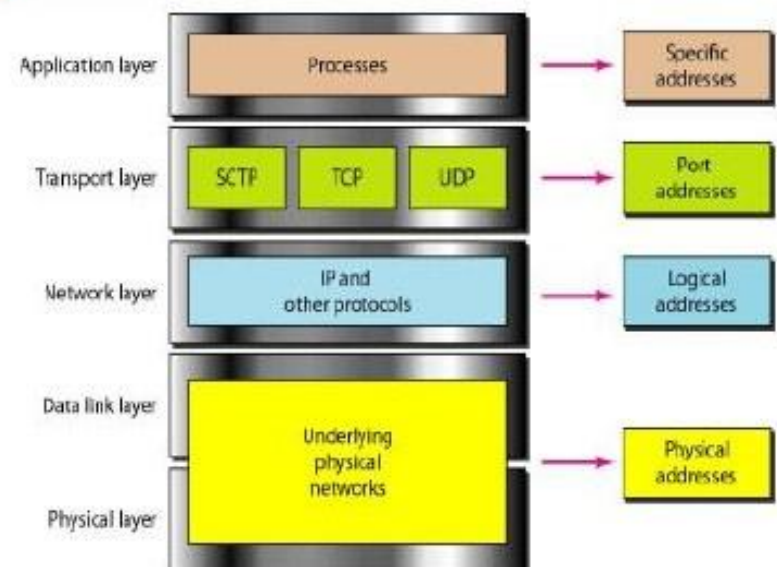
Frame

Data-link layer

Link-layer addresses

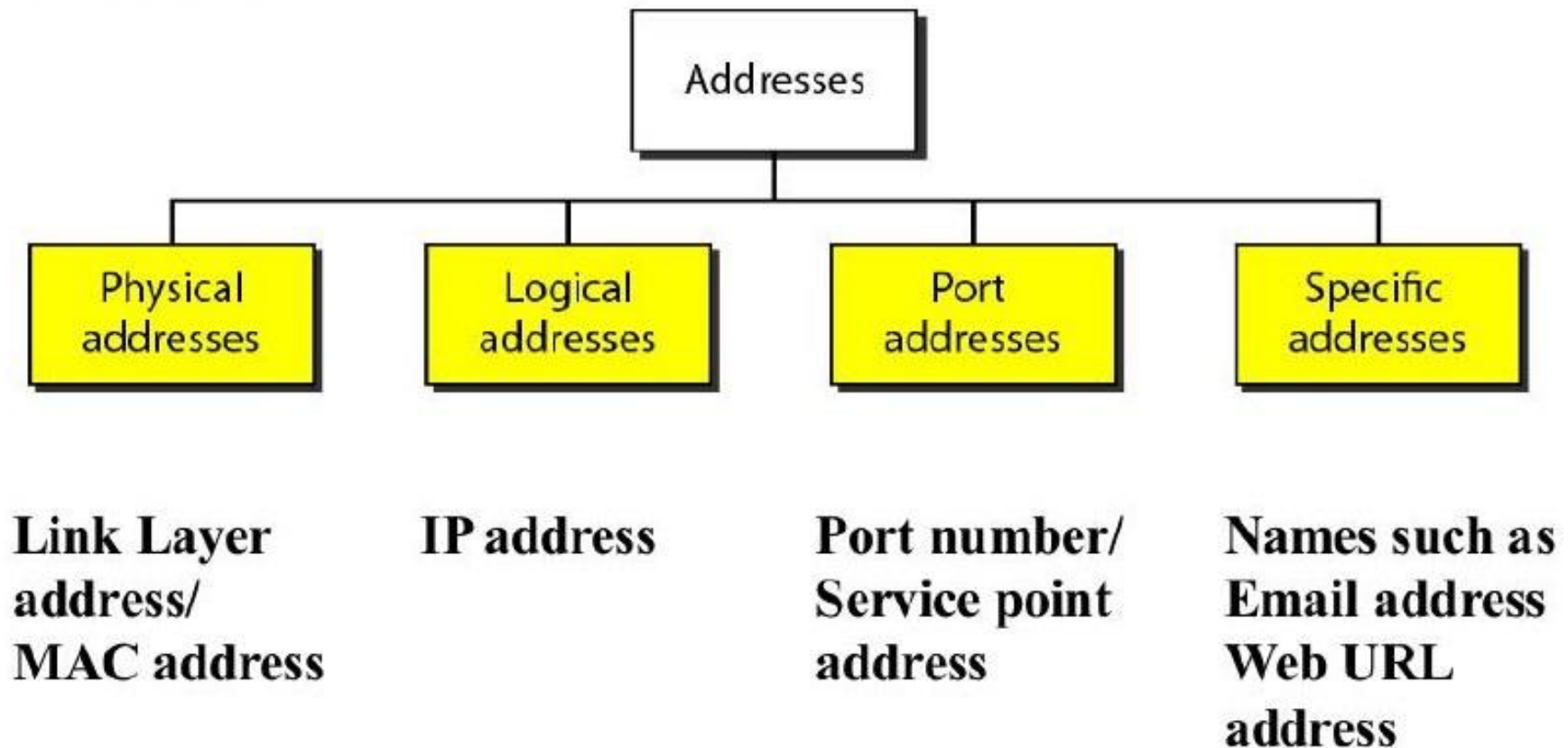
Bits

Physical layer



ADDRESSING

- Four levels of addresses are used in an internet employing the TCP/IP protocols
- Each address is related to a one layer in the TCP/IP architecture



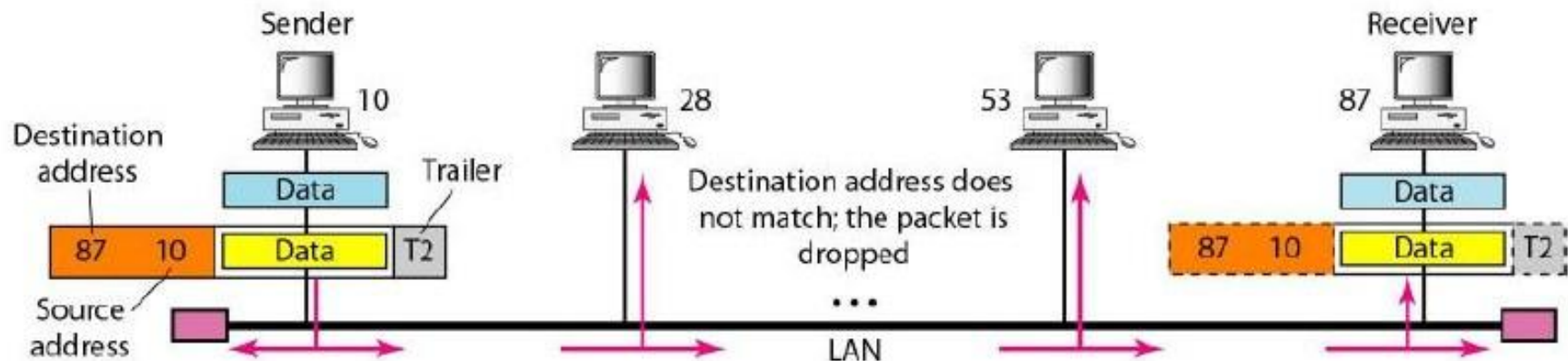
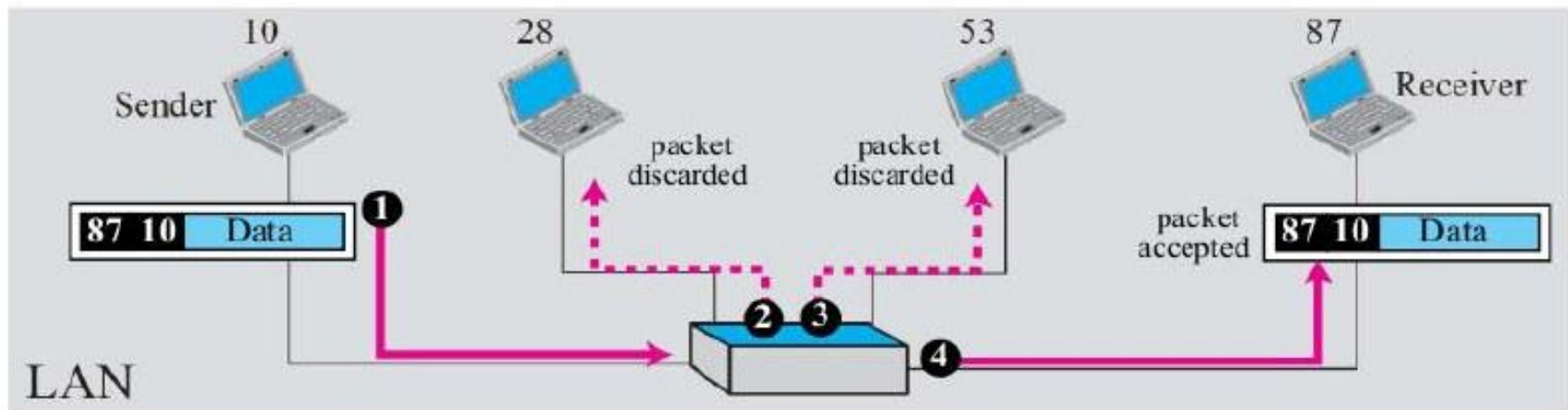
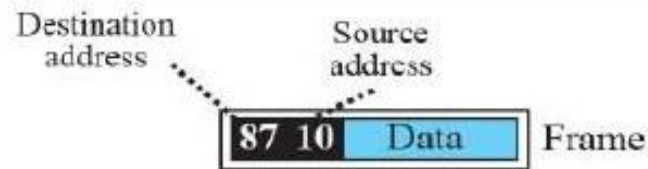
Addresses in TCP/IP – Physical Address

- Physical – 48 bit (6 byte) address
- Defined by LAN or WAN
- Physical address is written as 12 hexadecimal digits
- Each byte (2 hexadecimal digits) is separated by a colon, as shown below:

07:01:02:01:2C:4B

6-byte (12 hexadecimal digits)

Physical Address - Example



Addresses in TCP/IP – Logical Address

- Logical – 32 bit or 128 bit address
- Assigned by ISP

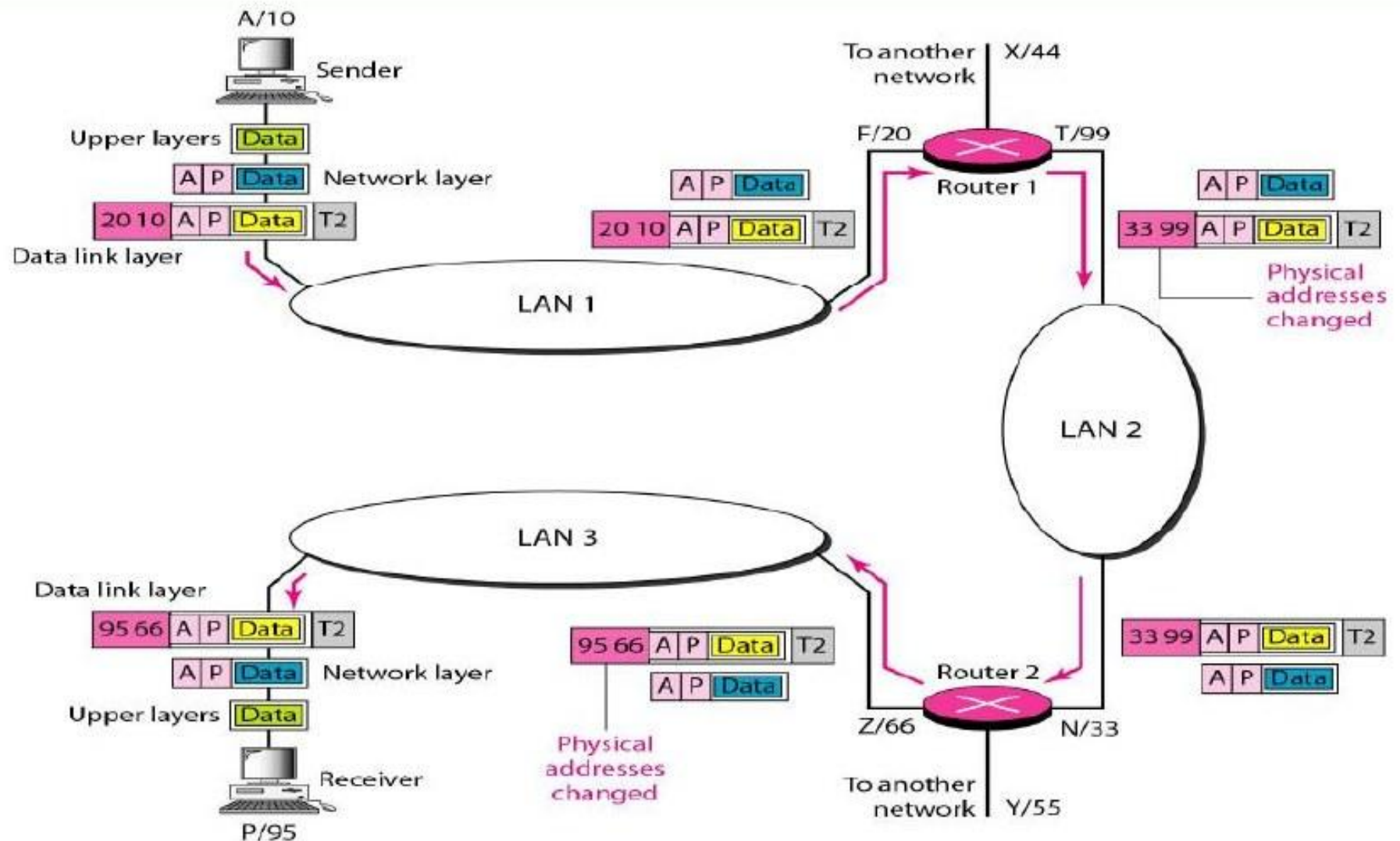
192.210.56.4

32 bit (4 byte) IPV4 Address

81AF:25B3:10FF:520D:5642:1234:9876:AB2F

128 bit (16 byte) IPV6 Address

Logical Address – Example



Note

The physical addresses will change from hop to hop; but the logical addresses usually remain the same.

Addresses in TCP/IP – Port Address

- Port – 16 bit address represented by one decimal number
- The purpose of ports is to uniquely identify different applications or processes running on a single computer

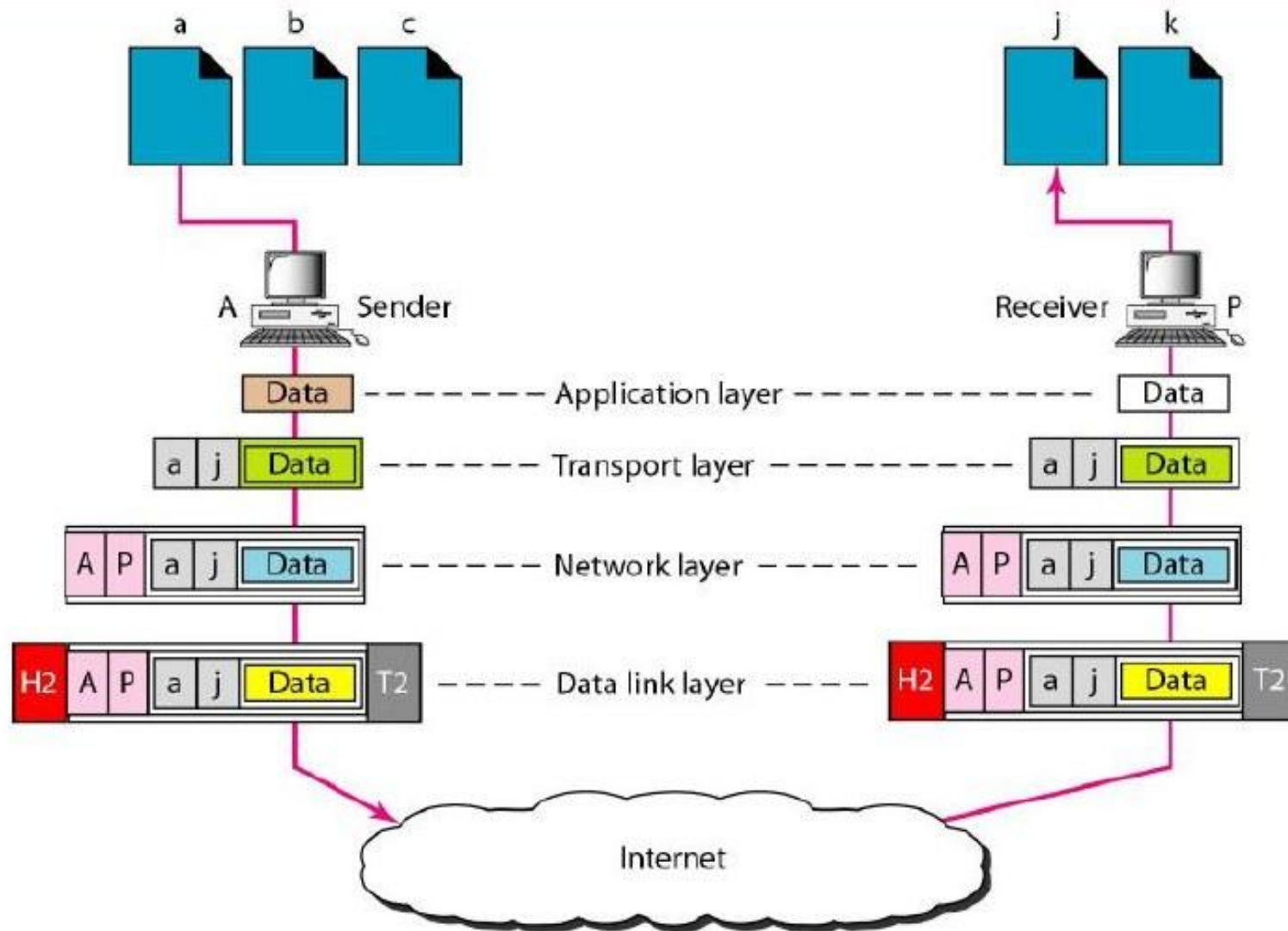
753

A 16-bit port address represented as one single number

20 – FTP

25 – SMTP

Port addresses – Example



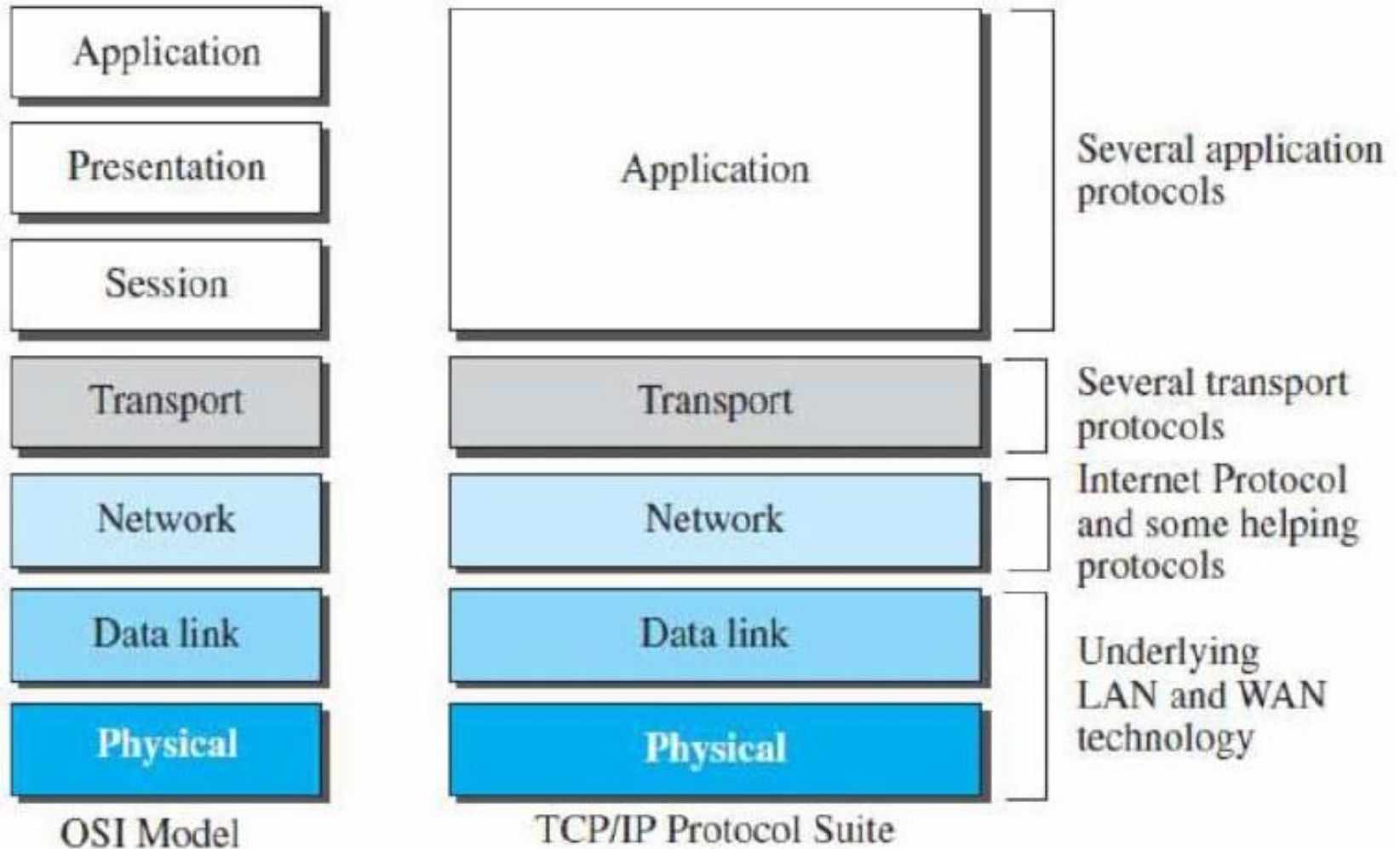
Note

The physical addresses will change from hop to hop; but the logical addresses and port address usually remain the same.

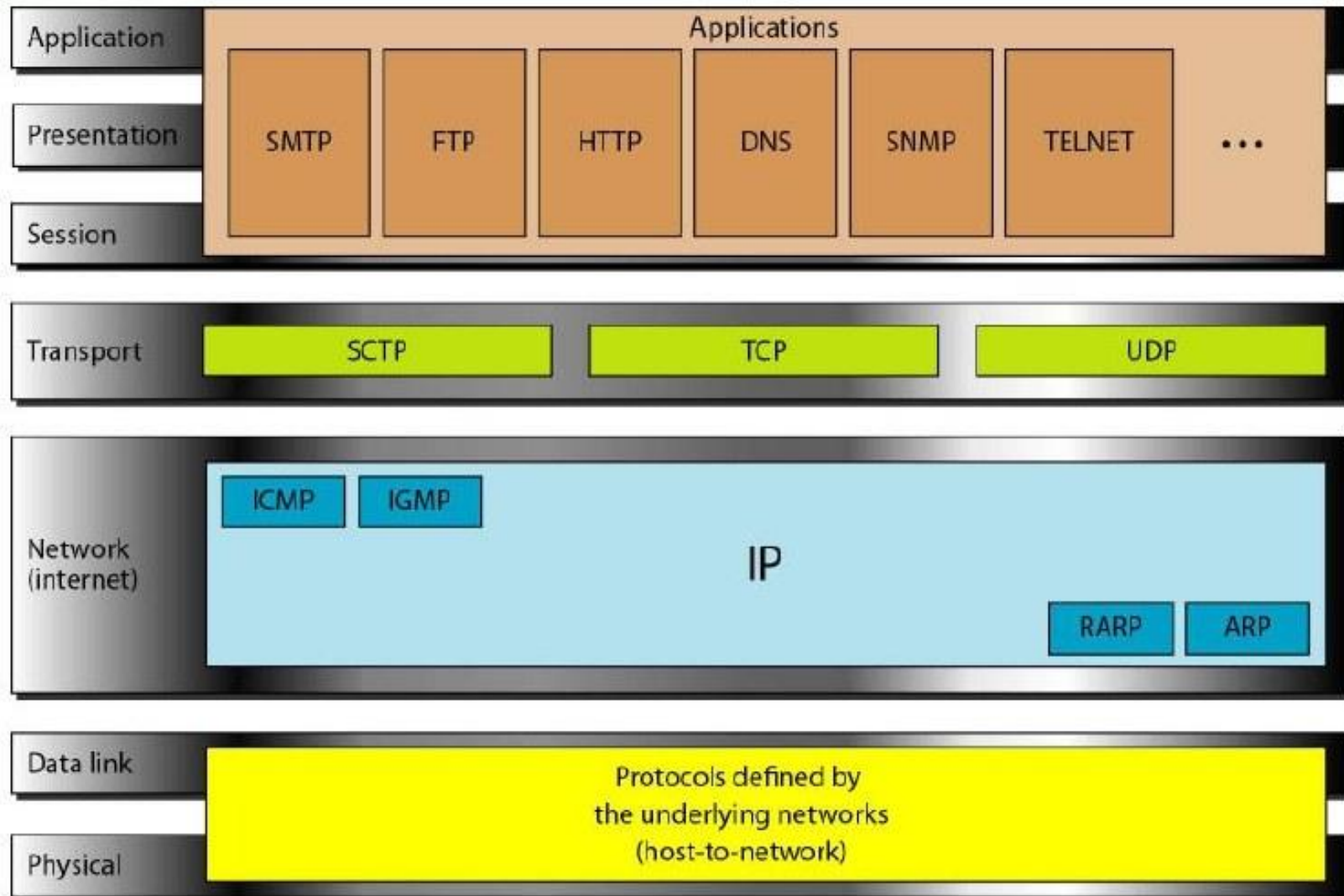
Addresses in TCP/IP – Specific Address

- User friendly address
- URL (Universal Resource Locator)
- Email Address

OSI model V/s TCP/IP



TCP/IP Protocol Suite



PHYSICAL AND DATA LINK LAYERS

- Together called as host to network layer
- Here TCP/IP does not define any specific protocol
- It uses all the standard and proprietary protocols
- A network in a TCP/IP internetwork can be a LAN or a WAN

NETWORK LAYER

- Also known as internetwork layer – host to host protocol (deliver a packet from one physical device to another)
- Supports Internetworking Protocol (IP)
- IP is an unreliable and connectionless protocol
- IP does not provide error checking
- IP uses 4 supporting protocols
 - ✓ ARP – Address Resolution Protocol
 - ✓ RARP – Reverse Address Resolution Protocol
 - ✓ IGMP – Internet Group Management Protocol
 - ✓ ICMP – Internet Control Message Protocol

- ARP is used to find the physical address of a node knowing its IP(logical) address
- RARP is used to discover the IP(logical) address of a node knowing its physical address
- ICMP is used to send notification of datagram problems to the sender. Sends query and error reporting messages
- IGMP is used to facilitate the simultaneous transmission of a message to group of recipients

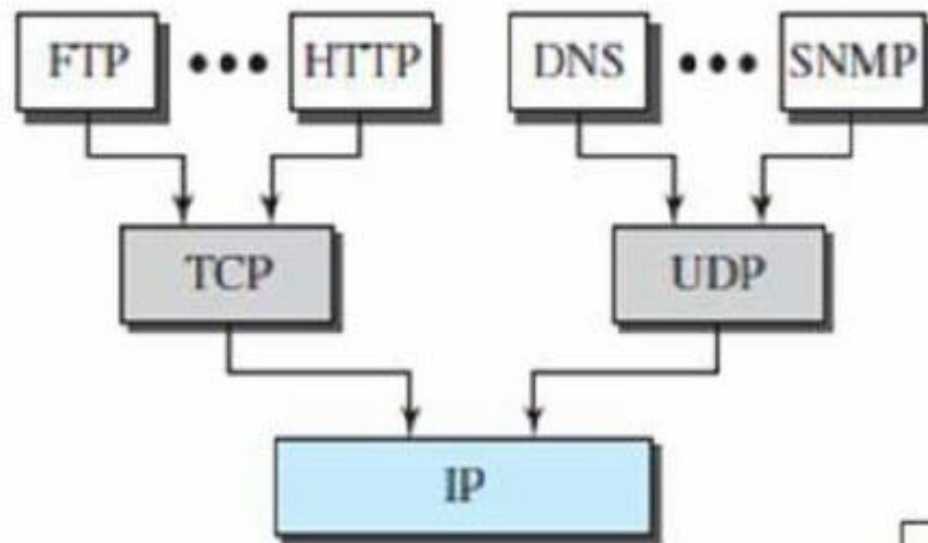
TRANSPORT LAYER

- Transport Layer is a process to process protocol
- It is traditionally represented by 2 protocols
 - ✓ TCP – Transmission Control Protocol
 - ✓ UDP – User Datagram Protocol
- It also defines a new protocol SCTP (Stream Control Transmission Protocol) to meet the needs of some newer applications
- UDP is connectionless and unreliable protocol
- TCP is connection oriented and reliable protocol
- SCTP combines the best features of TCP and UDP

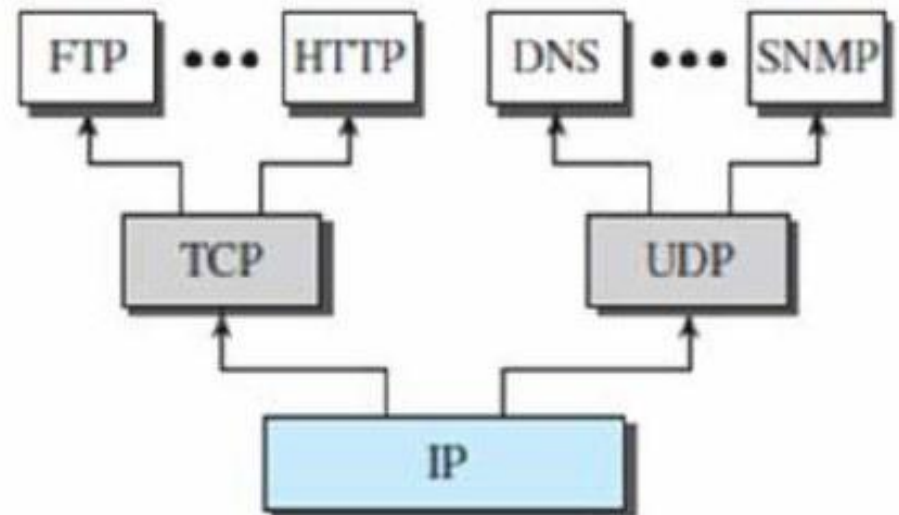
APPLICATION LAYER

- Application layer in TCP/IP is equivalent to the combined session, presentation and application layers in the OSI model
- Many protocols are defined at this layer
 - ✓ DNS – Domain Name System (Converts specific address to IP address)
 - ✓ SNMP – Simple Network Management Protocol
 - ✓ SMTP – Simple Mail Transfer Protocol (Email Services)
 - ✓ FTP – File Transfer Protocol
 - ✓ HTTP – Hyper Text Transfer Protocol (Retrieving a web document)
 - ✓ TELNET – Terminal Network (Remote Login)

Multiplexing and Demultiplexing



a. Multiplexing at source



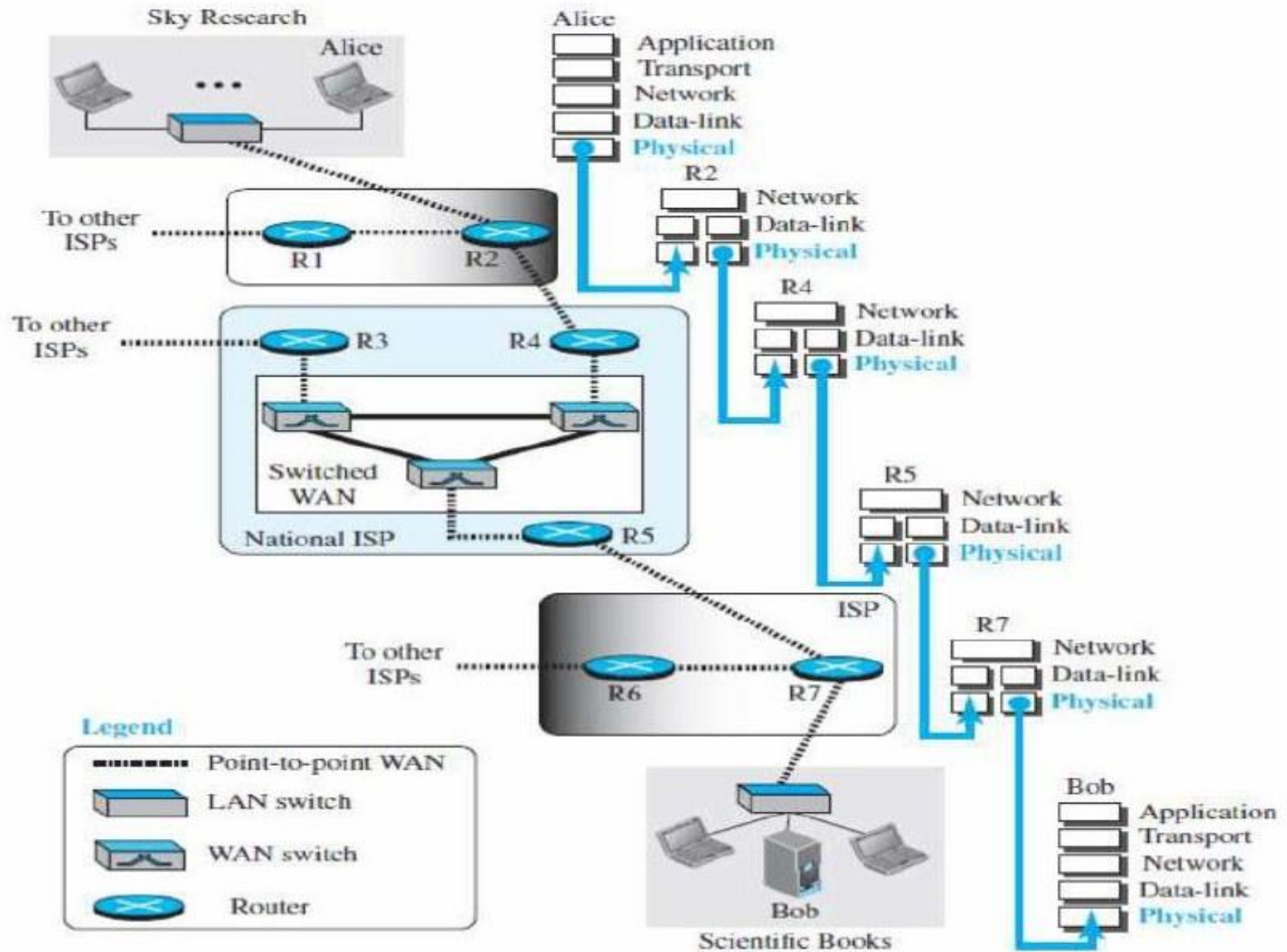
b. Demultiplexing at destination

Lack of OSI Model's Success

- First, OSI was completed when TCP/IP was fully in place and a lot of time and money had been spent on the suite; changing it would cost a lot.
- Second, some layers in the OSI model were never fully defined.
- Third, when OSI was implemented by an organization in a different application, it did not show a high enough level of performance to entice the Internet authority to switch from the TCP/IP protocol suite to the OSI model.

Physical Layer

Data and Signals



Data is an entity that conveys some meaning based on some mutually agreed upon rules/conventions between a sender and a receiver

Data can be Analog or Digital

Signals are the electric or electromagnetic encoding of data

Signal is an electrical impulse or radio wave transmitted or received

Signals can be Analog or Digital

ANALOG AND DIGITAL

Data can be *analog* or *digital*

- ❑ **Analog data** refers to information that is continuous
- ❑ Analog data take on continuous values
- ❑ Analog signals can have an infinite number of values in a range

- ❑ **Digital data** refers to information that has discrete states
- ❑ Digital data take on discrete values
- ❑ Digital signals can have only a limited number of values

Analog and Digital Data

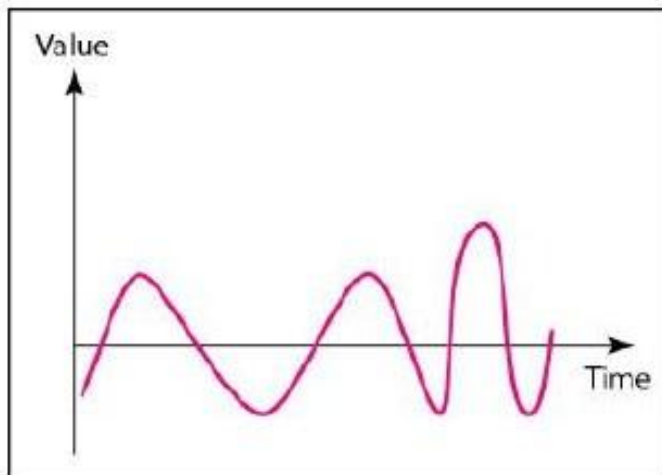
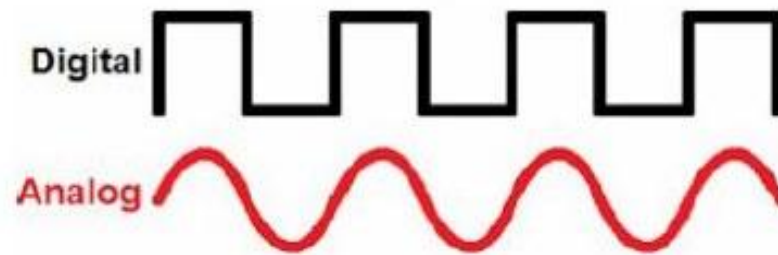


a. Analog

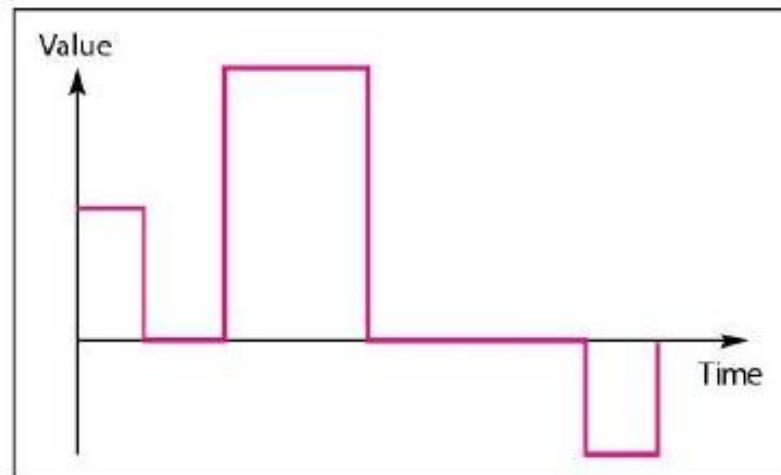


b. Digital

Analog and Digital Signals



a. Analog signal



b. Digital signal

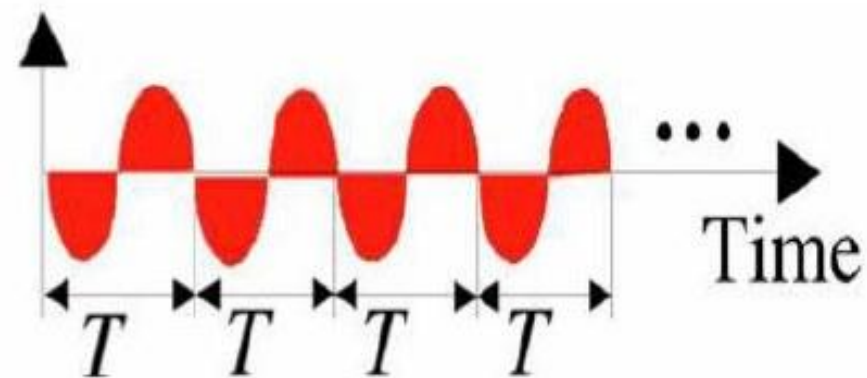
PERIODIC AND NONPERIODIC SIGNALS

- ✓ Periodic signals completes a pattern within a measurable time frame called period and repeats that pattern over subsequent identical periods.
- ✓ Completion of one full pattern is called a cycle.
- ✓ NonPeriodic signals changes without exhibiting a pattern or cycle that repeats over time
- ✓ NonPeriodic is also referred as Aperiodic
- ✓ Both analog and digital signals can be periodic or nonperiodic

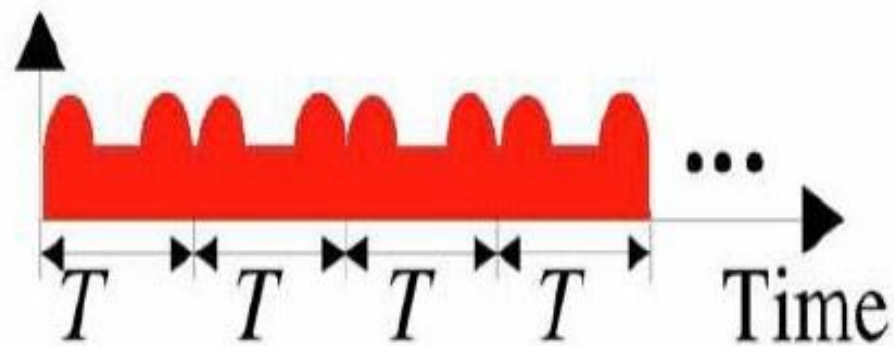
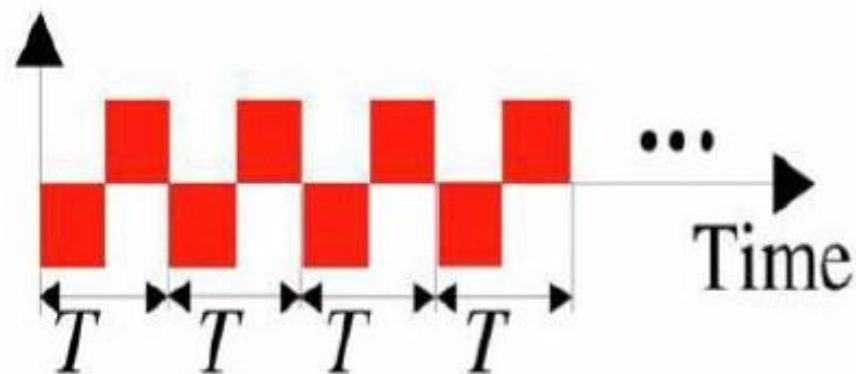
**In Data Communications, we commonly use
Periodic Analog Signals
and
Nonperiodic Digital Signals**

PERIODIC SIGNALS

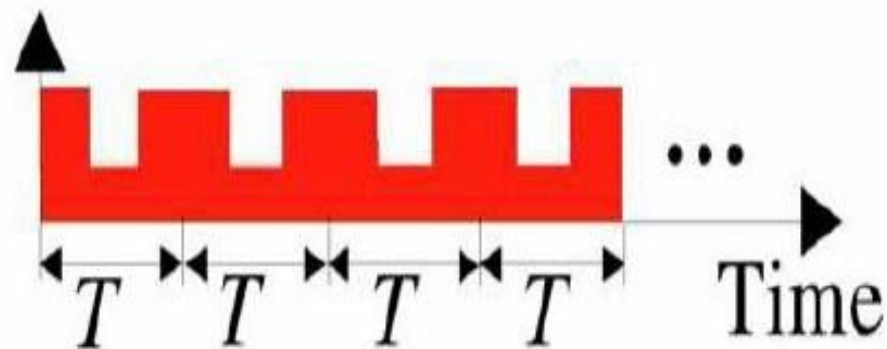
Value



Value

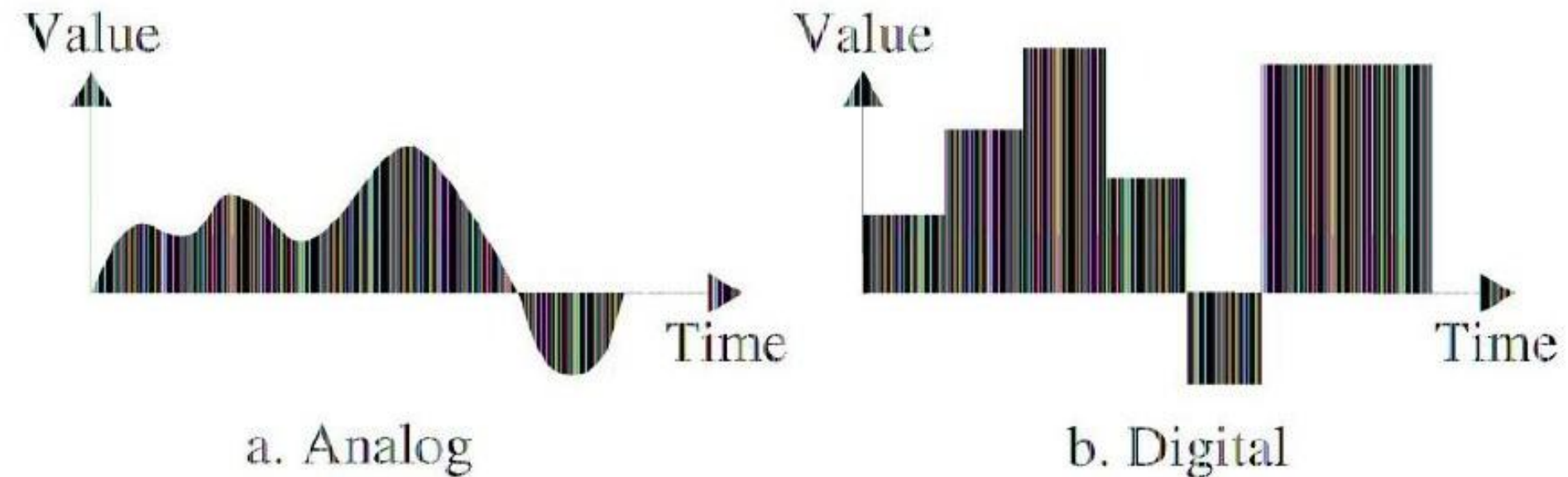


a. Analog



b. Digital

NONPERIODIC SIGNALS



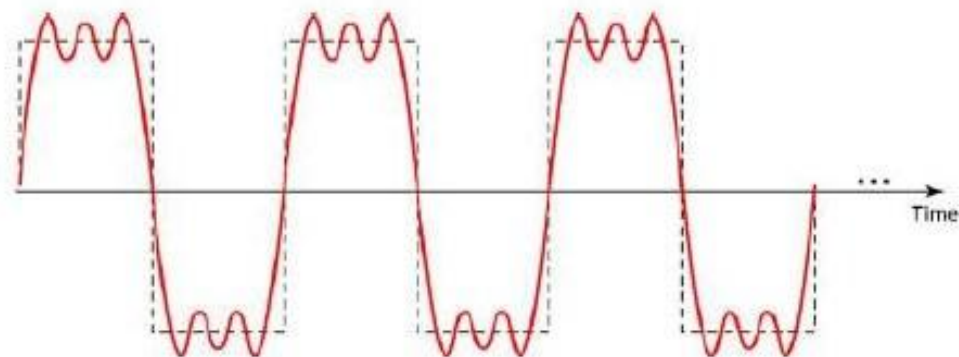
PERIODIC ANALOG SIGNALS

PERIODIC ANALOG SIGNALS

*Periodic analog signals can be classified as **simple** or **composite**.*

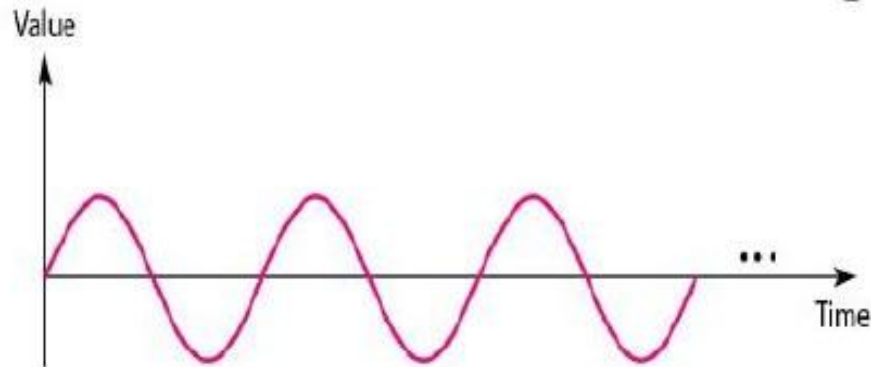
❑ A simple periodic analog signal, a **sine wave**, cannot be decomposed into simpler signals.

❑ A composite periodic analog signal is composed of multiple sine waves.



SINE WAVE

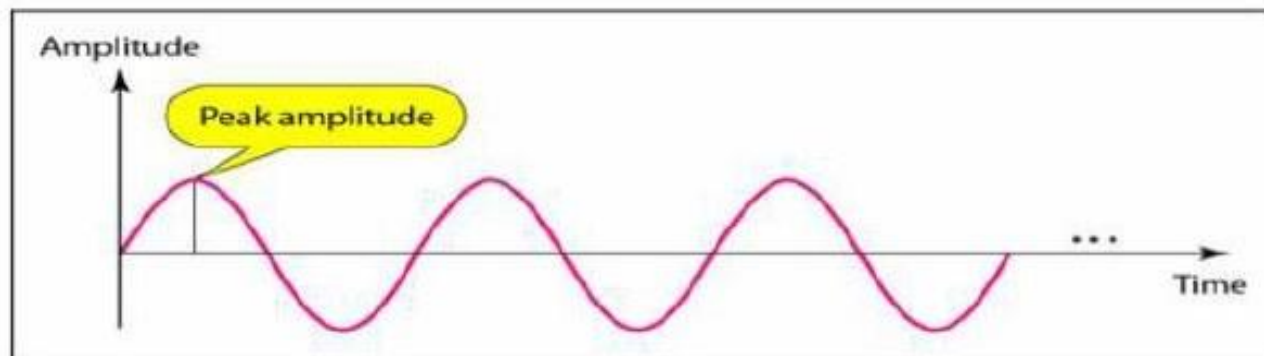
- ❑ Sine is the most fundamental form of a periodic analog signal



- ❑ It is a simple oscillating curve that change over a course of a cycle
- ❑ The cycle of a sine wave consists of a single arc above the time axis followed a single arc below it
- ❑ Sine wave can be represented by three parameters:
 - The Peak amplitude
 - The frequency
 - The phase

Peak Amplitude

- ✓ Amplitude is the vertical distance of any point on the sine wave curve from the time axis and the highest value is the peak amplitude
- ✓ The Peak amplitude is Absolute value of its highest intensity, proportional to the energy it carries
- ✓ Measured in volts



Frequency

Frequency is the rate of change of the signal with respect to time

- ✓ Frequency is associated with another parameter called period
- ✓ Period refers to the amount of time in seconds, a signal takes to complete one cycle
- ✓ Frequency refers to the number periods in on one second
- ✓ Measured in Hertz

- ❖ Change in a short span of time means high frequency
- ❖ Change over a long span of time means low frequency

- If a signal does not change at all, its frequency is zero
- If a signal changes instantaneously, its frequency is infinite

Frequency and Period

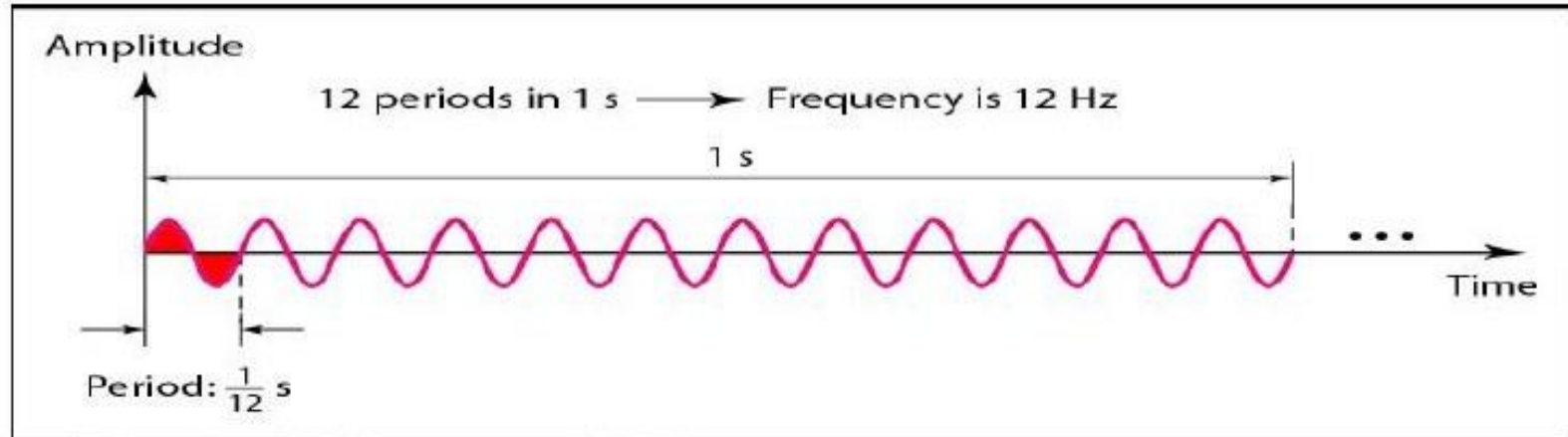
Frequency and period are the inverse of each other

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

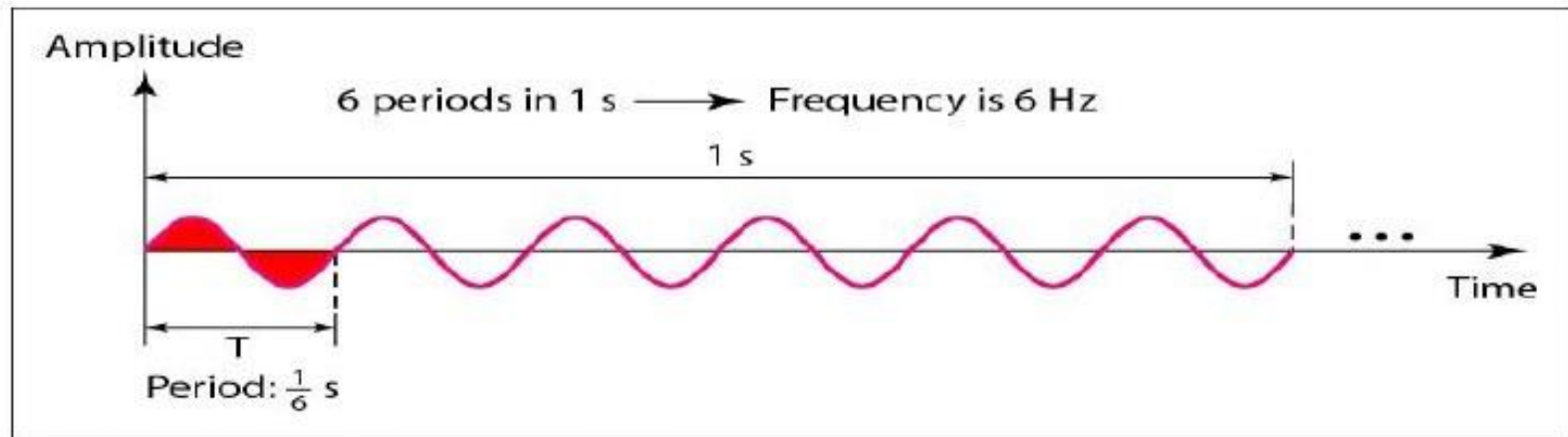
Units of period and frequency

<i>Unit</i>	<i>Equivalent</i>	<i>Unit</i>	<i>Equivalent</i>
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^3 Hz
Microseconds (μ s)	10^{-6} s	Megahertz (MHz)	10^6 Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^9 Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

Two signals with the same amplitude, but different frequencies



a. A signal with a frequency of 12 Hz



b. A signal with a frequency of 6 Hz

Examples

The power we use at home has a frequency of **60 Hz**. What is the period of this sine wave?

$$T = \frac{1}{f} = \frac{1}{60} = 0.0166 \text{ s} = 0.0166 \times 10^3 \text{ ms} = 16.6 \text{ ms}$$

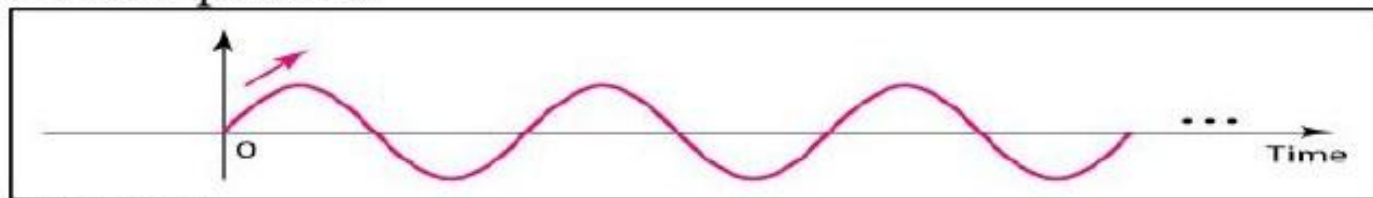
The period of a signal is 100 ms. What is its frequency in kilohertz?

$$100 \text{ ms} = 100 \times 10^{-3} \text{ s} = 10^{-1} \text{ s}$$
$$f = \frac{1}{T} = \frac{1}{10^{-1}} \text{ Hz} = 10 \text{ Hz} = 10 \times 10^{-3} \text{ kHz} = 10^{-2} \text{ kHz}$$

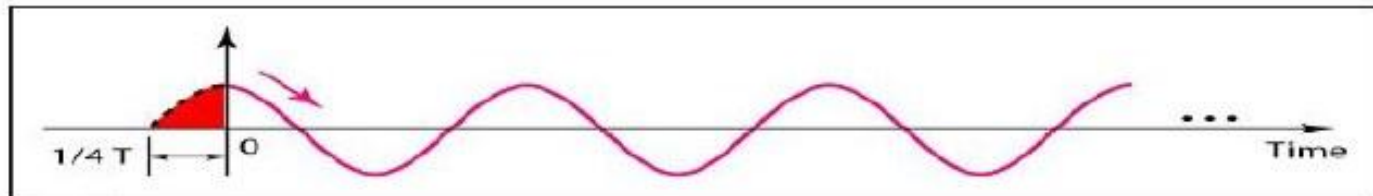
Phase

Phase describes the position of the waveform relative to time zero

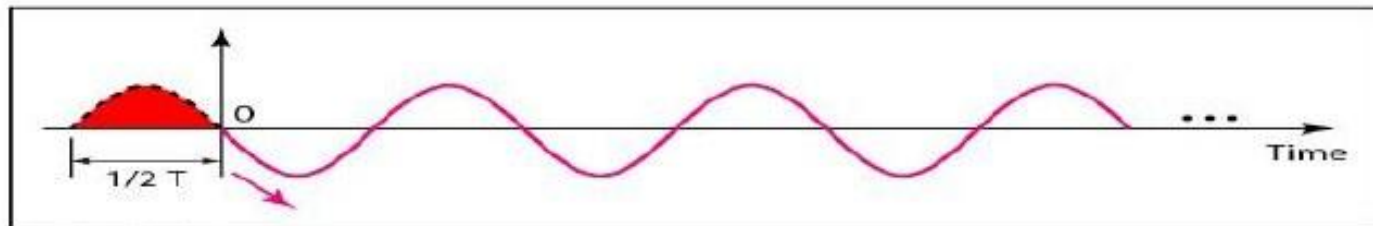
- ✓ Measured in degrees or radians
- ✓ Three sine waves with the same amplitude and frequency, but different phases



a. 0 degrees



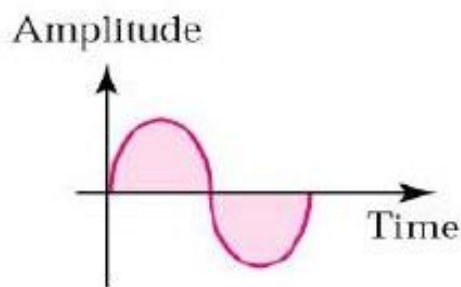
b. 90 degrees



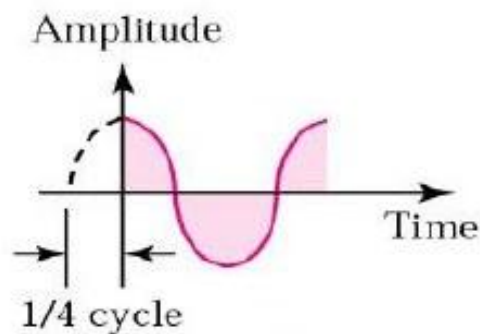
c. 180 degrees

Phase

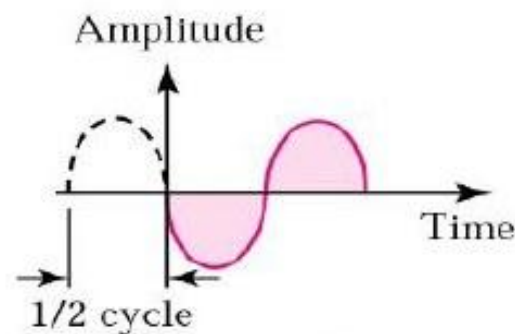
- ✓ A sine wave with a phase of 0° starts at time 0 with a zero amplitude. The amplitude is increasing.
- ✓ A sine wave with a phase of 90° starts at time 0 with a peak amplitude. The amplitude is decreasing.
- ✓ A sine wave with a phase of 180° starts at time 0 with a zero amplitude. The amplitude is decreasing.



a. 0°



b. 90°



c. 180°

- ✓ A sine wave with a phase of 0° is not shifted.
- ✓ A sine wave with a phase of 90° is shifted to the left by $\frac{1}{4}$ cycle. However, note that the signal does not really exist before time zero.
- ✓ A sine wave with a phase of 180° is shifted to the left by $\frac{1}{2}$ cycle. However, note that the signal does not really exist before time zero.

Example

A sine wave is offset 1/6 cycle with respect to time 0. What is its phase in degrees and radians?

Solution

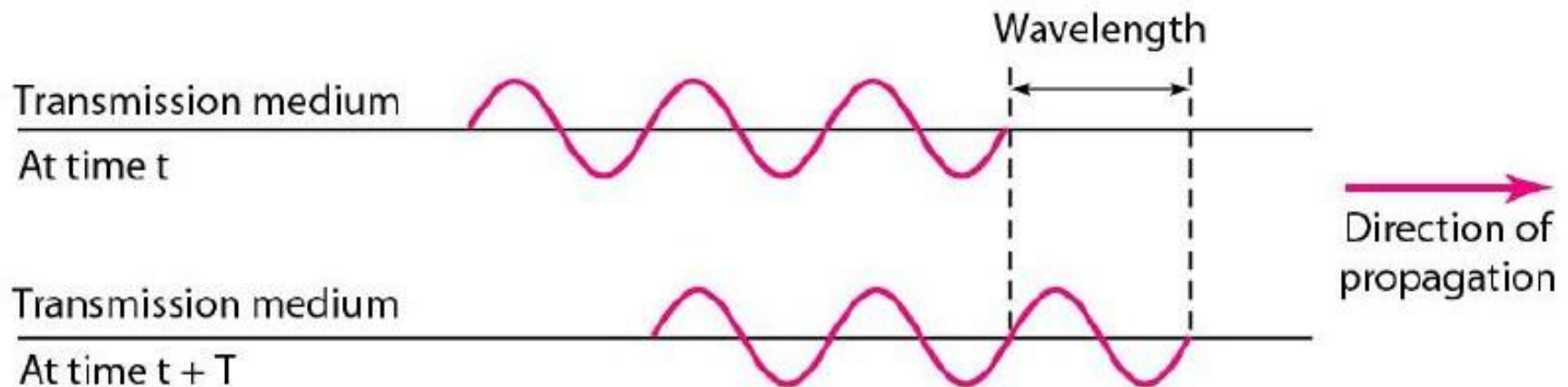
We know that 1 complete cycle is 360°. Therefore, 1/6 cycle is

$$\frac{1}{6} \times 360 = 60^\circ = 60 \times \frac{2\pi}{360} \text{ rad} = \frac{\pi}{3} \text{ rad} = 1.046 \text{ rad}$$

Wavelength

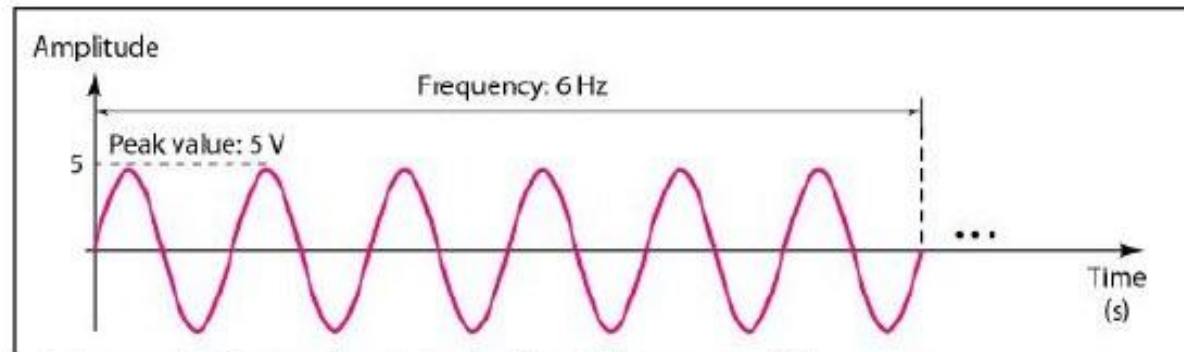
- ✓ It is another characteristic of a signal
- ✓ It binds the period or frequency of a simple sine wave to the propagation speed

$$\begin{aligned}\text{Wavelength } (\lambda) &= \text{Propagation speed} \times \text{Period} \\ &= \text{Propagation speed} / \text{Frequency}\end{aligned}$$

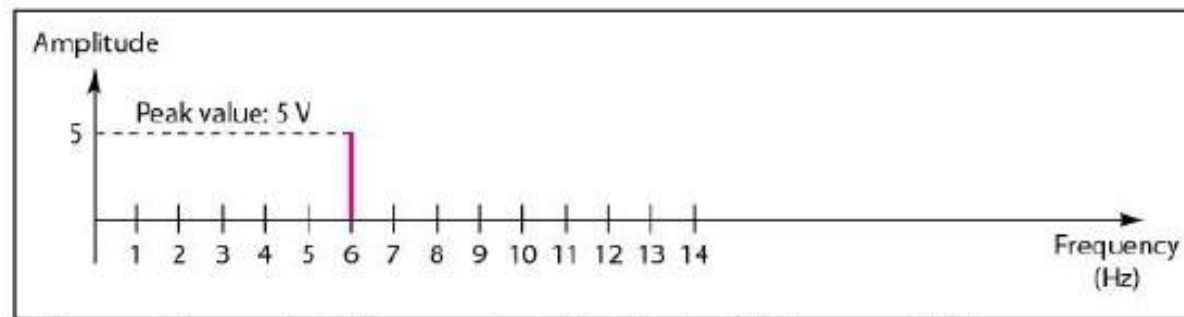


Time-domain and frequency-domain plots of a sine wave

A complete sine wave in the time domain can be represented by one single spike in the frequency domain

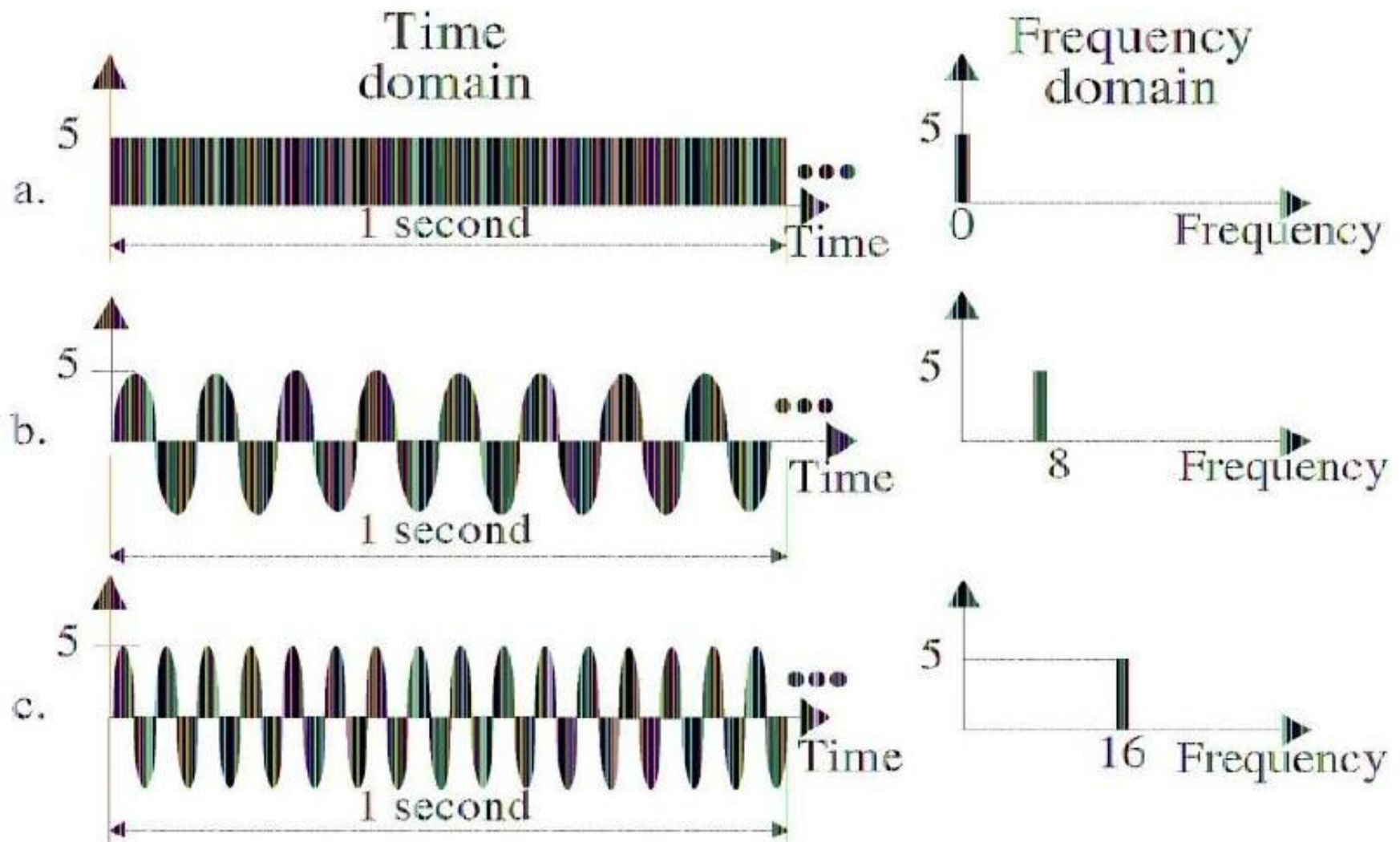


a. A sine wave in the time domain (peak value: 5 V, frequency: 6 Hz)

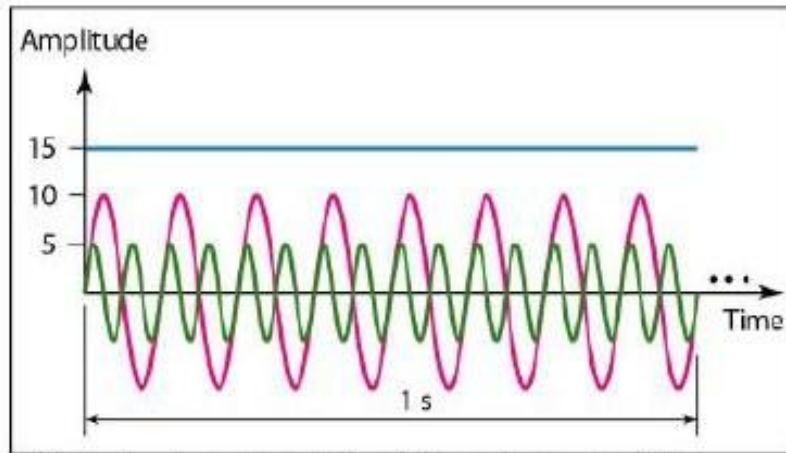


b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)

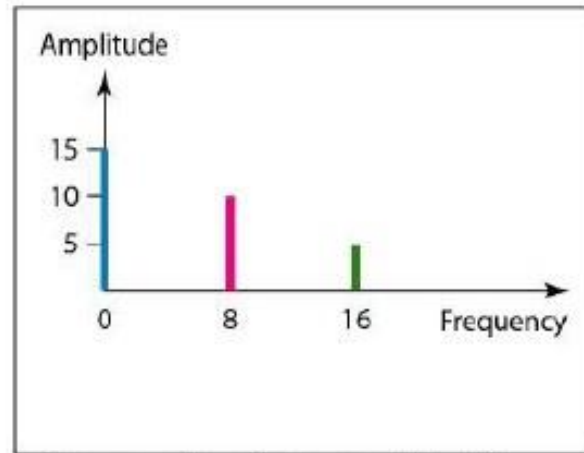
Time and Frequency Domain



Time and Frequency Domain



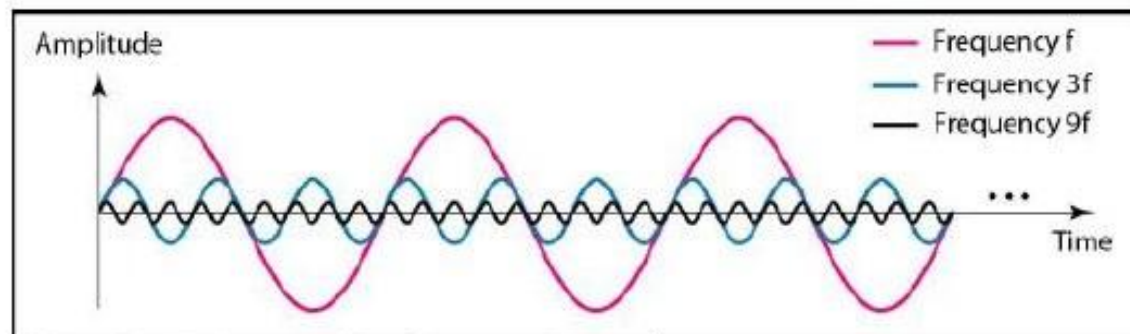
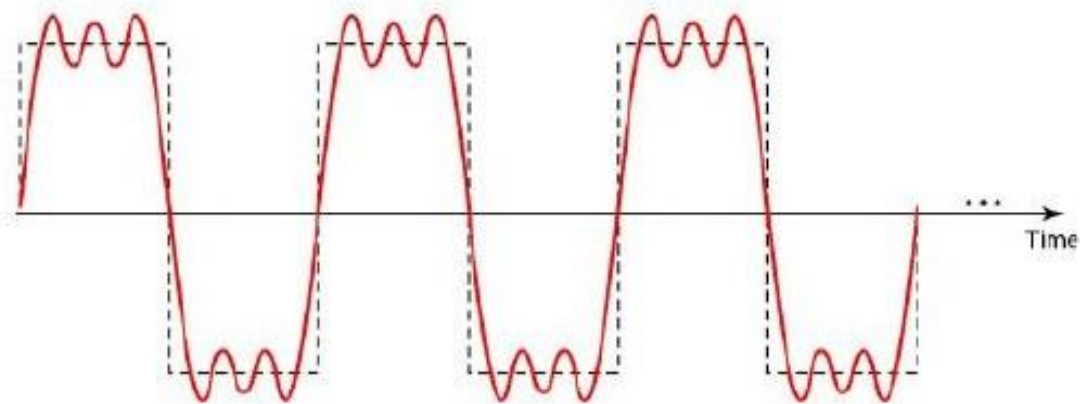
a. Time-domain representation of three sine waves with frequencies 0, 8, and 16



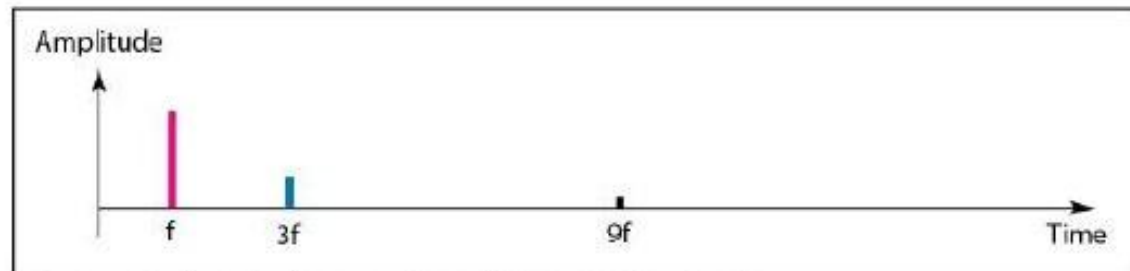
b. Frequency-domain representation of the same three signals

- ❑ The frequency domain is more compact and useful when we are dealing with more than one sine wave.
- ❑ A single-frequency sine wave is not useful in data communication
 - We need to send a **composite signal**, a signal made of many simple sine waves.

A composite periodic signal



a. Time-domain decomposition of a composite signal

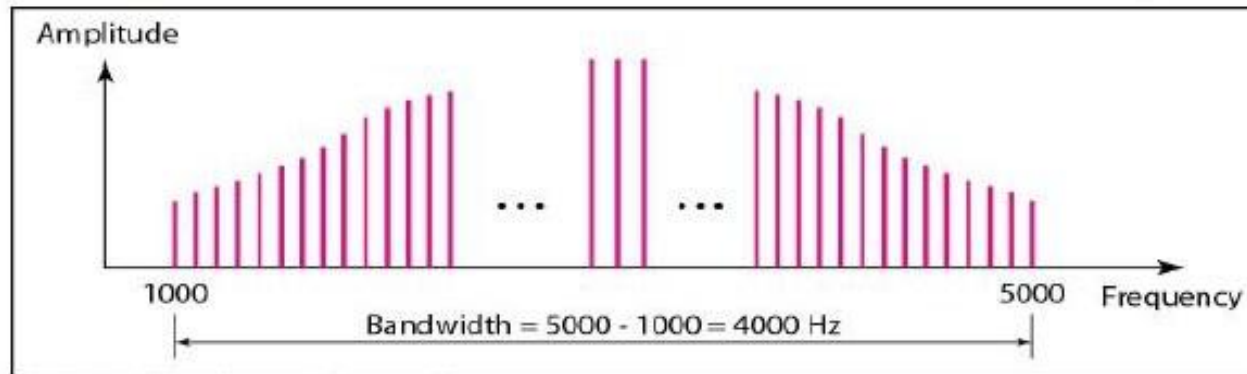


b. Frequency-domain decomposition of the composite signal

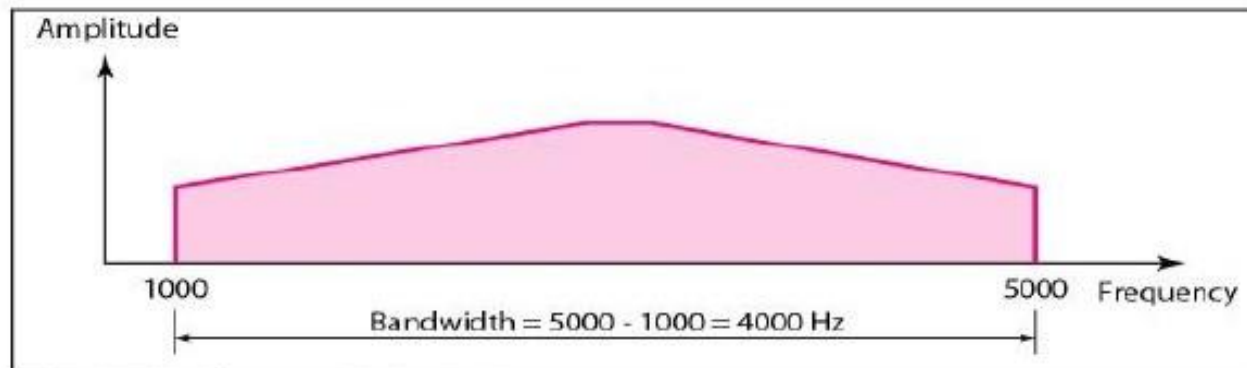
Decomposition of the composite periodic signal in the time and frequency domains

Bandwidth

The bandwidth of a composite signal is the difference between the highest and the lowest frequencies contained in that signal.



a. Bandwidth of a periodic signal



b. Bandwidth of a nonperiodic signal

Example

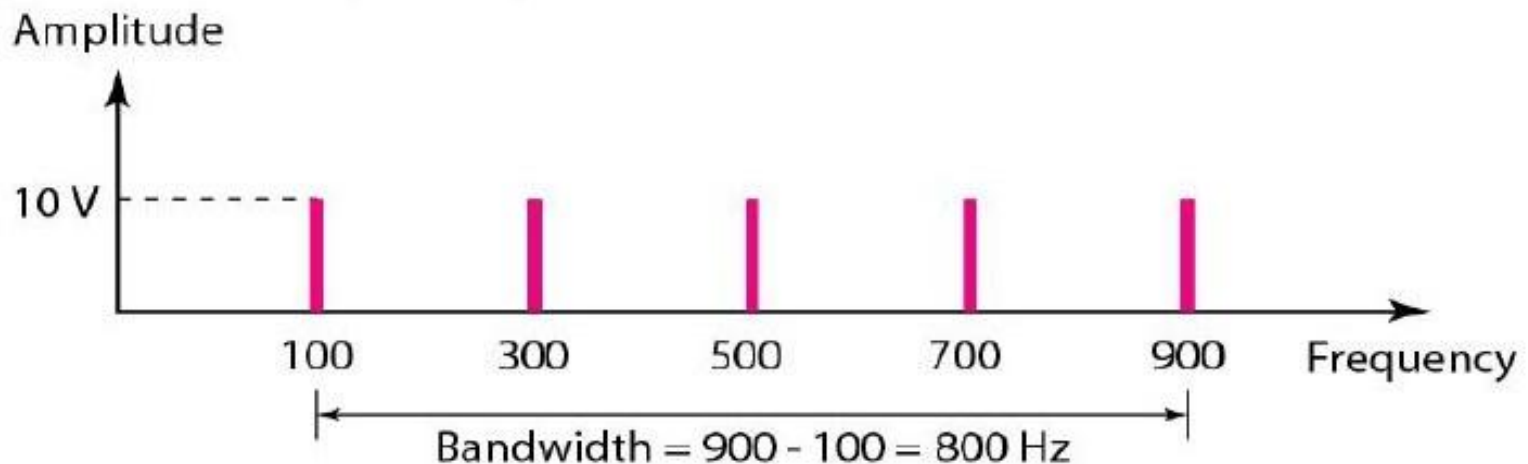
If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum (range of frequencies), assuming all components have a maximum amplitude of 10 V.

Solution

Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$B = f_h - f_l = 900 - 100 = 800 \text{ Hz}$$

The spectrum has only five spikes, at 100, 300, 500, 700, and 900 Hz



Example

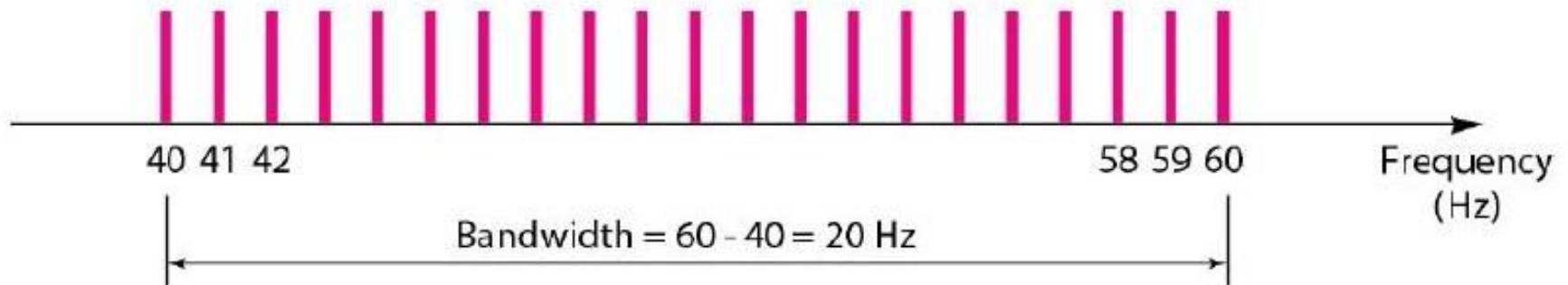
A periodic signal has a bandwidth of 20 Hz. The highest frequency is 60 Hz. What is the lowest frequency? Draw the spectrum if the signal contains all frequencies of the same amplitude.

Solution

Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$B = f_h - f_l \Rightarrow 20 = 60 - f_l \Rightarrow f_l = 60 - 20 = 40 \text{ Hz}$$

The spectrum contains all integer frequencies. We show this by a series of spikes

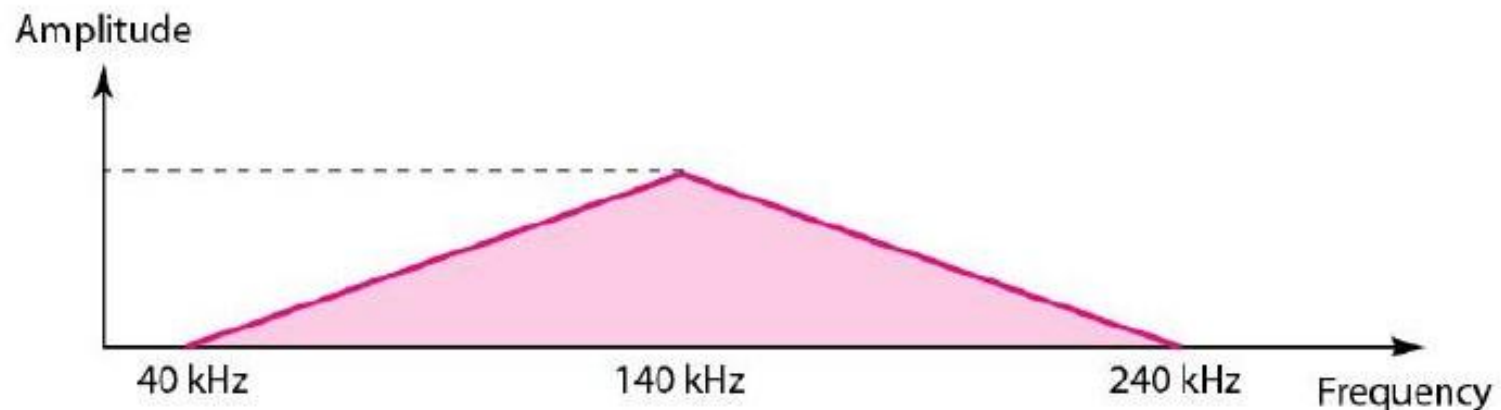


Example

A nonperiodic composite signal has a bandwidth of 200 kHz, with a middle frequency of 140 kHz and peak amplitude of 20 V. The two extreme frequencies have an amplitude of 0. Draw the frequency domain of the signal.

Solution

The lowest frequency must be at 40 kHz and the highest at 240 kHz.

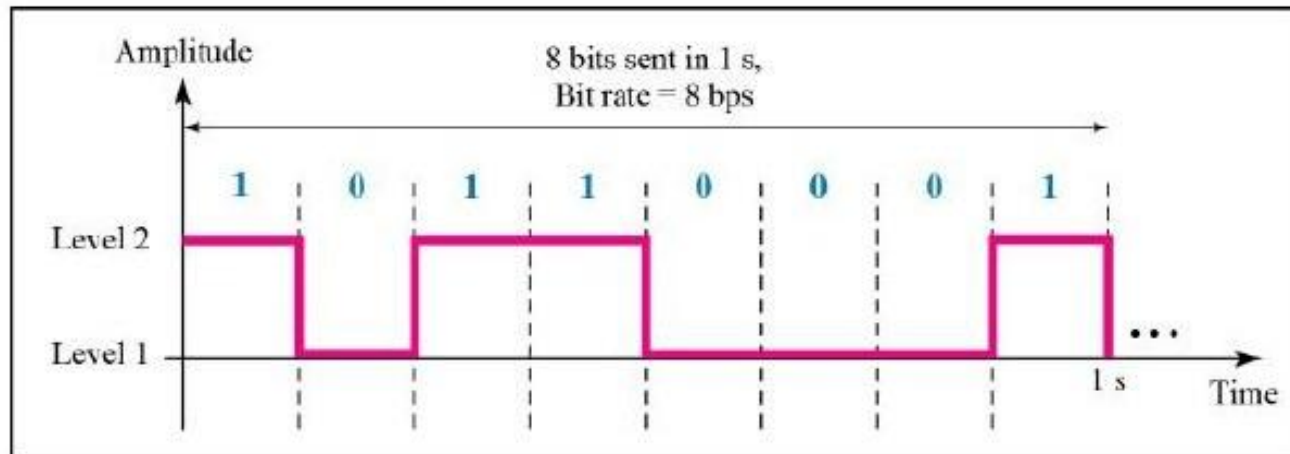


DIGITAL SIGNALS

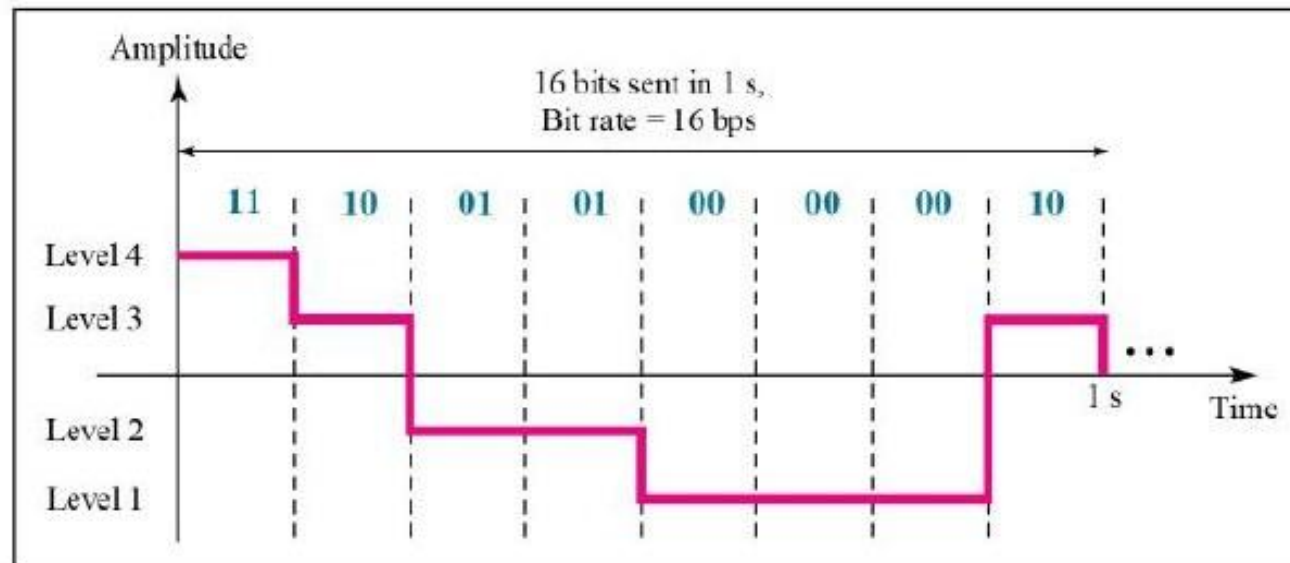
DIGITAL SIGNALS

- ✓ In addition to being represented by an analog signal, information can also be represented by a **digital signal**.
- ✓ For example, a one (1) can be encoded as a positive voltage and a zero (0) as zero voltage.
- ✓ A digital signal can have more than two levels.
- ✓ In this case, we can send more than 1 bit for each level.
- ✓ Represented by 2 major parameters
 - ✓ Bitrate – bits per second
 - ✓ Bitlength – distance one bit occupies on transmission medium; $\text{Bit Length} = \text{Propagation Speed} \times \text{Bit Duration}$

Two digital signals: one with two signal levels and the other with four signal levels



a. A digital signal with two levels



b. A digital signal with four levels

Examples

A digital signal has 8 levels. How many bits are needed per level?

We calculate the number of bits from the formula

$$\text{Number of bits per level} = \log_2 8 = 3$$

Each signal level is represented by 3 bits.

A digital signal has 9 levels. How many bits are needed per level?

Each signal level is represented by 3.17 bits.

The number of bits sent per level needs to be an integer as well as a power of 2.

Hence, 4 bits can represent one level.

Examples

Assume we need to download files at a rate of 100 pages per minute. A page is an average of 24 lines with 80 characters in each line where one character requires 8 bits. What is the required bit rate of the channel?

$$100 \times 24 \times 80 \times 8 = 1,536,000 \text{ bps} = 1.536 \text{ Mbps}$$

A digitized voice channel is made by digitizing a 4-kHz bandwidth analog voice signal. We need to sample the signal at twice the highest frequency (two samples per hertz). Assume that each sample requires 8 bits. What is the required bit rate?

$$2 \times 4000 \times 8 = 64,000 \text{ bps} = 64 \text{ kbps}$$

Example

HDTV uses digital signals to broadcast high quality video signals. There are 1920 by 1080 pixels per screen, and the screen is renewed 30 times per second. Also, 24 bits represents one color pixel.

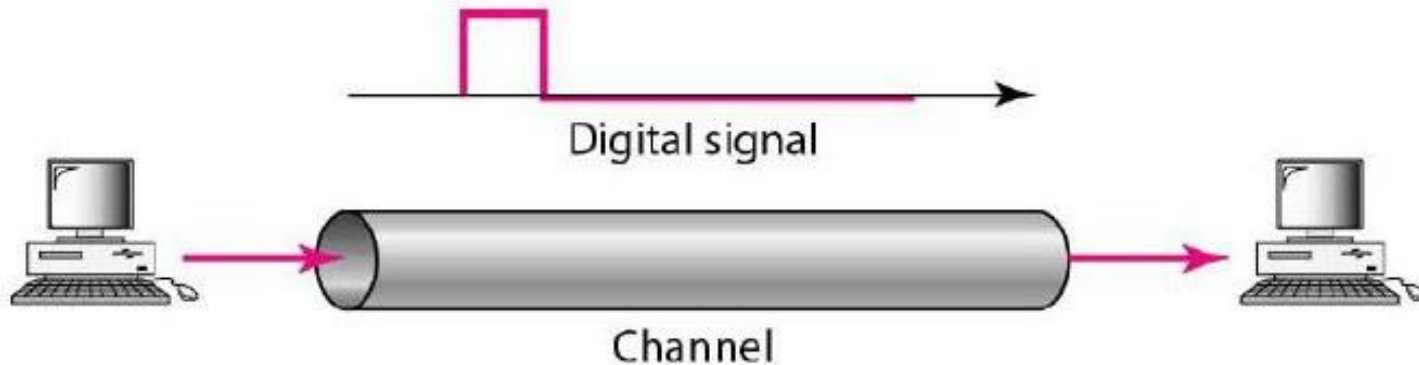
What is the bit rate for high-definition TV (HDTV)?

$$1920 \times 1080 \times 30 \times 24 = 1,492,992,000 \text{ or } 1.5 \text{ Gbps}$$

The TV stations reduce this rate to 20 to 40 Mbps through compression.

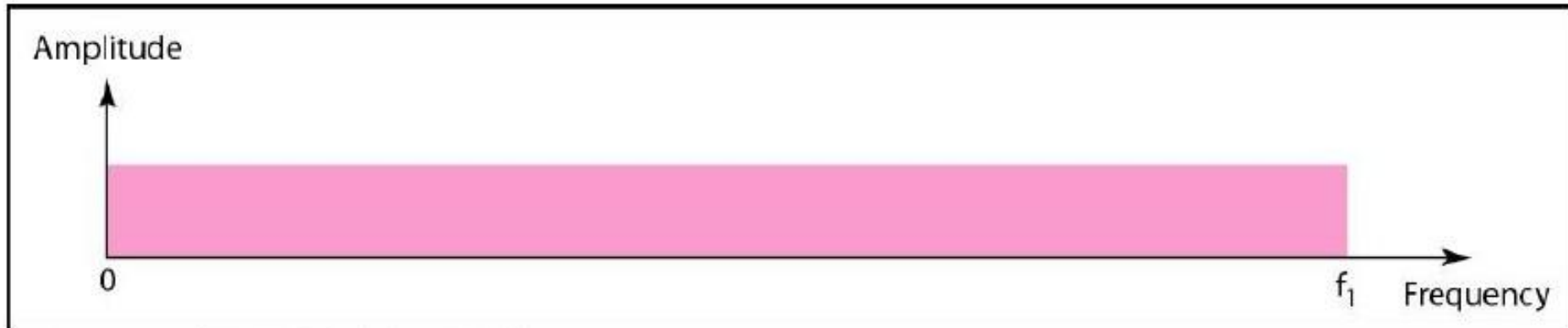
TRANSMISSION OF DIGITAL SIGNALS

Baseband transmission



A digital signal is a composite analog signal with an infinite bandwidth.

Bandwidths of two low-pass channels



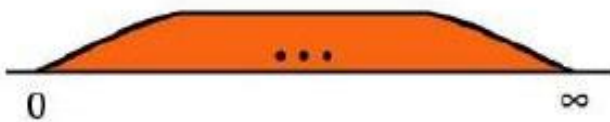
a. Low-pass channel, wide bandwidth



b. Low-pass channel, narrow bandwidth

Baseband transmission using a dedicated medium

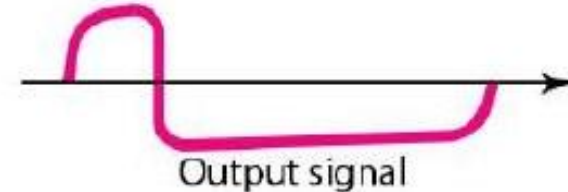
Input signal bandwidth



Bandwidth supported by medium

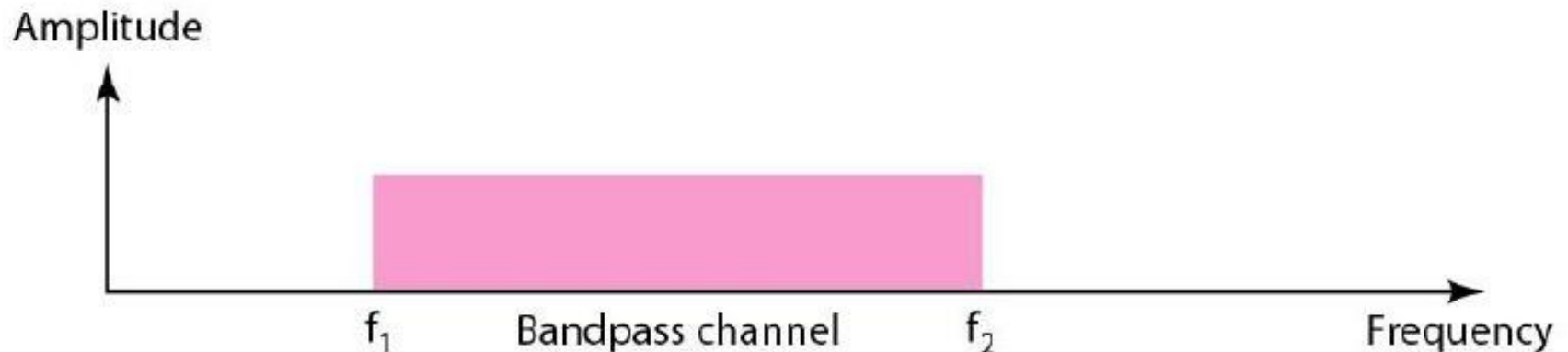


Output signal bandwidth



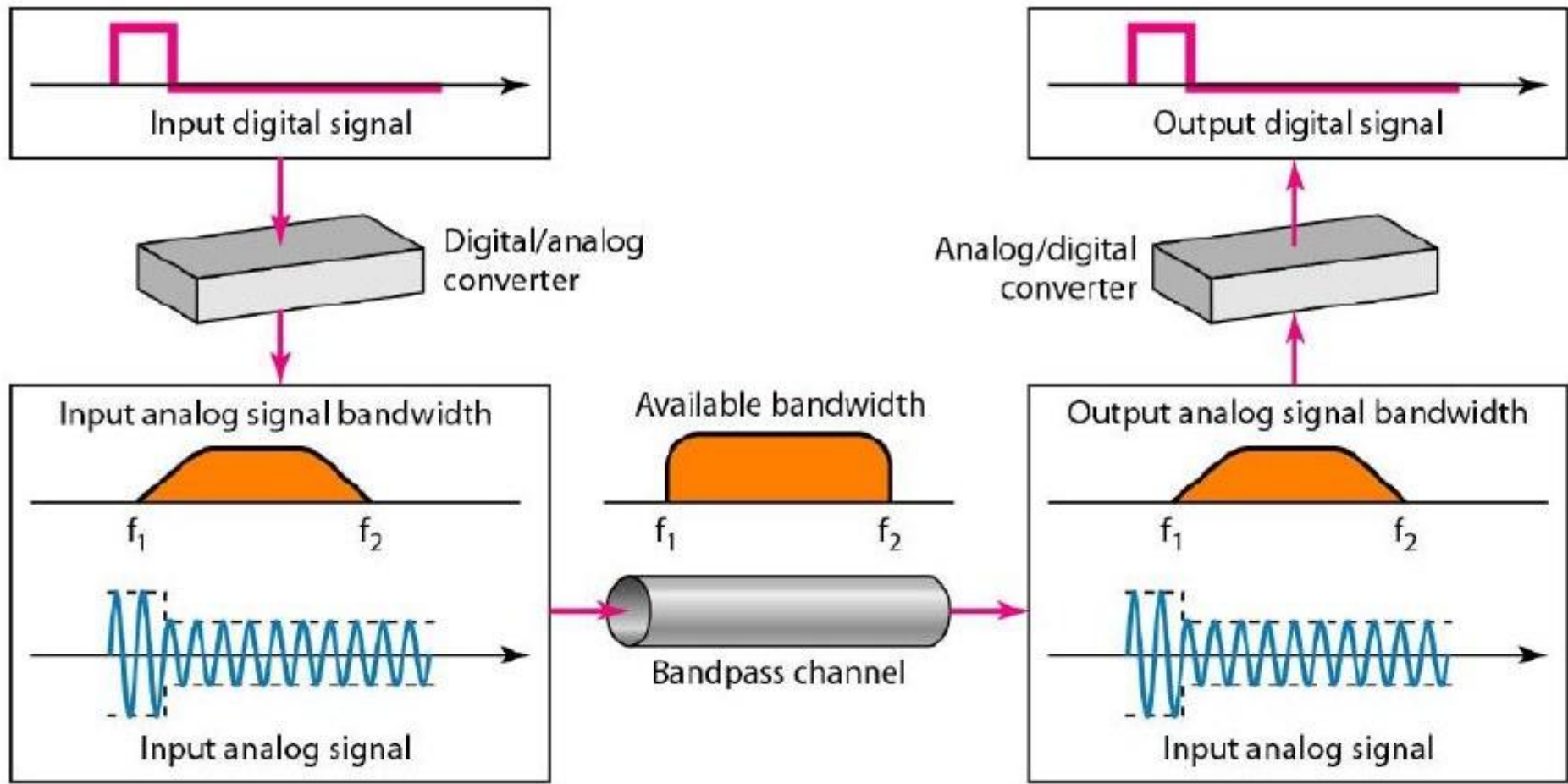
Baseband transmission of a digital signal that preserves the shape of the digital signal is possible only if we have a low-pass channel with an infinite or very wide bandwidth.

Bandwidth of a bandpass channel



If the available channel is a bandpass channel, we cannot send the digital signal directly to the channel; we need to convert the digital signal to an analog signal before transmission.

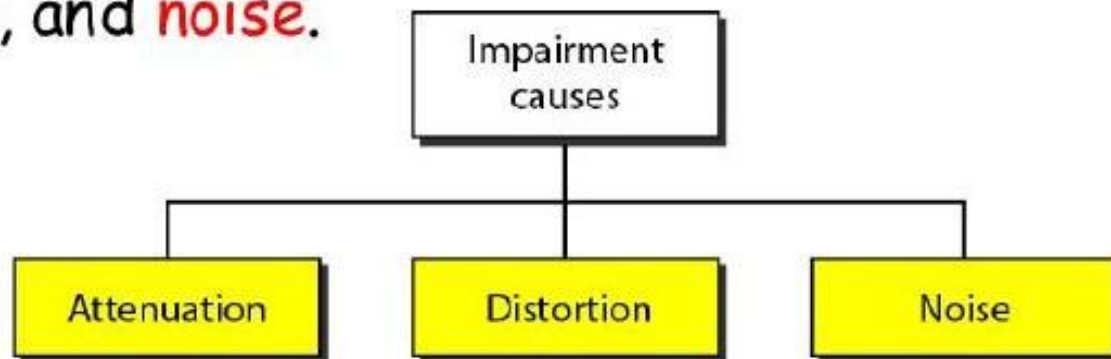
Modulation of a digital signal for transmission on a bandpass channel



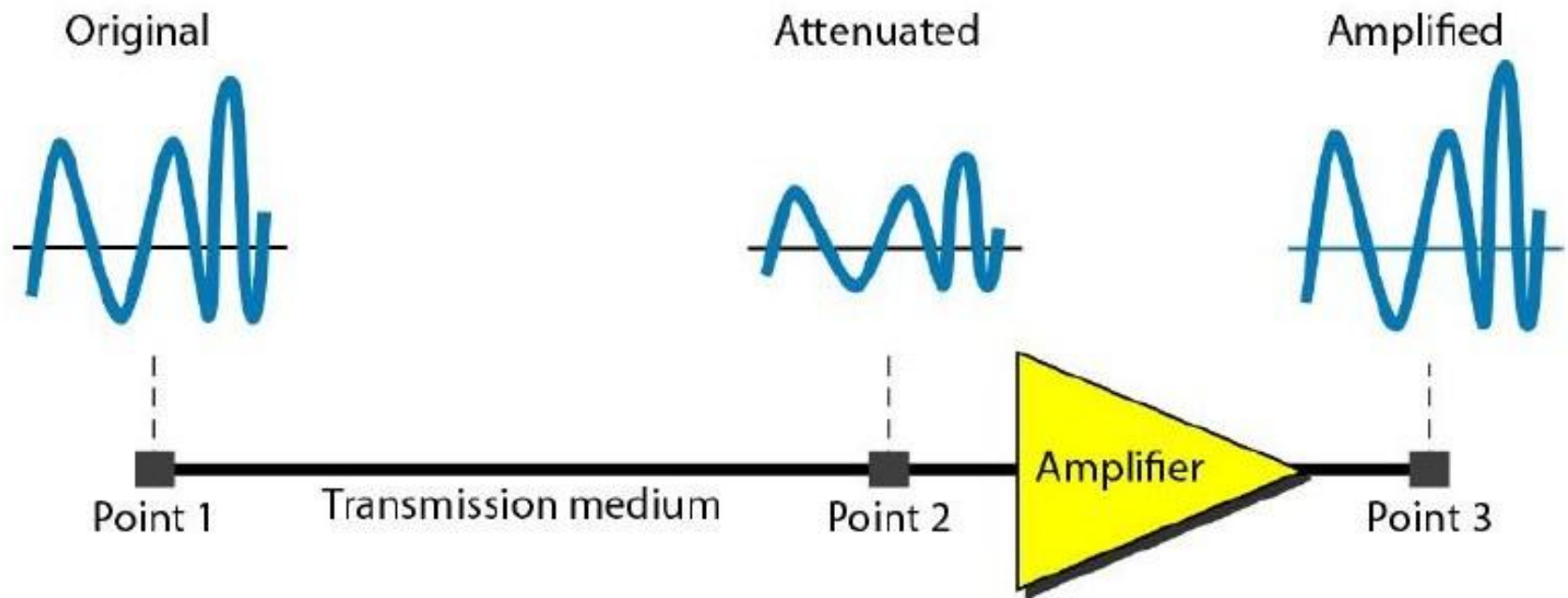
TRANSMISSION IMPAIRMENT

TRANSMISSION IMPAIRMENT

- ❑ Signals travel through transmission media, which are not perfect.
- ❑ The imperfection causes signal impairment.
- ❑ This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium.
- ❑ What is sent is not what is received.
- ❑ Three causes of impairment are **attenuation**, **distortion**, and **noise**.



Attenuation



Attenuation

- ❑ Means loss of energy -> weaker signal
- ❑ When a signal travels through a medium it loses energy overcoming the resistance of the medium
- ❑ Amplifiers are used to compensate for this loss of energy by amplifying the signal
- ❑ To show the loss or gain of energy the unit "decibel" is used.

$$dB = 10\log_{10}(P_2/P_1)$$

P_1 - input signal

P_2 - output signal

Example

Suppose a signal travels through a transmission medium and its power is reduced to one-half.

This means that P_2 is $(1/2)P_1$.

In this case, the attenuation (loss of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5 P_1}{P_1} = 10 \log_{10} 0.5 = 10(-0.3) = -3 \text{ dB}$$

A loss of 3 dB (-3 dB) is equivalent to losing one-half the power.

Example

A signal travels through an amplifier, and its power is increased 10 times.

This means that $P_2 = 10P_1$.

What is the amplification (gain of power)?

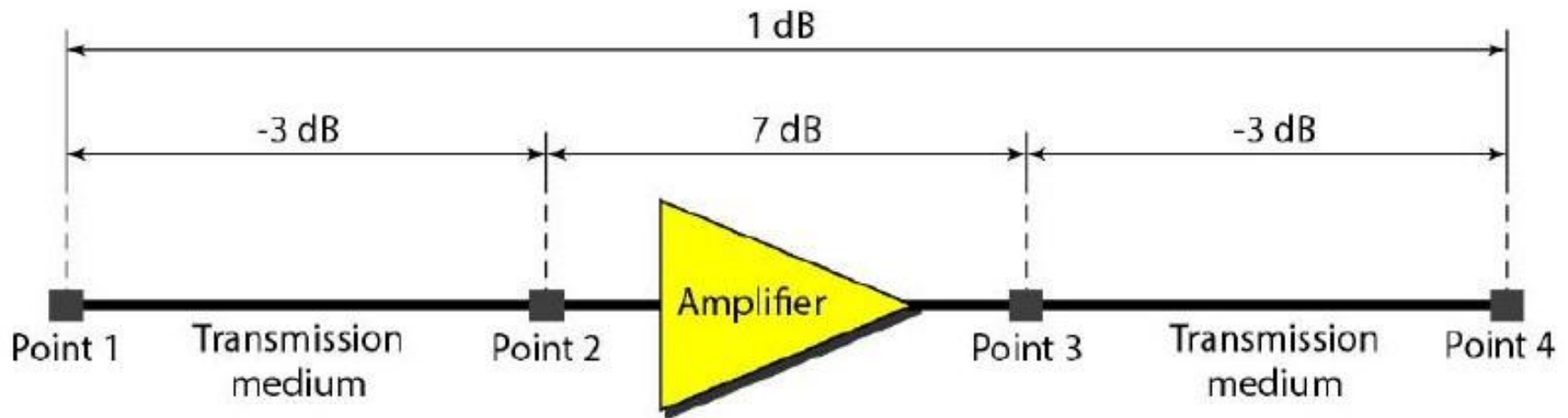
$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1}$$

$$= 10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

Example

One reason that engineers use the decibel to measure the changes in the strength of a signal is that decibel numbers can be added (or subtracted) when we are measuring several points (cascading) instead of just two.

A signal travels from point 1 to point 4.



In this case, the decibel value can be calculated as

$$\text{dB} = -3 + 7 - 3 = +1$$

Example

Sometimes the decibel is used to measure signal power in milliwatts.

In this case, it is referred to as dB_m and is calculated as $\text{dB}_m = 10 \log_{10} P_m$, where P_m is the power in milliwatts.

Calculate the power of a signal with $\text{dB}_m = -30$.

Solution

We can calculate the power in the signal as

$$\begin{aligned}\text{dB}_m &= 10 \log_{10} P_m = -30 \\ \log_{10} P_m &= -3 & P_m &= 10^{-3} \text{ mW}\end{aligned}$$

Example

The loss in a cable is usually defined in decibels per kilometer (dB/km).

If the signal at beginning of a cable with -0.3 dB/km has a power of 2 mW, what is the power of the signal at 5 km?

Solution

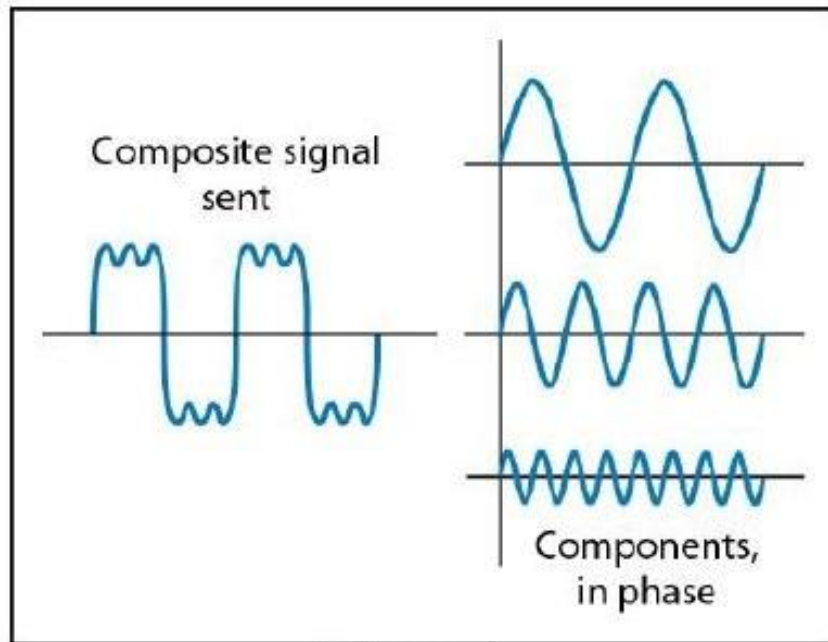
The loss in the cable in decibels is $5 \times (-0.3) = -1.5$ dB. We can calculate the power as

$$\text{dB} = 10 \log_{10} \frac{P_2}{P_1} = -1.5$$

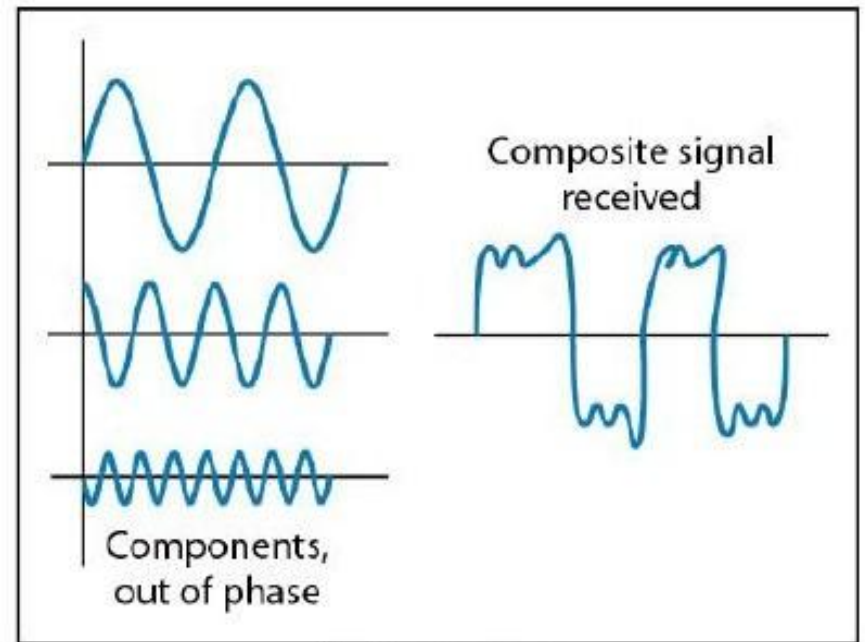
$$\frac{P_2}{P_1} = 10^{-0.15} = 0.71$$

$$P_2 = 0.71 P_1 = 0.7 \times 2 = 1.4 \text{ mW}$$

Distortion



At the sender

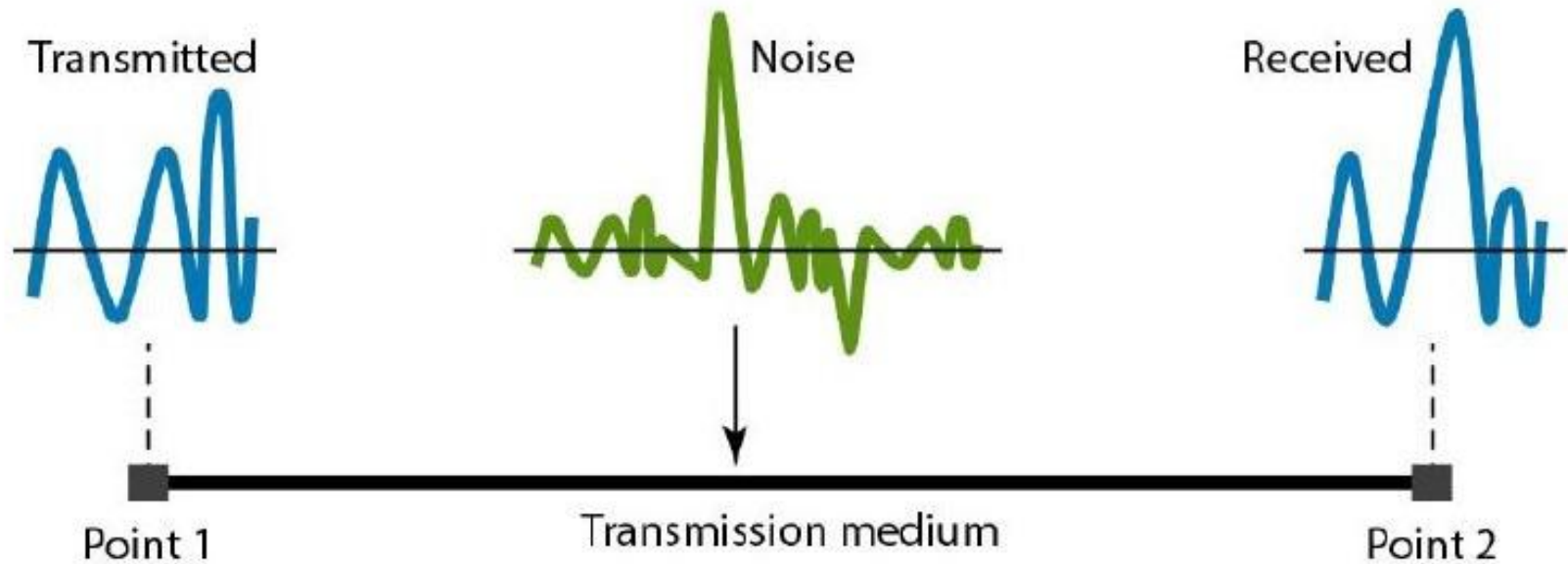


At the receiver

Distortion

- ❑ Means that the signal changes its form or shape
- ❑ Distortion occurs in composite signals
- ❑ Each frequency component has its own propagation speed traveling through a medium.
- ❑ The different components therefore arrive with different delays at the receiver.
- ❑ That means that the signals have different phases at the receiver than they did at the source.

Noise



Noise

- ❑ There are different types of noise
 - **Thermal** - random noise of electrons in the wire creates an extra signal
 - **Induced** - from motors and appliances, devices act as transmitter antenna and medium as receiving antenna.
 - **Crosstalk** - same as above but between two wires.
 - **Impulse** - Spikes that result from power lines, lightning, etc.

Signal to Noise Ratio - SNR

- ❑ To measure the quality of a system the SNR is often used.
- ❑ It indicates the strength of the signal with respect to the noise power in the system.
- ❑ It is the ratio between two powers.
- ❑ $\text{SNR} = \text{avg. signal power} / \text{avg. noise power}$
- ❑ It is sometimes given in dB and referred to as SNR_{dB} .
- ❑ $\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$

Example

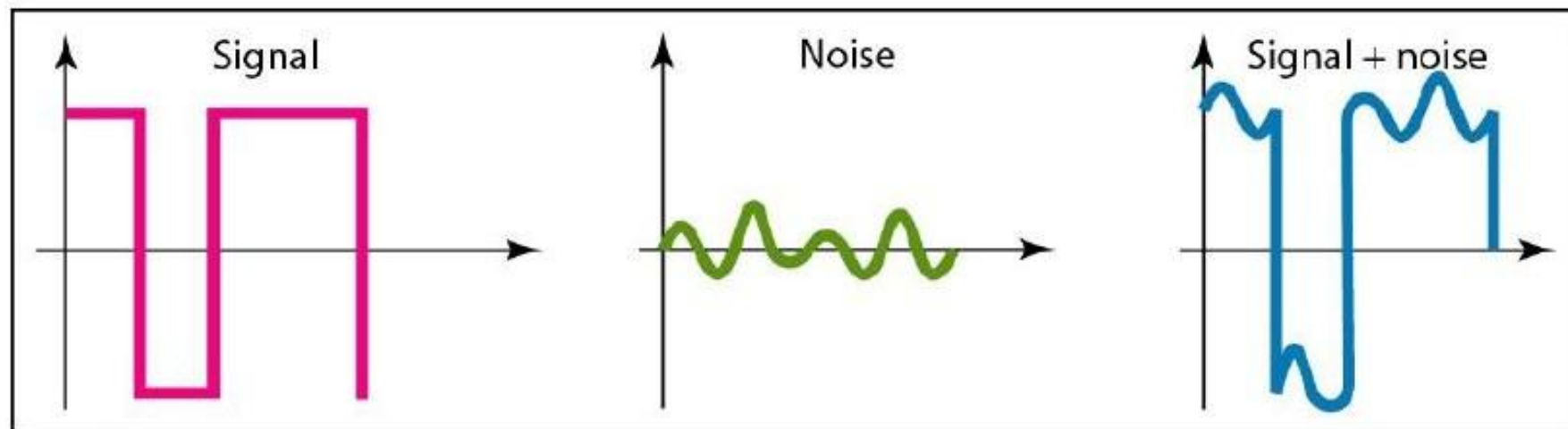
The power of a signal is 10 mW and the power of the noise is 1 μ W; what are the values of SNR and SNR_{dB}?

Solution

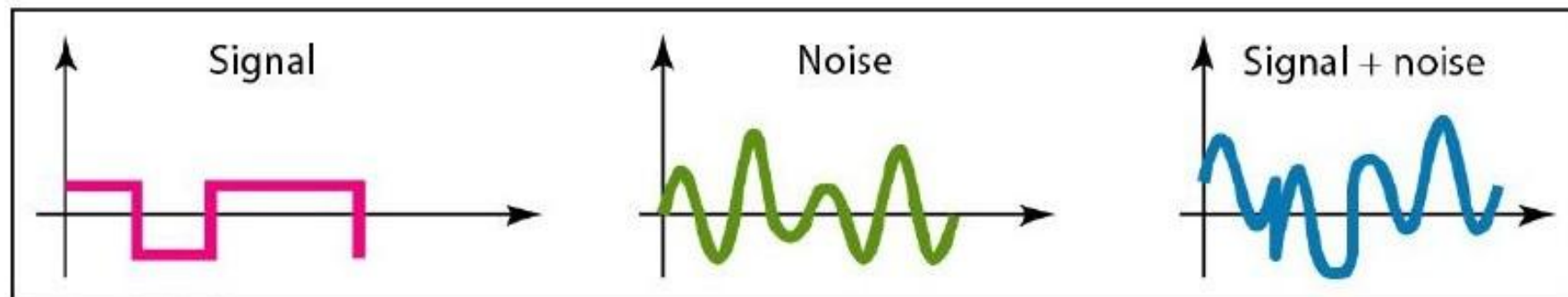
The values of SNR and SNR_{dB} can be calculated as follows:

$$\text{SNR} = \frac{10,000 \mu\text{W}}{1 \text{ mW}} = 10,000$$
$$\text{SNR}_{\text{dB}} = 10 \log_{10} 10,000 = 10 \log_{10} 10^4 = 40$$

Two cases of SNR: a high SNR and a low SNR



a. Large SNR



b. Small SNR

Example

The values of SNR and SNR_{dB} for a noiseless channel are

$$\text{SNR} = \frac{\text{signal power}}{0} = \infty$$
$$\text{SNR}_{\text{dB}} = 10 \log_{10} \infty = \infty$$

We can never achieve this ratio in real life; it is an ideal.

DATA RATE LIMITS

DATA RATE LIMITS

A very important consideration in data communications is how fast we can send data, in bits per second, over a channel.

Data rate depends on three factors:

1. The bandwidth available
2. The level of the signals we use
3. The quality of the channel (the level of noise)

Increasing the levels of a signal may reduce the reliability of the system.

Noiseless Channel: Nyquist Bit Rate

For noiseless channel,

$$\text{BitRate} = 2 \times \text{Bandwidth} \times \log_2 \text{Levels}$$

It can be applied when we have two or more levels of signals.

Increasing the levels of a signal may reduce the reliability of the system.

Examples

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. What is the maximum bit rate?

$$\text{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$

Consider the same noiseless channel transmitting a signal with four signal levels (for each level, we send 2 bits). What is the maximum bit rate?

$$\text{BitRate} = 2 \times 3000 \times \log_2 4 = 12,000 \text{ bps}$$

Example

We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

Solution

We can use the Nyquist formula as

$$\begin{aligned} 265,000 &= 2 \times 20,000 \times \log_2 L \\ \log_2 L &= 6.625 \quad L = 2^{6.625} = 98.7 \text{ levels} \end{aligned}$$

Since this result is not a power of 2, we need to either increase the number of levels or reduce the bit rate.

If we have 128 levels, the bit rate is 280 kbps.

If we have 64 levels, the bit rate is 240 kbps.

Noisy Channel: Shannon Capacity

In reality, we can not have a noiseless channel

For noisy channel,

$$\text{Capacity} = \text{Bandwidth} \times \log_2(1 + \text{SNR})$$

The Shannon capacity gives us the upper limit;
the Nyquist formula tells us how many signal levels we need.

Example

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero.

In other words, the noise is so strong that the signal is faint. What is the channel capacity?

Solution

$$C = B \log_2 (1 + \text{SNR}) = B \log_2 (1 + 0) = B \log_2 1 = B \times 0 = 0$$

This means that the capacity of this channel is zero regardless of the bandwidth.

In other words, we cannot receive any data through this channel.

Example

Let's calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000. The signal-to-noise ratio is usually 3162.

What is the channel capacity?

Solution

$$\begin{aligned} C &= B \log_2 (1 + \text{SNR}) = 3000 \log_2 (1 + 3162) = 3000 \log_2 3163 \\ &= 3000 \times 11.62 = 34,860 \text{ bps} \end{aligned}$$

This means that the highest bit rate for a telephone line is 34.860 kbps.

If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.

Example

The signal-to-noise ratio is often given in decibels.

Assume that $\text{SNR}_{\text{dB}} = 36$ and the channel bandwidth is 2 MHz.

What is the theoretical channel capacity?

Solution

$$\begin{aligned}\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR} &\quad \rightarrow \quad \text{SNR} = 10^{\text{SNR}_{\text{dB}}/10} \quad \rightarrow \quad \text{SNR} = 10^{3.6} = 3981 \\ C = B \log_2 (1 + \text{SNR}) &= 2 \times 10^6 \times \log_2 3982 = 24 \text{ Mbps}\end{aligned}$$

Example

For practical purposes, when the SNR is very high, we can assume that $\text{SNR} + 1$ is almost the same as SNR.

In these cases, the theoretical channel capacity can be simplified to

$$C = B \times \frac{\text{SNR}_{\text{dB}}}{3}$$

For example, we can calculate the theoretical capacity for the previous problem as follows:

$$C = 2 \text{ MHz} \times \frac{36}{3} = 24 \text{ Mbps}$$

Example

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63.

What are the appropriate bit rate and signal level?

Solution

First, we use the Shannon formula to find the upper limit.

$$C = B \log_2 (1 + \text{SNR}) = 10^6 \log_2 (1 + 63) = 10^6 \log_2 64 = 6 \text{ Mbps}$$

The Shannon formula gives us 6 Mbps, the upper limit. For better performance we choose something lower, 4 Mbps, for example.

Then we use the Nyquist formula to find the number of signal levels.

$$4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L \quad \rightarrow \quad L = 4$$

PERFORMANCE

PERFORMANCE

One important issue in networking is the **performance** of the network—how good is it?

- ✓ Bandwidth
- ✓ Throughput
- ✓ Latency
- ✓ Jitter

BANDWIDTH

In networking, we use the term bandwidth in two contexts

The first, bandwidth in **hertz**, refers to the range of frequencies in a composite signal or the range of frequencies that a channel can pass.

The second, bandwidth in **bits per second**, refers to the speed of bit transmission in a channel or link.

THROUGHPUT

Throughput is the measure of how fast we can send data through a network

A link may have a bandwidth of B bps but we can only send T bps through the link where T is always less than B

LATENCY

Latency or delay defines how long it takes for an entire message to completely travel to the destination from time the first bit is sent out from the source

It is the sum of propagation time, transmission time, queuing time and processing time

Propagation time = Distance / Propagation speed

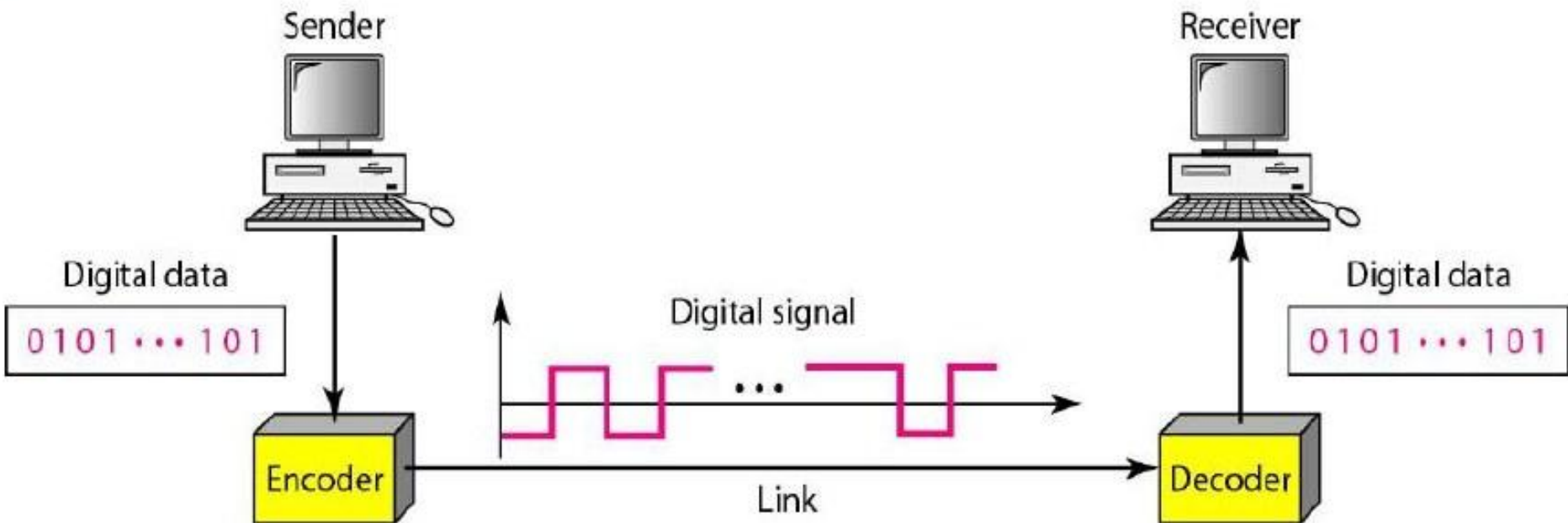
Transmission time = Message size / Bandwidth

Line Coding Schemes in Digital Transmission

Line Coding

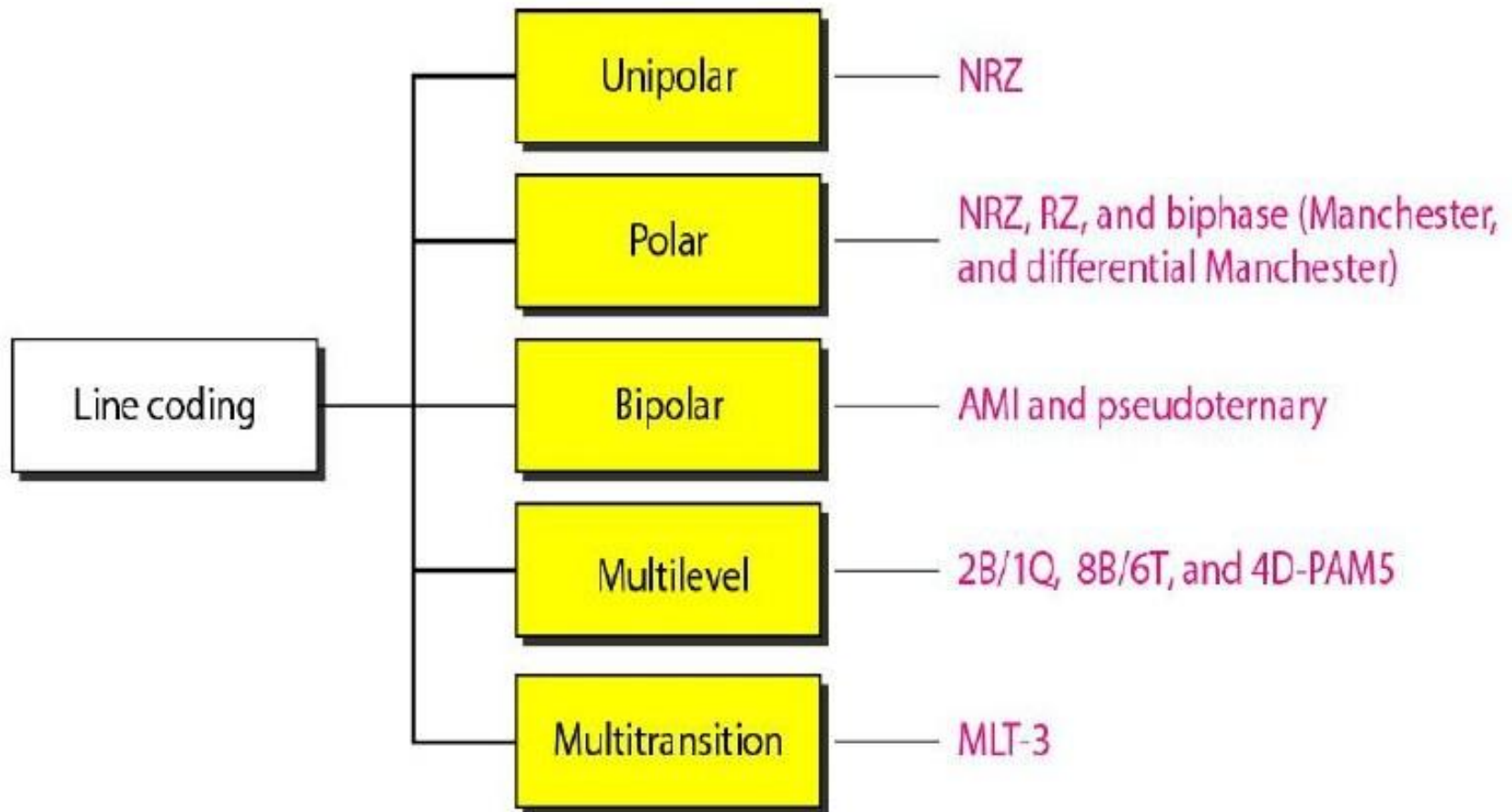
- Process of converting digital data to digital signals
- Converting a string of 1's and 0's (digital data) into a sequence of signals that denote the 1's and 0's.
- For example a high voltage level (+V) could represent a “1” and a low voltage level (-V) could represent a “0” or viceversa

Figure 4.1 *Line coding and decoding*



- At sender, digital data are encoded into digital signal
- At receiver, digital data are recreated by decoding the digital signal

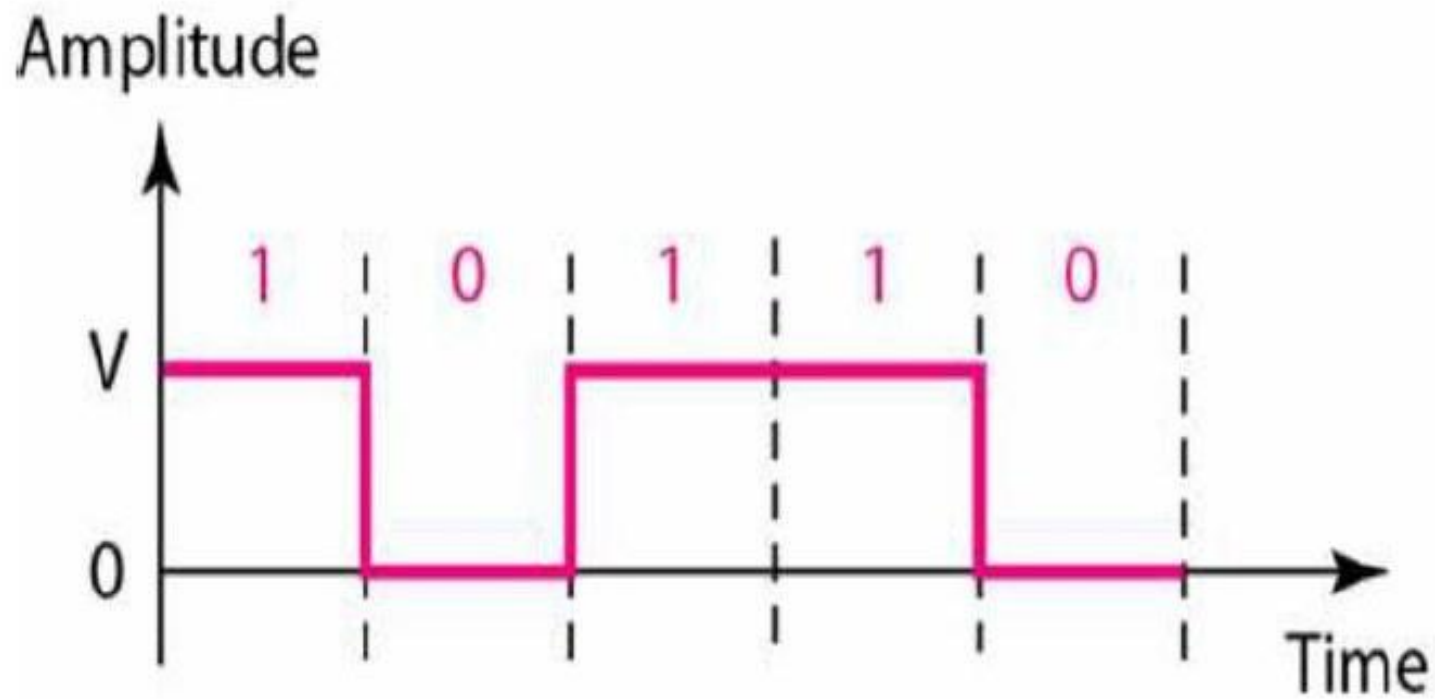
Line coding schemes



Unipolar NRZ

- All signal levels are on one side of the time axis - either above or below
- NRZ - Non Return to Zero scheme is an example of this code. The signal level does not return to zero during a symbol transmission.

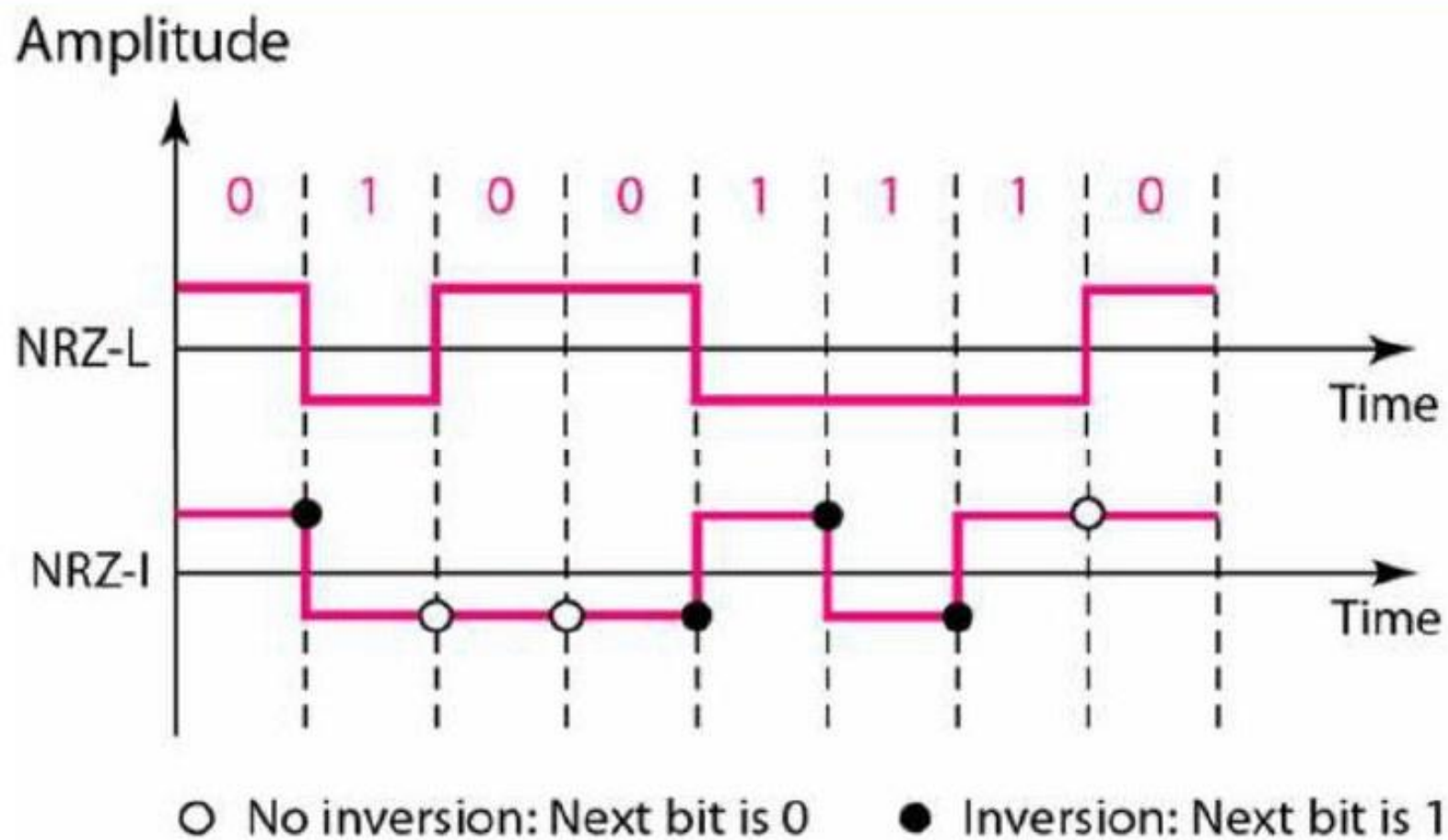
Figure 4.5 *Unipolar NRZ scheme*



Polar - NRZ

- The voltages are on both sides of the time axis.
- There are two versions:
 - NZR - Level (NRZ-L) - positive voltage for one symbol and negative for the other
 - NRZ - Inversion (NRZ-I) - the change or lack of change in polarity determines the value of a symbol. E.g. a “1” symbol inverts the polarity a “0” does not.

Figure 4.6 *Polar NRZ-L and NRZ-I schemes*



Note

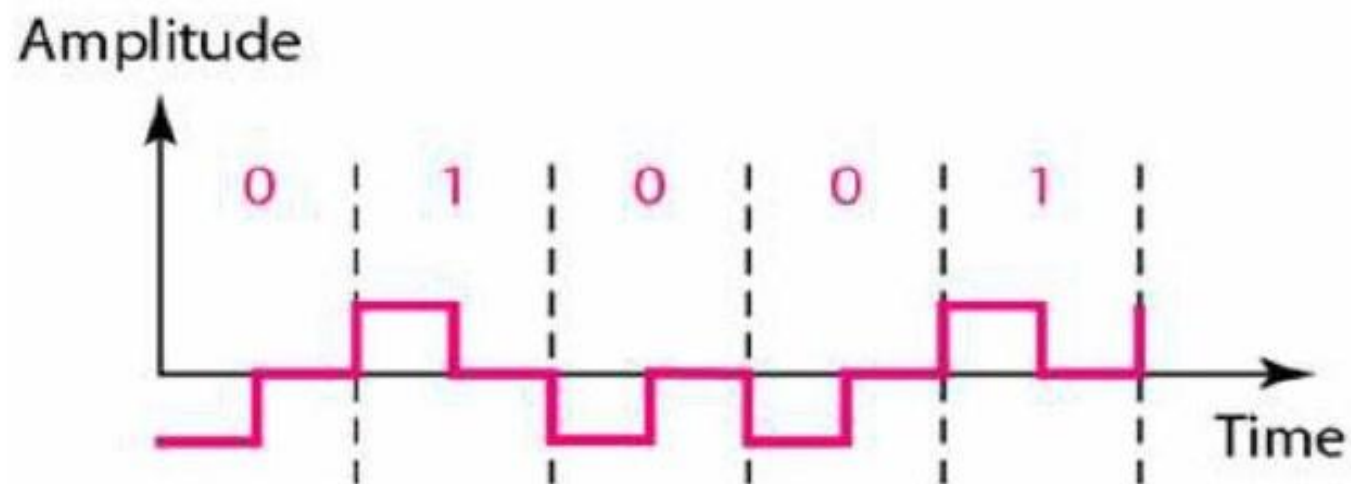
In NRZ-L the level of the voltage determines the value of the bit.

In NRZ-I the inversion or the lack of inversion determines the value of the bit.

Polar - RZ

- The Return to Zero (RZ) scheme uses three voltage values. +, 0, -.
- Each symbol has a transition in the middle. Either from high to zero or from low to zero.

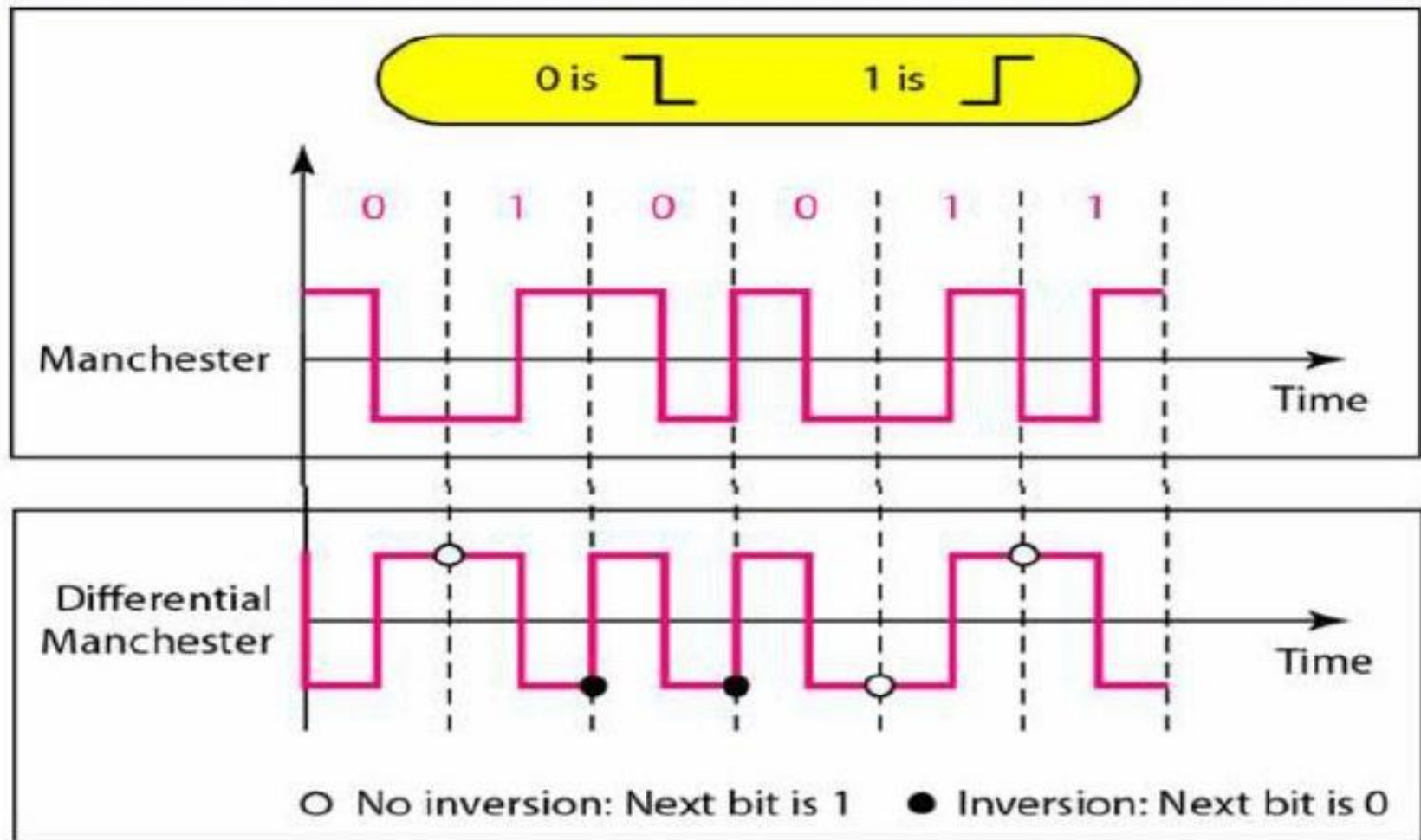
Figure 4.7 *Polar RZ scheme*



Polar - Biphase: Manchester and Differential Manchester

- **Manchester** coding consists of combining the NRZ-L and RZ schemes.
 - Every symbol has a level transition in the middle: from high to low or low to high. Uses only two voltage levels.
- **Differential Manchester** coding consists of combining the NRZ-I and RZ schemes.
 - Every symbol has a level transition in the middle. But the level at the beginning of the symbol is determined by the symbol value. One symbol causes a level change the other does not.

Figure 4.8 *Polar biphase: Manchester and differential Manchester schemes*



Note

In Manchester and differential Manchester encoding, the transition at the middle of the bit is used for synchronization.

Bipolar - AMI and Pseudoternary

- Code uses 3 voltage levels: $+$, 0 , $-$, to represent the symbols (note not transitions to zero as in RZ).
- Voltage level for one symbol is at “0” and the other alternates between $+$ & $-$.
- Bipolar Alternate Mark Inversion (AMI) - the “0” symbol is represented by zero voltage and the “1” symbol alternates between $+V$ and $-V$.
- Pseudoternary is the reverse of AMI.

Figure 4.9 *Bipolar schemes: AMI and pseudoternary*

